OpenText™ Service Manager

Troubleshoot

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1. Troubleshoot

This section helps you identify and resolve some issues that you might come across while using Service Manager.

After detecting a problem, follow these steps:

- Check the Known issues section in the Release notes.
- Check for the availability of patches that may have fixed some known issues. You can also check the Knowledge Base:

  https://www.microfocus.com/en-us/support
- Refer to the troubleshoot topics to find solutions.
- Check if the issue relates to a third-party product. Contact the respective vendor for support.
- If you are still unable to resolve the issue on your own, then analyze the Service Manager logs and contact support.

Related topics

- For more information about the supported hardware and software that you must have to successfully install and run Service Manager, see Support matrix.
- To check the list of known issues, see Known issues.
- To check if the latest patch is available for Service Manager, see Release notes.
- For more information about how to access logs, see Access logs.
- To contact support, see Contact support.

1.1. Troubleshoot server

This section provides some basic information about anticipating and correcting issues related to Service Manager server.

Some of these issues include:

- Server doesn't start
- Bad version number in .class file
- Failed to initialize or attach to shared memory environment (Windows only)
- Unable to attach to shared memory (Linux only)
- The sm -reportlbstatus command returns nothing
- The sm -reportlbstatus command returns itself
- Error 1053 occurs when starting Service Manager
- Errors occur when restarting all nodes on a host
- The "no physical address" error occurs
- The display of user information in system status is incorrect

1.1.1. Server doesn't start

You are unable to start Service Manager server.

Possible causes

- Cause 1: Incorrect database connection configuration.
- Cause 2: Unsupported version of JRE.
- Cause 3: License issues. The license has expired or is invalid.
- Cause 4: Tomcat connector failed.
- Cause 5: The `smmasterkey` key data is corrupted or doesn't match the configurations.
- Cause 6: The `encryptionkey` value is inconsistent.
- Cause 7: The JGroups cluster state transfer failed.

Solutions

Solution for Cause 1: Incorrect database connection configuration

If you are unable to start Service Manager from the command prompt, review your configuration in the `sm.ini` file and confirm that you have the correct database connection information, login, and password, using the instructions in Install.



Tip

Make sure that the following parameters are configured correctly in the sm.ini file: sqldb, sqllogin, and sqldictionary.

The Service Manager server starts as a Windows service by default. If the server starts from the command line, but not as a Windows service, uncomment the following line in the `sm.cfg` file to produce an output file which may contain some information from the Java JVM.

```
#cmd /c "sm" > C:\\process_stdout 2>&1
```

You can change the name and location of the output file by changing "C:\\process_stdout" to the path and file name you prefer. Using "C:\\process_stdout.log" creates a file called `process_stdout.log` at the root of the C drive. Once you have done this, try starting the server as a service. Any Java errors that occur will show up in the file and folder you specified.

Solution for Cause 2: Unsupported version of JRE

If you have installed a version of Java Runtime Environment (JRE) that's unsupported, the server displays the following error message:

`Exception in thread "main" java.lang.UnsupportedClassVersionError: Bad version number in .class file`

Install a supported version of the JRE as indicated in the Support matrix.

Solution for Cause 3: License issues

The IP address of the first host on which the Service Manager Server starts up must be defined in the grouplicenseip parameter in the `sm.ini` file, and the license must be bound to this IP address.

Review the server log file (sm.log) to check for license issues. A typical eye-catcher is that shared memory could not be created, which may be a consequence of license issues.

Solution for Cause 4: Tomcat connector failed

1. Check the **sm.log** file to identify whether the SM Servlet started successfully or not:
   - If started successfully: Tomcat - HTTPS/HTTP port is 13080
   - If failed to start: Tomcat - HTTPS/HTTP port 13080 is unavailable
2. Analyze the **sm.log** file or the error log, for example, "sm_13052_stdouterr.log" to check the error details.
3. Any error causing Tomcat connector fail to start will cause SM Servlet fail to start.For example, Address already in use, certificate problems (which may not cause SM Servlet fail to start in the past), and so on.

Solution for Cause 5: The `smmasterkey` key data is corrupted or doesn't match the configurations

1. If `smmasterkey` is stored in Server\RUN\sm.ini:
   Remove the `smmasterkey` parameter from sm.ini.
   If `smmasterkey` is stored in the local file system:
   Remove the file folder which includes `smmasterkey` .
2. Reset all encrypted parameter values in the sm.ini configuration file to clear text.
3. Start the Service Manager server. A new master key is automatically generated; the parameter values are encrypted again.

Solution for Cause 6: The `encryptionkey` value is inconsistent.

If the `encryptionkey` value is inconsistent with the one used by this database, the server displays the following error message:

`System starting failed. Another encryptionkey is already used in this database. Refer to the troubleshooting document for more information.`

1. Check the `encryptionkey` value in the **sm.ini** file for each host. Make sure the `encryptionkey` value is correct and consistent on each host. Correct `encryptionkey` for all nodes as required.
2. Run the **`sm -unlockdatabase`** command to clean up the database lock record.
3. Restart the Service Manager server.

Note: To guarantee that the cluster uses a same `encryptionkey` , Service Manager stores the hash value of `encryptionkey` in the database by the first start node, then the follow-up start nodes will check if the hash value of `encryptionkey` is consistent with the first start node.

However, Service Manager doesn't check the correctness of `encryptionkey`. So, be cautious when you maintain your `encryptionkey`.

Solution for Cause 7: The JGroups cluster state transfer failed

If you have not opened the necessary ports in your firewall, the server cluster is probably unstable and even fails to start with the following error message.

`JRTE F Exception in initializing cluster clustername(clustername)state transfer failed`

Review the JGroups configuration in the **tcpcluster.xml** and **udpcluster.xml** files and confirm that the necessary ports are opened in your firewall. Refer to the following rules for the necessary ports.

1. No bind port specify in JGroups config (OOB UDP)

   OS picks a random listen port.

2. Specify `bind_port` in JGroups config (OOB TCP)

   1. Listen port of Transport protocol (UDP/TCP)

      Transport protocol picks the available port in `[bind_port,bind_port+port_range]` as listen port.

      For example:

      OOB TCP: `bind_port="${tcp.bind_port:7800}", port_range="50"`

      Transport protocol picks available ports in `[7800, 7850]`. You need to open these ports in your firewall.

   2. Listen port of Failure detection protocol (FD_SOCK/FD_SOCK2)

      FD_SOCK is used in JGroups 4.2, it's independent of Transport protocol and uses a random listen port by default.

      FD_SOCK2 is used in JGroups 5.2, it picks available ports in `[Transport protocol port + offset, Transport protocol port + offset + port_range]`. In which:

      `Transport protocol port`: Transport protocol actual listen port.

      `offset`: 100 by default.

      `port_range`: 3 by default

      For example:

      OOB TCP: `bind_port="${tcp.bind_port:7800}", port_range="50"`

      For FD_SOCK2 in JGroups 5.2, it picks available ports in the range `[7900, 7953]`. You need to open these ports in your firewall.

1.1.2. Bad version number in .class file

The server doesn't start and you receive a `Bad version number in .class file` error message.

Cause

The Service Manager server requires a proper Java run-time environment (JRE) to start. If the server doesn't start and you receive a `Bad version number in .class file` error message, then your system doesn't have the proper version of the JRE installed.

Service Manager installations on Windows should already include the proper JRE necessary to start the server. If you receive this error on a Windows system, you may need to reinstall the server to get the proper JRE version.

Solution

Installations on Linux require you to provide your own JRE. Install a proper version of JRE as directed by Install and the Support matrix.

1.1.3. Failed to initialize or attach to shared memory environment (Windows only)

The Service Manager server doesn't start and you receive a "Failed to initialize or attach to shared memory environment" error message. When reviewing the sm.log file, you can find one of the following RTE error messages:

Type of error	Error Message
Insufficient free memory	RTE E Error 8 in call MapViewOfFileEx - Not enough storage is available to process this command.
Address already in use	RTE E Error 487 in call MapViewOfFileEx - Attempt to access invalid address.

Contents

- 1 Possible causes
- 2 Solutions
  - 2.1 Solution for Cause 1: There is not enough free contiguous memory available to initialize Service Manager
  - 2.2 Solution for Cause 2: The address Service Manager attempted to use for shared memory was already in use by a shared library (DLL), dynamic memory (heap), or another data structure.

Possible causes

- Cause 1: There is not enough free contiguous memory available to initialize Service Manager.

  Out-of-box, Service Manager analyzes the memory layout of the process address space and identifies free memory ranges big enough to hold shared memory. If Service Manager could not find any contiguous piece of memory large enough to contain your Service Manager shared memory, the system fails to start with the following error messages:

  ```
  RTE E Error 8 in call MapViewOfFileEx - Not enough storage is
  available
  to process this command.
  ...
  RTE E No valid shared_memory_address parameter setting was found for a
  shared memory size of nnnnnnnnnn (NNNN MB)
  ```

- Cause 2: The address Service Manager attempted to use for shared memory was already in use by a shared library (DLL), dynamic memory (heap), or another data structure.

When a Service Manager process attaches to shared memory, it has to use the same address as the shared memory was originally created with. The process might find that that address is already occupied by another data structure such as a DLL or dynamic memory (heap). The same scenario might happen if you use the shared_memory_address parameter, but the address specified in the parameter is already occupied.

If you had previously specified a shared memory address with the shared_memory_address parameter, we recommend you remove the parameter and have Service Manager automatically determine a suitable memory address. The current best practice is to only specify a shared_memory_address value if the automatic assignment fails.

After this type failure, the system will scan your system memory searching for a memory range that's large enough to contain Service Manager's shared memory. If the scan finds a suitable memory range it lists it in the log file with the following RTE information messages:

```
6116(  2716) 03/05/2010 11:06:41  RTE E shmat: MapViewOfFileEx failed
6116(  2716) 03/05/2010 11:06:41  RTE E Error 487 in call
MapViewOfFileEx - Attempt to access invalid address.
6116(  2716) 03/05/2010 11:06:41  RTE E sm_init: shmat( 516,
0x73000000, 0 ) failed for size 67108864 and key 0x2A02E500
6116(  2716) 03/05/2010 11:06:41  RTE I List of possible
shared_memory_address parameter settings for a shared memory size of
67108864 (64 MB)
6116(  2716) 03/05/2010 11:06:41  RTE I # Free range 0x7FFF0000-
0xFFFAD000 size=2147209216 (2047 MB)
6116(  2716) 03/05/2010 11:06:41  RTE I Preferred        :
shared_memory_address:0x80000000
6116(  2716) 03/05/2010 11:06:41  RTE I Lowest available :
shared_memory_address:0x80000000
6116(  2716) 03/05/2010 11:06:41  RTE I Highest available:
shared_memory_address:0xFBFA0000
6116(  2716) 03/05/2010 11:06:41  RTE I For a complete list of
available shared_mempory_address settings, run with the debugvmmap
parameter.
6116(  2716) 03/05/2010 11:06:41  RTE W !!! Use the preferred
shared_memory_address parameter above
6116(  2716) 03/05/2010 11:06:41  RTE W !!! or remove the
shared_memory_address parameter completely.
6116(  2716) 03/05/2010 11:06:41  RTE E HP Service Manager is unable
to start. Failed to initialize or attach to shared memory environment
```

The scan lists three possible shared_memory_address values. The first memory address is the preferred shared_memory_address value. This address is the least likely to already be in use by a Windows or Service Manager shared library (DLL), dynamic memory (heap), or another data structure. The next address is the lowest possible address within this range you can use for shared memory. The last address is the highest possible address within this

range you can use for shared memory. Pick one of these values for your shared_memory_address and test to see if the system starts.

Solutions

Solution for Cause 1: There is not enough free contiguous memory available to initialize Service Manager

In order to fix this issue, you must reduce the amount of shared memory.

Solution for Cause 2: The address Service Manager attempted to use for shared memory was already in use by a shared library (DLL), dynamic memory (heap), or another data structure.

If your system continues to fail with the address already in use error, you can start the server with the debugvmmap parameter to see a complete list of all address ranges large enough to contain shared memory as well as an address map of all the files that are currently using system memory. The list of memory ranges are in order of suggested use. The address ranges at the top of the list are better candidates than those at the bottom of the list. Within each address range there is a recommended address, lowest possible address, and highest possible address.

```
2724(  3572) 03/05/2010 11:06:38  RTE I List of all possible
shared_memory_address parameter settings for a shared memory size of
67108864 (64 MB)
2724(  3572) 03/05/2010 11:06:38  RTE I # Free range 0x7FFF0000-0xFFFAD000
size=2147209216 (2047 MB)
2724(  3572) 03/05/2010 11:06:38  RTE I Preferred        :
shared_memory_address:0x80000000
2724(  3572) 03/05/2010 11:06:38  RTE I Lowest available :
shared_memory_address:0x80000000
2724(  3572) 03/05/2010 11:06:38  RTE I Highest available:
shared_memory_address:0xFBFA0000
2724(  3572) 03/05/2010 11:06:38  RTE D # Free range 0x122B5000-0x30000000
size= 500477952 ( 477 MB)
2724(  3572) 03/05/2010 11:06:38  RTE D
shared_memory_address:0x20000000
2724(  3572) 03/05/2010 11:06:38  RTE D Lowest available :
shared_memory_address:0x122D0000
2724(  3572) 03/05/2010 11:06:38  RTE D Highest available:
shared_memory_address:0x2C000000
...
2724(  3572) 03/05/2010 11:06:38  RTE D # Free range 0x77DF0000-0x7C360000
size=  72810496 (  69 MB)
2724(  3572) 03/05/2010 11:06:38  RTE D
shared_memory_address:0x77E00000
2724(  3572) 03/05/2010 11:06:38  RTE D Lowest available :
shared_memory_address:0x77E00000
2724(  3572) 03/05/2010 11:06:38  RTE D Highest available:
shared_memory_address:0x78360000
2724(  3572) 03/05/2010 11:06:38  RTE D # Free range 0x039B0000-0x10000000
size= 207945728 ( 198 MB)
2724(  3572) 03/05/2010 11:06:38  RTE D
shared_memory_address:0x039C0000
2724(  3572) 03/05/2010 11:06:38  RTE D Lowest available :
shared_memory_address:0x039C0000
2724(  3572) 03/05/2010 11:06:38  RTE D Highest available:
shared_memory_address:0x0C000000
2724(  3572) 03/05/2010 11:06:38  RTE W !!! Use the preferred
shared_memory_address parameter above
2724(  3572) 03/05/2010 11:06:38  RTE W !!! or remove the
shared_memory_address parameter completely.
2724(  3572) 03/05/2010 11:06:38  RTE E HP Service Manager is unable to
start. Failed to initialize or attach to shared memory environment
```

If the system can't find any suitable memory ranges, it lists the following RTE error message:

```
RTE E No valid shared_memory_address parameter setting was found for a
shared memory size of nnnnnnnnnn (NNNN MB)
```

If there are no suggested shared memory parameters, you must reduce the amount of shared memory.

1.1.4. Unable to attach to shared memory (Linux only)

When executing Service Manager (SM) commands on Linux with a non-SM user account or using the crontab of an SM user account, an error occurs:

" `Unable to attach to shared memory. HP Service Manager may be damaged.` "

Below is an example.

You have installed Service Manager with a specific account (adminsc4 - groups: adminsc4), and you have some verification processes that are scheduled and executed by another account (sdi - groups: dba and adminsc4). Since the group "adminsc4" is the same for both accounts, you are able to execute sm command lines as follows:

1. Connect to the SM server with the account "sdi".
2. Run the following command lines:

```
su -adminsc4
cd /soft/sc400/RUN
sm –reportshm
```

The above described error occurs.

Cause

The system parameters in /etc/security/limits.conf of Linux control the resources that are available to a user or group, and may affect Service Manager’s behavior. When certain parameter values defined for a Service Manager user or for the group to which the user belongs are too low, Service Manager may fail to execute commands or even fail to start the servlet.

Since a non-SM user may execute the “su” command to run shell with a substituted SM user, the resource control of the non-SM user may or may not affect Service Manager’s behavior after the user substitutes the SM user, depending on the implementation of the Linux kernel.

Solution

We recommend that you use a Service Manager user directly to perform any SM related operations on Linux.

If failures occur when starting an SM servlet or executing SM commands, you are recommended to check the parameters in /etc/security/limits.conf.

To do a diagnostic check:

1. Set a parameter in /etc/security/limits.conf to “unlimited”.
2. Restart the system.

3. Perform the operation. If this solves the previous issue, the conclusion can be made that this limit parameter was set to a value that was too low and caused the failure.

To identify which parameter caused the problem, repeat the above steps for the parameters one by one until the problem is solved.

When the problematic parameter is identified, set it to a higher value appropriate to your Service Manager system. It is difficult to provide an accurate value for each parameter. The right value only depends on your system load. For this reason, you may need to repeat the step for each parameter and repeat until you get the correct value.

1.1.5. The sm -reportlbstatus command returns nothing

The **sm -reportlbstatus** command returns nothing when package loss is higher than 5%.

Cause

This issue is caused by a JVM bug.

Solution

If you experience this issue, run the **sm -reportlbstatus** command again.

1.1.6. The sm -reportlbstatus command returns itself

Some Service Manager processes are running, but the **sm -reportlbstatus** command only returns the command ifself:

```
------------------Server Instances--------------------------- ProcessID
ClusterAddress HttpPort HttpsPort Sessions DbgMode QMode LB State LowMem
JAVA_USED/MAX/PERCENT HEAP_USED/MAX/PERCENT ----------Non Server
Instances---------------------------- ProcessID ClusterAddress State
LowMem JAVA_USED/MAX/PERCENT HEAP_USED/MAX/PERCENT Command Line 14196
LIHAOS2-14196 RUN N (3763512/67108864/5) (338305024/4294836224/7.) -
reportlbstatus
```

Possible causes

This issue happens when you enable Jgroups TCP mode. The main causes are:

1. GossipRouter is not started normally or it is down;
2. GossipRouter is not configured in sm.ini.

Solution

Check whether the GossipRouter is started and is working as expected.

If the GossipRouter is started and working as expected, check whether a GossipRouter is configured in sm.ini but is not started or is down. In this situation, messages that resemble the following are printed to the sm.log file:

```
failed reconnecting stub to GR at
peterenvlx1.cpe.sm.com/192.168.0.8:12001: java.lang.Exception: Could not
connect to peterenvlx1.cpe.sm.com/192.168.0.8:12001
```

In this situation, start the GossipRouter process.

1.1.7. Error 1053 occurs when starting Service Manager

When you start Service Manager from Service Console on Windows and there are more than 30 servlets configured on one host, you receive the following error message:

```
Error 1053: The service did not respond to the start or control request in a timely fashion
```

Cause

The Microsoft Windows Service Control Manager controls the state (For example, started, stopped or paused) of all installed Windows services.

By default, the Service Control Manager will wait 30,000 milliseconds (30 seconds) for a service to respond.

However, certain configurations, technical restrictions, or performance issues may result in the service taking more than 30 seconds to start and report ready to the Service Control Manager.

Solution

This issue doesn't affect the startup process. You can ignore this error message.

1.1.8. Errors occur when restarting all nodes on a host

When you shut down and then immediately restart all the nodes on a host, errors that resemble the following are logged in the sm.log file.

```
JRTE I address: 59cfb5bb-82ba-1e02-b42a-7a043a29b3a1 is suspected of
having crashed.
JGRP000032: IWFVM02454-55985: no physical address for 59cfb5bb-82ba-1e02-
b42a7a043a29b3a1, dropping message
```

Cause

Jgroups detects some member process to be crashed. This process may be terminated by the previous shutdown.

Solution

This issue doesn't affect functionality. You can ignore this error message.

1.1.9. The JS E SetProperty() error occurs

The `JS E SetProperty()` error message appears in the **sm.log** file.

Cause

An error occurs when setting the property value of an object. This error is due to code errors in JavaScript.

Solution

Read the error message and check your JavaScript to debug and fix the code.

Here is an example message:

```
JS E SetProperty() called on Object 0x2A0811F0 of class SCFile('cm3r')
with an ID of String 'testfield' failed, the property value is 'Value
'test'', the JS source info: 'CustomerScript':4:1
```

In the sample error, the message indicates that at line 4 of the `CustomerScript`, an error occurred when setting the `testfield` field as "test".

In this case, possible causes can be lack of the `testfield` field in the `cm3r` table, or unmatched field type. You may need to further go and check your code as required.

1.1.10. The "no physical address" error occurs

The following message is logged in the sm.log file on a Linux-based system:

```
JGRP000032: IWFVM02439-49391: no physical address for 5f31f288-5c4f-b2f6-
974216693a1bcc65, dropping message
```

Cause

1. The process number for the specific user reaches to the maximum value;
2. The process stack configuration fails.

Solution

Increase the maximum number of processes for the Service Manager user until this message no longer appears.

For example, run the following command to increase the maximum number of user processes:

ulimit -S -u 95536

If this configuration fails due to lack of permissions, contact your System Administrator.

1.1.11. The display of user information in system status is incorrect

When the Service Manager server is running in horizontal scaled (HS) clusters, the display of user information in the system status is incorrect. When you go to **System Status** > **Display Options** > **User**s or **Scorecard** > **User Information**, the system status displays only part of the users.

Cause

1. Not all servlet processes are alive as expected.
2. There are duplicate hostnames in HS clusters.

Solution

First, run the `sm -reportlbstatus` command to check the HS cluster status. If the result displays all servlet processes, check the hostnames in the clusters to keep them unique.

1.1.12. CI relationship mass update failures

Cause

The root cause is in Tomcat versions (specifically from 9.0.74 onwards) where the default `maxParameterCount` in `server.xml` was reduced from 10,000 to 1,000. This caused requests with more than 1,000 parameters to be truncated, leading to incorrect application behavior, particularly after the customer upgraded Service Manager from 9.70 (Tomcat 9.0.65) to 9.80 (Tomcat 9.0.98).

Solution

Update the connector configuration in `TOMCAT_HOME/conf/server.xml` to set `maxParameterCount="10000"` and then restart Tomcat.

1.2. Troubleshoot common database errors

This section helps you identify and resolve some of the issues that you might come across while working with Service Manager databases. Some of these issues include:

- Cannot find SQL server
- Corresponding Join Not Defined error occurs
- Oracle_createDirectory failed during RTE startup
- Power outages and hardware failures occur
- Error occurs after you add the "Service Display Name" field to a Change view
- Cannot unload the records in the patchrelresults and scmessage tables that have a Primary Key

1.2.1. Cannot find SQL server

The system displays the `Cannot Find SQL Server` and `Error Connecting to servername` error messages.

These generic connection failure messages indicate that the Run-time Environment (RTE) attempted to log in to Service Manager and either failed or received no reply. See the following list for common causes and possible solutions.

Contents

- 1 Possible causes
- 2 Solutions
  - 2.1 Solution for Cause 1: The Service Manager server is not running
  - 2.2 Solution for Cause 2: Your login information is incorrect
  - 2.3 Solution for Cause 3: The number of valid Service Manager licenses are already in use

Possible causes

- Cause 1: The Service Manager server is not running.
- Cause 2: Your login information is incorrect.
- Cause 3: The number of valid Service Manager licenses are already in use.

Solutions

Solution for Cause 1: The Service Manager server is not running

Ensure that the server is running and then attempt to connect using the Service Manager client.

Solution for Cause 2: Your login information is incorrect

On failure, Service Manager prompts you to re-enter your login information. Ensure that you enter the correct case-sensitive login and password. For future login attempts, you can optionally modify program options to automatically set them to the correct information.

Solution for Cause 3: The number of valid Service Manager licenses are already in use

Service Manager accepts a set number of concurrent users, depending on your license agreement. If all available licenses are in use, Service Manager rejects your login. Contact your Service Manager administrator to confirm the number of licenses available.

1.2.2. Corresponding Join Not Defined error occurs

The system displays the `Corresponding Join Not Defined` error message.

Cause

This message indicates that the report is attempting to join multiple tables that have not been properly joined in the database.

When designing reports, linking tables using the application is not sufficient to create a join. You must also configure Service Manager with the equivalent values.

Solution

To join the tables:

1. Inspect the joined tables in your application. Arrows should connect to only the fields that you want to join between the tables.
2. Make an unload of both tables before making modifications so that you can return to the original configuration if needed.
3. Open Service Manager and compare the settings in the JOINDEF and ERDDEF tables to this linking scheme. You may need assistance from your administrator to confirm this task. Ensure that the following criteria apply:
    - The joins match exactly.
    - The join in Service Manager is unique, no other join connects the same tables in a different manner.
    - Stop, and then restart, the Service Manager server after creating the join.



Note

Modify the JOINDEF and ERDDEF tables in Service Manager with caution and only with prior approval from an administrator.

1.2.3. Oracle_createDirectory failed during RTE startup

A warning message is displayed during RTE startup when performing IR search:
`RTE W oracle_createDirectory failed with error: CREATE DIRECTORY SM_FULLTEXT_DIRECTORY was failed because of oracle_home is empty.`

Cause

RTE get oracle home by reading the record of library "DBMS_SUMADV_LIB" in the system table: dba_libraries.This aims to read the stop words files on the oracle home to generate the world stop list. For Oracle 19c installed on Linux platform, the oracle home is empty because there no such record, but there is $ORACLE_HOME environment for Oracle 19c.

Solution

1. Check if $ORACLE_HOME exists in the system table by running the following command:
   `SELECT DIRECTORY_PATH FROM DBA_DIRECTORIES WHERE UPPER(DIRECTORY_NAME)='ORACLE_HOME'`
2. Check if the record of library "DBMS_SUMADV_LIB" exists in the system table by running the following command:
   `SELECT FILE_SPEC FROM dba_libraries WHERE LIBRARY_NAME='DBMS_SUMADV_LIB'`
3. Analysis and resolve the problem as required.
4. Run the `sm -upgradeir` command to regenerate the IR index. Check if this problem is resolved.

1.2.4. Power outages and hardware failures occur

The system has power outages and hardware failure.

Cause

Power outages and hardware failure occur.

Solution

Ensure that procedures are in place that provide continuous power in the event of a power outage or hardware failure. In the event of a power outage or hardware failure, transactions written to the log file are compromised and can't be retrieved. If a backup process is running when the outage occurs, use your RDBMS tools to check the validity of the backup.

1.2.5. Error occurs after you add the Service Display Name field to a Change view

After you add the "Service Display Name" field to a Change view and specify the **Service** field as a group-by field, an error occurs and the view doesn't load when you sort the view by Service Display Name.

Cause

Group-By columns need to be the columns which lies in the **SELECT** columns.

When you add the **Service Display Name** field to a Change view and specify the **Service** field as a group-by field, the **Service Display Name** field will be as the **SELECT** column, and the **Service** field will be as the **Group-By** column but not lied in the **SELECT** column.

Solution

In this case, add either the **Service** field or the **Service Display Name** field to both the view and the list of group-by fields. Note that the **Service** field is preferred because this field is populated with CI Display Name values. There is no need to add the second level field, Service Display Name.

1.2.6. Cannot unload the records in the patchrelresults and scmessage tables that have a primary key

Cannot unload the records in the patchrelresults and scmessage tables that have a primary key.

Cause

Due to a limitation in Oracle 11g, records in the patchrelresults and scmessage tables that have a primary key can't be unloaded correctly in an Oracle 11g database.

Solution

In this case, you can manually change the Primary Key of the patchrelresults and scmessage tables to a Unique Key before you perform the unload.

1.3. Troubleshoot Knowledge Management

This section helps you identify and resolve some issues that you might come across while using Service Manager Knowledge Management. Some of these issues include:

- Service Manager doesn't support true multilingual capabilities when using Microsoft SQL Server
- The exported article displays the There is no content for this KM Article message
- Can't export customized fields to KM articles
- The article count in metadata.xml is inconsistent with that in the articles folder
- Can't find KM articles in the expected directory
- Stop word list isn't working

1.3.1. Service Manager doesn't support true multilingual capabilities when using Microsoft SQL Server

Service Manager doesn't support true multilingual capabilities when using Microsoft SQL Server.

Cause

MSSQL Server 2019 is now UTF-8 compliant with collation suffix as _UTF8. However, Service Manager does not support such collation.

Solution

For Service Manager 9.40 and later versions, when you run the server configuration utility to load the Applications data, you can select the Unicode data type option to store characters from multiple languages on Microsoft SQL Server. In the Unicode mode, SQL Server uses the NVARCHAR data type, which is capable of storing characters from many different languages. This Unicode support for SQL Server is available only for new installations of the Service Manager 9.40 or later applications, but not available if you upgrade the applications from a version earlier than 9.40.

If your applications are upgraded from a version earlier than 9.40, be aware that SQL Server doesn't offer a UTF-8 code page. Regular data types, such as VARCHAR, CHAR, or TEXT use single-byte code pages (such as, Windows 1252) or double-byte code pages (such as Shift-JIS), but they can't hold characters out of multiple regions. SQL Server uses a specific collation per language and allows one collation per database. For example, if you set collation for Chinese, you can't store Japanese characters in the database.



Note

This is not a problem for Oracle database, since they use plain UTF-8.

If you have a requirement to store special characters from many different languages, you must use the Microsoft SQL Server data types NVARCHAR, NCHAR, or NTEXT instead of VARCHAR, CHAR, or TEXT. These data types use UTF-16 as a code page and, therefore, are capable of storing special characters from many different languages.

Example: Shift-JIS allows you to store English and Japanese characters, but not Cryllic or Portuguese characters. On the other hand, Microsoft Windows 1251 allows you to store Cryllic characters, but not Japanese. If you want to store characters from Japanese, English, and Cryllic languages, you must use the Microsoft SQL Server data types NVARCHAR, NCHAR, or NTEXT.

Caution

Use these data types only for fields that need to be localized, not for fields that only use English characters, because these types of data take up twice as much storage space. In addition, Microsoft SQL Server has a limitation of 8060 bytes per row in a single table.

1.3.2. The exported article displays the There is no content for this KM Article message

The exported article displays the `There is no content for this KM Article` message.

Possible causes

1. The `hasView` argument in `doExport(path, hasView, exportStatus)` has the value of `false`, whereas the out-of-the-box article content fields (such as problem, solution, question, answer, error, cause, fix, summary, and reference) are all empty.
2. The `hasView` argument in `doExport(path, hasView, exportStatus)` has the value of `true`, whereas the fields defined in the template view are empty.

Solutions

Solution for cause 1:

Use the exporting view to export KM articles.

Solution for cause 2:

Check the Auto Created View for KM Export view of the corresponding **kmdoc** type. Make sure that all content fields have their definitions in this view before exporting.

1.3.3. Cannot export customized fields to KM articles

Cannot export customized fields to KM articles.

Possible causes

1. The `hasView` argument in `doExport(path, hasView, exportStatus)` is set to `false`. If the `hasView` argument is set to `false`, only OOB article content fields (such as problem, solution, question, answer, error, cause, fix, summary, and reference) can be exported, whereas other customized fields can't be exported.
2. The `hasView` argument in `doExport(path, hasView, exportStatus)` is set to `true`, but the template view lacks some customized fields.

Solutions

Solution for cause 1:

Use the exporting view if any customized fields need to be exported.

Solution for cause 2:

Check the Auto Created View for KM Export view of the corresponding kmdoc type to make sure that all content fields to be exported are defined in this view.

1.3.4. The article count in metadata.xml is inconsistent with that in the articles folder

The article count in metadata.xml is inconsistent with that in the articles folder.

Cause

The metadata.xml file is overwritten with the latest results after each exporting. However, the files created by previous exporting in both the articles folder and the resources folder aren't removed during the next exporting process. This may cause the article count inconsistency issue when comparing the article count in metadata.xml with that in the articles folder.

Solution

Use a new export path and a new folder each time.

1.3.5. Cannot find KM articles in the expected directory

You can't find KM articles in the expected directory.

Cause

Possible causes:

- KM articles are exported in the Service Manager server side instead of the client side.
- The `path` argument in `doExport(path, hasView, exportStatus)` is wrong.

  For example, if you have entered the `path` argument as `C:\Work\KMExport` for the Windows server, you can't find the exported articles in the expected directory because the correct format is `C:\\Work\\KMExport`.

Solution

Find the exported articles in the `<SM_install_location>\Server\RUN` directory.

1.3.6. Stop word list is not working

When you perform a content search using some stop words, the search result still displays the records that include these words.

Cause

The stop word list may be missing, or the stop word may be missing from the stop word list.

Solution

1. Check if the custom stop list or SM global stop list exists:
    - check if the customer stop list exists by running the following command: `select * from CTX_STOPLISTS where spl_owner='$dbuser' and spl_name='$custome_list';`
    - check if the SM global stop list exists by running the following command: `select * from CTX_STOPLISTS where spl_owner='$dbuser' and spl_name='SM_FULLTEXT_STOP_LIST';`
2. Check if the stop word exists in the stop list by running the following command: `SELECT * FROM ctx_stopwords where spw_stoplist='$stoplist';`
3. Analysis and resolve the problem as required.
4. Run the `sm -upgradeir` command to regenerate the IR index. Check if this problem still exists.

1.4. Troubleshoot Event Services

This section helps you identify and resolve some of the issues that you might come across while using Service Manager Event Services. Some of these issues include:

- Cannot process incidents events separately
- Cannot test notifications to external programs after installing SCAuto
- Cannot open problem records when there are pmo records in Event Input queue
- Cannot receive emails after opening a problem
- Cannot only send emails when opening emergency problems
- Cannot set the category for incidents that are opened through email
- Cannot verify a mail sent to myself is received
- Cannot send test reports to external programs after installing SCAuto

1.4.1. Cannot process incidents events separately

Can I have my incident events processed separately, so they're not held up by other events?

Solution

1. Copy the event agent to a new agent called (for example) probevent.
2. Copy the associated info record, substituting probevent for event.
3. Modify the query field for the event agent to read `evtype~#"pm"`.
4. Modify the query field for the probevent agent to read `evtype#"pm"`.

1.4.2. Cannot send test notifications to external programs after installing SCAuto

Need instructions about how to send test notifications to external programs after installing SCAuto

Solution

To send a test notification of a new device to my external program after installing SCAuto:

1. Click **System Administration** > **Ongoing Maintenance** > **Communication Utilities** > **Write an Output Event**.
2. Select Configuration Management.
3. Click **Write Event**.
4. Note the event type.
5. Click **Queues > Output Event**.
6. Search for an event with a event code of `icma`.

   The Event Services Output Queue should include the test item you generated.

1.4.3. Cannot open problem records when there are pmo records in Event Input queue

Cannot open problem records even though there are pmo records in the Event Input queue.

Cause

The cause of this issue may vary. See the following troubleshoot information.

Solution

To troubleshoot this problem:

1. Verify the records in the queue have processed. If the records have processed, there should be no Event Time value.
    - The Status field should contain a value.
    - Any messages should appear in the Messages field.
2. Verify there is an active event agent.
    - In the event agent, the Stop button should be enabled and a start time and an idle time should appear.
    - Click the Refresh button to reset the idle time to 00:00:00. It should begin increasing again.
    - If the Start button is enabled and there is no start and idle time, click Start and wait until the problem agent recycles.
3. Verify the following, and then wait for the event processor to recycle:
    - The event schedule record exists.
    - The Class field has a value of event.
    - The Status field has a value of rescheduled.
4. If there is an active event agent, check the Event Registration table.
    - Are there entries for Event pmo with a Type of input?
    - Is the Execute Condition true?
    - Verify the content of the pmo registration.
5. Verify there are event maps matching the Event Map Name values in the registration record. The same rules apply to all event types, not just pmo.
6. Verify that you have provided an active category.

1.4.4. Cannot receive emails after opening a problem

Cannot receive email even after installing Service Manager and opening a problem.

Cause

The cause of this issue may vary. See the following troubleshoot information.

Solution

To troubleshoot this problem:

1. Verify you are a member of the assignment group for the problem. If not, you will not receive notification of any kind.
2. Determine whether you are attempting to send email to yourself when you open a problem. Service Manager doesn't send email to the individual who is opening, updating or closing a problem, regardless of their membership in the assignment group.
3. Sign in to Service Manager as someone else.
4. Open a new problem.
5. Determine whether the operator to whom you are sending email has an email address specified in the operator record.
6. Ensure it is correct.
7. Check the Message Class file for External Email records.
8. Is there one for problem open?
9. If not, add one.
10. Verify there are records in the event output queue with a type of email.
11. If so, determine whether the JavaMail agent or another email agent is active.

    Use Agent Status to:

    - Check that in the event agent, the Stop button should be enabled, and a start time and an idle time should appear.
    - Click the Refresh button to reset idle time to 00:00:00. It should begin increasing again.
    - If the Start button is enabled and there is no start and idle Time, click Start and wait until the problem agent recycles.
12. Determine if there is an output type event registration record for email.
13. If the JavaMail agent or another email agent is active and you still don't receive mail, stop the agent.
14. Open a problem and check the event output queue for new events with a type of email.
15. If a new email event is added to the queue, restart the JavaMail agent or another email agent.
    When the email has been sent, the event is deleted only if the `keepmai` l parameter is turned off.

Note

Always check the Service Manager Message Log and any external log files for errors. All SCAutomate errors are logged with a class of `event management errors`.

1.4.5. Cannot only send emails when opening emergency problems

Cannot only send emails when a user opens problems with a priority code of emergency.

Cause

The cause of this issue is unknown.

Solution

To solve this problem:

1. Click **Tailoring** > **Notifications** > **External Email Message Class**.
2. Remove any External Email record for problem open.
3. Click **Tailoring** > **Tailoring Tools** > **Macros**.
4. Select the incidents macro that sends the email.
5. Change the Condition field value to `nullsub(priority.code in $L.new, "")="1"` .

1.4.6. Cannot set the category for incidents that are opened through email

Cannot set the category for the message when a user opens problems through email.

Cause

The cause of this issue is unknown.

Solution

1. Put each field assignment on a separate line in your mail message, uniquely identified by a label.
2. Use mapping expressions to extract the information and populate the appropriate fields in the incident.

```
The mail message looks like this:
      Fri, 12 Jan 05 14:40:41 -08:00
    Re: Test to assign a category
     John Jones <john@mac.acme.com>
      CATEGORY: example
 This is line 1 of the text of mail.
       This is line 2 of the text of mail.
```

3. In the eventin record, the `evfields` should appear as follows:

```
xjohn^^^^Fri, 12 Jan 01 14:40:41 -08:00|Re: Test to assign a
category|John Jones<john@hp800.hp.com>|CATEGORY:
example||This is line 1 of the text of mail.|This is line 2
of the text of mail.|^^^^^^^^^^^^^John Jones
<john@mac.acme.com>^^
```

4. In the problem open event map record for the category field, enter the following Initialization statements:

```
$axtype=type in $axces.target
if (index("axmail", evuser in $axces)>0) then $axtype=type
in $axces.target
if (index("axmail", evuser in $axces)>0) then
($ax.action=denull(action in $axces.target);$axl=lng($ax.action))
if (index("axmail", evuser in $axces)>0) then for $axpos = 1 to $axl
do ($axt=$axpos in $ax.action;if $axt#"CATEGORY
then ($axtype=substr($axt, 10, lng($axt) - 9);
$ax.action=delete($ax.action, $axpos);
action in $axces.target=$ax.action))
```

5. Enter the following Instructions:

```
if (index("axmail", evuser in $axces)>0) then category in
$axces.target=$axtype
cleanup($axtype);cleanup($axt);cleanup($axpos);cleanup($axl)
cleanup($ax.action)
```

This procedure (substituting other field names) allows specification of any incident field values within the body of the email message as long as the map record in which the instructions are entered has a higher sequence number than that of the action (or `update.action`) field.

1.4.7. Cannot verify if a mail sent to oneself is received

A user can't verify if a mail sent to oneself is received.

Cause

The cause of this issue is unknown.

Solution

To troubleshoot this problem:

Click **Mail > All Mail**.

Your message should appear in the list of email messages.

1.4.8. Cannot send test reports to external programs after installing SCAuto

Cannot send test reports to external programs after installing SCAuto.

Cause

The cause of this issue is unknown.

Solution

To test sending an incident report to my external program once SCAuto/SDK is installed:

1. Click **System Administration** > **Ongoing Maintenance** > **Communication Utilities** > **Write an Output Event**.
2. Select Incident Management.
3. Click **Write Event**.
4. Note the incident number.
5. Click **Queues > Output Event**.
6. Search for an event with a event code of `pmo`.

   The Event Services Output Queue should include the test incident report you generated.

1.5. Troubleshoot clients

This section helps you identify and resolve some issues that you might come across while using Service Manager clients.

Some of these issues include:

- Fail to sign in to Service Manager

Web client

- Unable to view an attachment from the web client
- Certain shortcut doesn't work in the web client

Windows client

- No trusted certificate found in the Windows client
- Windows client crash due to multiple auto login attempt

Mobility client

- Mobility client doesn't support Dynamic View Dependencies (DVD)

1.5.1. Fail to sign in to Service Manager

When you attempt to sign in to Service Manager, the login page always displays the following error message.

`You have been logged out due to session timeout.`

When you check the log file, you can find the `Authentication failure` error.

Possible causes

- Cause 1: This issue occurs due to the user access and security control
- Cause 2: This issue occurs due to the RFC 2617 authentication scheme requirement

Solutions

Solution for Cause 1: User access and security control

You may contact the System Administrator to check the access permission to Service Manager. See Controlling user access and security.

Solution for Cause 2: RFC 2617 authentication scheme

Based on the RFC 2617 authentication scheme, the colon (":") character cannot be used in usernames starting from 9.72. If existing users have the colon (":") character included in their usernames, they will fail to sign in to Service Manager.

In such cases, the System Administrator needs to update their user profiles and remove the colon (":") character from these usernames.

1.5.2. Unable to view an attachment from the web client

Cannot open an attachment in a record (for example, an incident record) from the web client.

Cause

The **getAbsolutePath()** function in Attachment.java returns the full paths of attachments. The web tier supports long full paths (greater than 255 characters) for attachments when running on Linux, or Windows with an NTFS file system (not a FAT file system).

If the web tier is running on Windows with a FAT file system, and if the attachment file full path is greater than 255 characters, you can't open the attachment. This issue is most likely to occur on WebSphere systems where the deployment path to the temp directory is prefixed by various WebSphere node name and server names, and attachment file full paths are dependent on the length of WAR file name and the installation path of the web application server.

Solution

The following is an example of a full path to an attachment that's greater than 255 characters, where the "x" string represents the unique directory created for each user session that handles attachment files:

```
/app/IBM/WebSphere7/AppServer/profiles/AppSrv01/temp/qasolaris01Node01/server1/PROD_webtier-7_02_013_P3_ear_SSO_HELP_SRVRPROD_webtier-7_02_013_P3_ear_SSO_HELP_SRVR_war/PROD_webtier-7_02_013_P3_ear_SSO_HELP_SRVRPROD_webtier-7_02_013_P3_ear_SSO_HELP_SRVR.war/attachments/xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx/sm_attachment.txt
```



Note

If you use a very long file name or folder name, which causes the full path of the file to exceed 255 bytes, you will not be able to delete the web application server's temporary file/folder that's generated when you upload the file (normally, this is located in the web application server installation directory). Deleting the temporary file/folder will fail with a system warning message that states the file name is too long. In this case,follow these steps to delete the file:

1. Rename the temporary file to a short name, and then delete it.
2. If the deletion still fails, open the system command prompt, type dir /x to display the file's short name, and then run the **rmdir {short name} /S** command to delete the folder.

1.5.3. Certain shortcut doesn't work in the web client

Cannot focus on the first button in a toolbar in main content page when pressing Alt+Ctrl+M in the Edge browser.

Cause

This is because this keyboard shortcut is already used as a browser shortcut.

Solution

Manually change the key values for these shortcuts in shortcut.xml to use other keys.

1.5.4. No trusted certificate found in the Windows client

Users running the Windows client from a redeployed installer may experience the following error.

`Error Connecting to server, server not up or wrong connection parameters: https://<hostname>:<port>`

`Cause: javax.xml.soap.SOAPException: Message send failed - sun.security.validator.ValidatorException: No trusted certificate found`

Cause

The path to the CA certificates file on the local Windows client installation is different from the path that the Client Packaging Utility originally saved. This is usually the result of the local user saving the Windows client to a different path than the Windows client used by the Client Configuration Utility.

Solution

Change the path to the CA Certificate file.

1. Sign in to the machine running the Windows client using a local administrator account.
2. Start the Windows Client. The Windows client displays the "No trusted certificate found" error message.
3. Click **OK**.
4. Close the Connections window.
5. Click **Window** > **Preferences** > **Service Manager** > **Security**.
6. In the **CA Certificates File** field, type or select the path to the CA certificates file.

1.5.5. Issues occur when viewing buttons

For some monitors, when the display screen is set to 1024×768 resolution, the toolbar buttons preview, approve internal, and submit for approval don't appear on the toolbar. Also, the buttons in the toolbar of the Contribute Document form don't fit properly on the toolbar. They overlap.

Cause

Particularly UI layouts aren't able to adapt to some screen resolutions.

Solution

On the affected forms, the drop-down option menu has the buttons available.

1.5.6. Mobility client doesn't support Dynamic View Dependencies (DVD)

In the Mobility client, some widgets on the forms don't support the Dynamic View Dependencies (DVD) feature of Forms Designer.

Cause

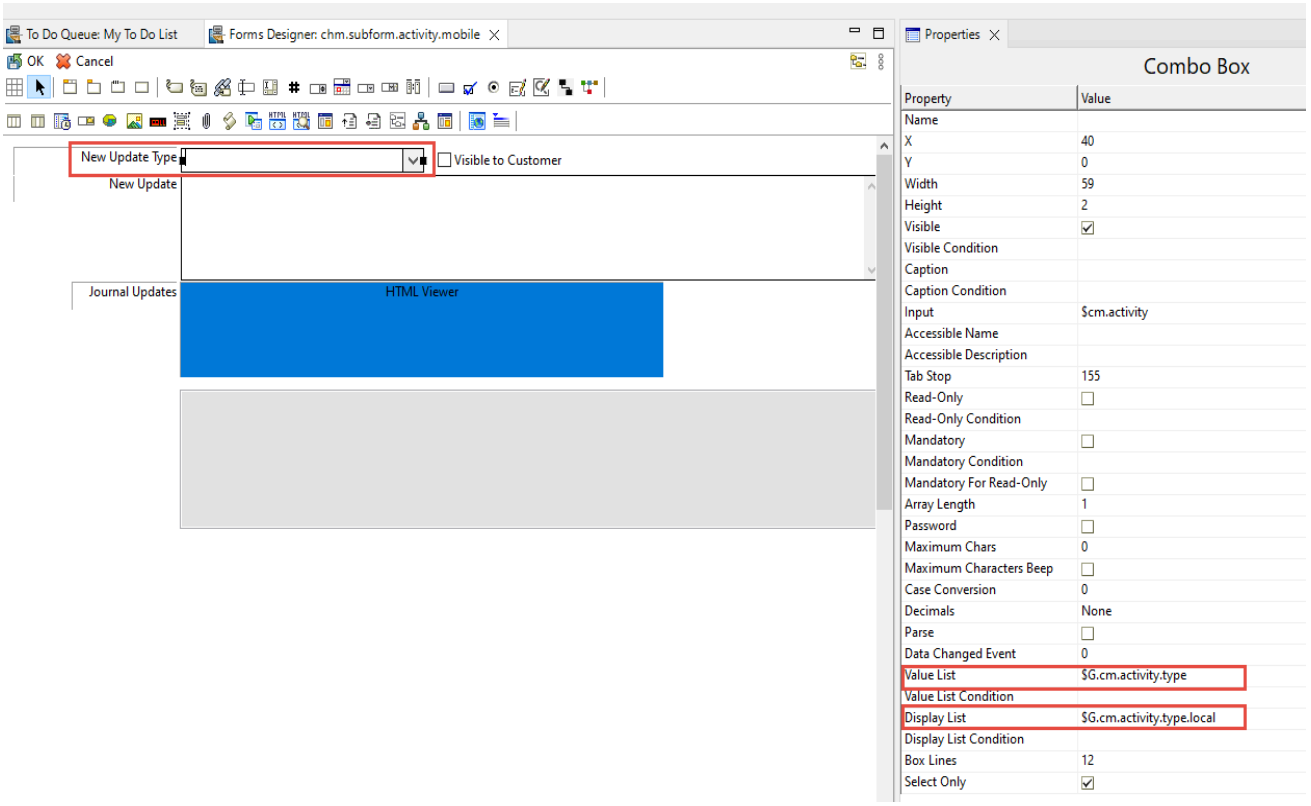
The current version of Service Manager Mobile Applications doesn't support the DVD feature.

Solution

Since the current version of Service Manager Mobile Applications doesn't support the DVD feature, change the properties of **Value List** and **Display List** as below to avoid using the DVD feature.

Property	Old Value	New Value
Value List	Null	$G.cm.activity.type
Display List	Null	$G.cm.activity.type.local

Refer to the following screenshot as an example:



1.5.7. Windows client crash due to multiple auto login attempt

When Windows client have **two configurations with Auto Login enabled**, the Service Manager (SM) Windows client may throw an error or crash.

Cause

When the two Service Manager windows client configurations are enabled simultaneously with auto login, it can lead to create conflict causing the service manager windows client to crash.

Solution

To resolve the issue follow the following steps:

1. Stop the Service Manager (SM) server.
2. Launch the Windows client.
3. Ensure only one client is auto login enabled per service manager server.
4. Start the server again.
5. Log in to the Windows client.

1.6. Troubleshoot integration

This section helps you identify and resolve some of the issues that you might come across while using Service Manager integration.

- Troubleshoot Teams integration
- Troubleshoot UCMDB integration
- Troubleshoot SAP integration
- Troubleshoot PPM integration
- Troubleshoot CIT integration

1.6.1. Troubleshoot Teams integration

This section helps you identify and resolve some of the issues that you might come across while using the Teams integration. Some of these issues include:

- New channels are not displayed when clicking Synchronize
- Cannot open the URL correctly after clicking Show in SM
- Errors occur when displaying the detail of an Incident
- Bot fails to find an operator
- Bot displays a maximum active login error

1.6.1.1. New channels are not displayed when clicking Synchronize

Service Manager doesn’t list the newly added channels when you click **Synchronize** immediately after adding new channels in Teams.

Cause

Due to a limitation of the Microsoft Graph API, it may take some time to synchronize the channel updates.

Solution

Wait a few minutes after adding a new team or channel in Teams, updating or removing the teams or channels in Teams, and then click the **Synchronize** button again.

1.6.1.2. Cannot open the URL correctly after clicking Show in SM

Cannot open the URL correctly after clicking the **Show in SM** button though you have correctly set the server's IP address in the **Web Server URL** field in **System Information Record**.

Cause

The Teams Card OpenUrl action doesn't support IP address.

Solution

Use the server’s Fully Qualified Domain Name (FQDN) instead of the IP address in the **Web Server URL** field.

1.6.1.3. Errors occur when displaying the detail of an Incident

After clicking the **Show in SM** button, the URL opens correctly but errors occur when Service Manager displays the detail of an Incident.

Cause

- You may not correctly configure query hash parameters.
- In **System Information Record**, you may not use the server's Fully Qualified Domain Name (FQDN) instead of the IP address in the Web Server URL field.

Solution

Correctly configure the query hash parameters. For more information, see Enable query hash and Add the web server URL.

1.6.1.4. Bot fails to find an operator

A bot returns the following error message during operation.

Agent bot:

`Operator xxx was not found.`

Virtual agent:

`Sorry, you are not a Service Manager user. Please contact the administrator.`

Cause

Service Manager can't find the corresponding operator record of the Teams user.

Solution

Troubleshoot this issue as follows:

- Check sm.log for detailed information.
- Check the **MicrosoftTeams** SMIS instance. Make sure that this integration instance has started and connected successfully.
- Make sure that the corresponding operator record of the Teams user exists in Service Manager. Either the operator's login name or email must be the same as the user's Teams login ID.

1.6.1.5. Bot displays a maximum active login error

Agent bot or virtual agent displays the following error message:

`Login failed. Maximum active logins for this user exceeded.`

Cause

There are too many concurrent operations of the integration operator.

Solution

Increase the value of **Max Logins** in the integration operator record in Service Manager.

1.6.2. Troubleshoot UCMDB integration

This section helps you identify and resolve some of the issues that you might come across while using the UCMDB integration. Some of these issues include:

- The Get Related and Impact Simulation widgets don't work
- Data push fails when using ServiceManagerEnhancedAdapter9.41
- The query is not configured in smPushConf.xml when using ServiceManagerEnhancedAdapter9.41
- The mapping file is not well-formed when using ServiceManagerEnhancedAdapter9.41
- Data population fails when using ServiceManagerEnhancedAdapter9.41
- The TQL query is not configured in smPopConf.xml when using ServiceManagerEnhancedAdapter9.41
- Nonexistent mapping file name is defined for a TQL query in smPopConf.xml when using ServiceManagerEnhancedAdapter9.41
- Data federation fails when using ServiceManagerEnhancedAdapter9.41
- The mapping file for the federation TQL query is not well-formed when using ServiceManagerEnhancedAdapter9.41
- Data push fails when using ServiceManagerAdapter9.41
- Query is not configured in smSyncConfFile.xml when using ServiceManagerAdapter9.41
- Nonexistent XSLT file name is defined for a query in smSyncConfFile.xml when using ServiceManagerAdapter9.41
- Request name is not found for a query in smSyncConfFile.xml when using ServiceManagerAdapter9.41
- Wrong Service Manager WS request name is defined in smSyncConfFile.xml when using ServiceManagerAdapter9.41
- The XSLT file is not well-formed when using ServiceManagerAdapter9.41
- The UCMDB attribute name is wrong in the XSLT file when using ServiceManagerAdapter9.41
- The Service Manager field name is wrong in the XSLT File when using ServiceManagerAdapter9.41
- The data push job passes with failures when using ServiceManagerAdapter9.41
- The CI display name is truncated when using ServiceManagerAdapter9.41
- Global ID and Customer ID are missing from XSLT when using ServiceManagerAdapter9.41
- Data population fails when using ServiceManagerAdapter9.41
- Query is not configured in smPopConfFile.xml when using ServiceManagerAdapter9.41
- Nonexistent XSLT file name is defined for a query in smPopConfFile.xml when using ServiceManagerAdapter9.41
- The "Retrieve" type request is undefined in smPopConfFile.xml when using ServiceManagerAdapter9.41
- Wrong request name of retrieveKeysQueryName is configured in smPopConfFile.xml when using ServiceManagerAdapter9.41
- Wrong request name of retrieveListQueryName is configured in smPopConfFile.xml when using ServiceManagerAdapter9.41
- The XSLT file is not well-formed when using ServiceManagerAdapter9.41

- The UCMDB attribute name is wrong in the XSLT File when using ServiceManagerAdapter9.41
- The Service Manager field name is wrong in the XSLT file when using ServiceManagerAdapter9.41
- The Universal CMDB attribute data type is wrong in the XSLT file when using ServiceManagerAdapter9.41
- sm_id is not mapped to the right Service Manager field in XSLT when using ServiceManagerAdapter9.41
- The CI Visualization page displays "Fail to render widget"

1.6.2.1. The Get Related and Impact Simulation widgets don't work

When the UCMDB UI embeds in UCMDB, the Get Related and Impact Simulation widgets that embed in Service Manager don't work.

Cause

The cause of this issue is unknown.

Solution

Follow these steps for a workaround:

1. Stop the UCMDB server.
2. Replace the `<*UCMDB installation path*>\UCMDBServer\deploy\ucmdb-browser.war` file with the file extracted from the `HPE-Browser-4.12.61-ucmdb-browser-embedded.zip` file in the browser 4.12 release build.
3. Delete `<*UCMDB installation path*>\<UCMDB Server\runtime\jetty-cache`.
4. Restart the UCMDB server.
5. Sign in to the UCMDB JMX Console and set the value of the `ui.framing.host` global setting to the desired Service Manager web tier host address from which Service Manager will access the embedded UCMDB browser. For example, `frame-ancestors https://<IP>:<port>` or `frame-ancestors https://<web tier FQDN>:<port>`.

1.6.2.2. Data push fails when using ServiceManagerEnhancedAdapter9.41

Data push fails when using **ServiceManagerEnhancedAdapter9.41**.

Cause

When data push errors or problems occur, you can check the error messages and the log file to figure out the root causes and then fix the errors.

This integration uses the following error codes for data push.

SM Error Code	Description	UCMDB Error Code
-1	Server Application Error from SM	900008
-4	401 Authorization error from SM	900001
3	Data Locked Error from SM	900003
9	Restful configuration not found	900009
51	Changed By Another Process Error from SM	900004
70	Invalid Action Error from SM	900005
71	Validation Check Failed error from SM	900002
881	Invalid Data Error from SM	900007
882	Can't find CI in SM for relationship operation	900010
	Failed to create messages in SM	900013
	Communication with SM Failed	900014
	Other Error from SM	900099

When a data push job has failed, the job status becomes **Failed**.

Solution

Troubleshoot the failed job as follows:

- Check the error messages of the failed job in the Universal CMDB studio.

  See *How to check the error message of a failed push job* below.

- Check the log file for more details.

  See *How to check the push log file* below.

When a data push job has completed with failures of partial records, the job status becomes **Completed**. Troubleshoot the failed records as follows:

- Check the error messages of failed CIs in the Universal CMDB studio.

  See *How to check the error messages of failed CIs or relationships in a push job* below.

- Check the log file for more details.

  See *How to check the push log file* below.

How to check the error message of a failed push job

To check the error message of a failed job:

1. Sign in to UCMDB as an administrator.
2. Navigate to **Data Flow Management** > **Integration Studio**.
3. Select the integration point for this integration from the Integration Point list.
4. Click the **Data Push** tab.
5. Select the job from Integration Jobs.
6. Click the **Job Errors** tab, and double-click the Severity of a message from the list.

   A popup window displays the detailed error message of this failed job. The following is an excerpt of a sample error message indicating that the system found no mapping configuration for the push TQL query.

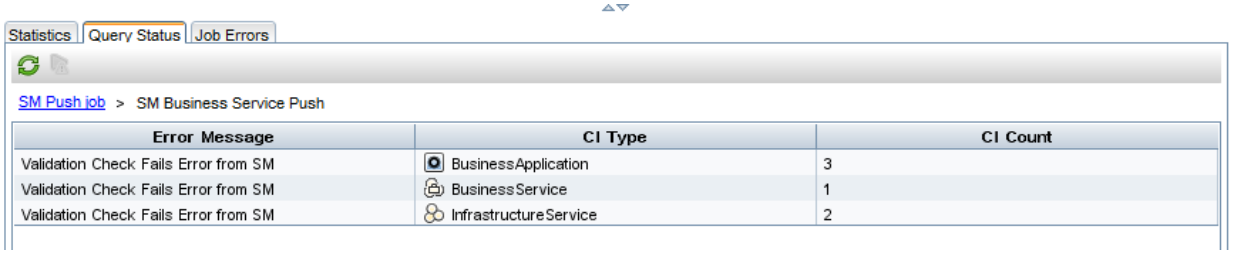
```
java.lang.RuntimeException: No mapping is found for TQL: "SM Business
Service Push 2.0", or Cannot retrieve the mapping from SM side by
QueryNodeName [bizservice", please configure in smPushConf.xml or SM
configuration
        at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.SmGenericPush
er.push(SmGenericPusher.java:119)
        ... 35 more
--- End of probe-side exception ---
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adhoctasks.AdHocProbeRequ
estOperation.convertThrowableToStringSafeException(AdHocProbeRequestOp
eration.java:86)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adhoctasks.AdHocProbeRequ
estOperation.performAction(AdHocProbeRequestOperation.java:77)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskdispatcher.AdHocTaskD
ispatcher.dispatchTask(AdHocTaskDispatcher.java:70)
        at sun.reflect.GeneratedMethodAccessor59.invoke(Unknown
Source)
        at
sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccess
orImpl.java:43)
        at java.lang.reflect.Method.invoke(Method.java:601)
        at
com.sun.jmx.mbeanserver.StandardMBeanIntrospector.invokeM2(StandardMBe
anIntrospector.java:111)
        at
com.sun.jmx.mbeanserver.StandardMBeanIntrospector.invokeM2(StandardMBe
anIntrospector.java:45)
        at
com.sun.jmx.mbeanserver.MBeanIntrospector.invokeM(MBeanIntrospector.ja
va:235)
        at
com.sun.jmx.mbeanserver.PerInterface.invoke(PerInterface.java:138)
        at
com.sun.jmx.mbeanserver.MBeanSupport.invoke(MBeanSupport.java:252)
        at
javax.management.StandardMBean.invoke(StandardMBean.java:405)
        at
com.sun.jmx.interceptor.DefaultMBeanServerInterceptor.invoke(DefaultMB
eanServerInterceptor.java:819)
        at
com.sun.jmx.mbeanserver.JmxMBeanServer.invoke(JmxMBeanServer.java:792)
```

```
        at
javax.management.MBeanServerInvocationHandler.invoke(MBeanServerInvoca
tionHandler.java:305)
        at
org.springframework.jmx.access.MBeanClientInterceptor.doInvoke(MBeanCl
ientInterceptor.java:405)
        at
org.springframework.jmx.access.MBeanClientInterceptor.invoke(MBeanClie
ntInterceptor.java:353)
        at
org.springframework.aop.framework.ReflectiveMethodInvocation.proceed(R
eflectiveMethodInvocation.java:172)
        at
org.springframework.aop.framework.JdkDynamicAopProxy.invoke(JdkDynamic
AopProxy.java:202)
        at com.sun.proxy.$Proxy57.dispatchTask(Unknown Source)
        at
com.hp.ucmdb.discovery.probe.agents.probegw.managementtasks.adhoctasks
.AdhocThread.run(AdhocThread.java:54)
        ... 3 more
```

How to check the error messages of failed CIs or relationships in a push job

When a data push job has completed with failures of partial records, you can check the error details for each failed record in the Universal CMDB studio. To check the error messages of failed records in a data push job:

1. Sign in to UCMDB as an administrator.
2. Navigate to **Data Flow Management** > **Integration Studio**.
3. Select the integration point for this integration from the Integration Point list.
4. Click the **Data Push** tab.
5. Select the job from Integration Jobs.
6. Click the **Query Status** tab.
7. Double-click a query with failures. The system displays the Error Message and CI Count for each failed CI Type are displayed.



8. Double-click an error message. The system displays a list of failed records.

9. Double-click a failed record. The system displays the detailed error message of the record.

How to check the push log file

You need to set the Development adapter log level to `DEBUG` or `TRACE`. Hence, you can check the push result tree nodes of UCMDB, and then send messages to and receive messages from Service Manager. Or, you can set the log level to `TRACE` so as to check the conversion period based on a mapping file.



Note

To troubleshoot push, population, and federation issues, you need to set the Development adapter log level to `DEBUG` or `TRACE` in the `fcmdb.properties` and `fcmdb.push.properties` files, which are located in the `\conf\log` folder of your Data Flow Probe or Integration Service installation. Additionally, you can view push, population and federation log information in the `fcmdb.push.all.log` file, which resides in the `\runtime\log` folder of your Data Flow Probe or Integration Service installation. You may install the Data Flow Probe or Integration Service on the same host as your UCMDB Server or on a separate host.

To set the Development adapter log level to `DEBUG` or `TRACE`:

1. Sign in as an administrator to the host on which you have installed UCMDB Data Flow Probe or Integration Service.
2. Navigate to the `<UCMDB installation folder>\DataFlowProbe\conf\log` folder or the `<UCMDB installation folder\UCMDBServer\integrations\conf\log`. For example:

   `C:\UCMDB\DataFlowProbe\conf\log\`

   Or, `C:\UCMDB\UCMDBServer\integrations\conf\log`
3. Open the `fcmdb.properties` file in a text editor, and change the log level to `DEBUG` or `TRACE`. For example:

   ```
   #loglevel can be any of TRACE DEBUG INFO WARN ERROR FATAL
   loglevel=DEBUG
   def.file.max.size=5000KB
   def.files.backup.count=10
   msg.layout=%d  %-5p  [%t] - %m%n
   ```
4. Open the `fcmdb.push.properties` files in a text editor, and change the `log4j` log level to `DEBUG` or `TRACE`. For example:

```
### UCMDB log4j Properties
log.file.path=log/${log.folder.path.output}
#loglevel can be any of TRACE DEBUG INFO WARN ERROR FATAL
loglevel=DEBUG
def.file.max.size=5000KB
def.files.backup.count=10
msg.layout=%d  %-5p  [%t] - %m%n
```

5. Save the files.

To check the push log file:

1. Sign in as an administrator to the host on which you have installed UCMDB Data Flow Probe or Integration Service.
2. Navigate to the `<*UCMDB installation folder*>\DataFlowProbe\runtime\log` folder or the `<*UCMDB installation folder*\UCMDBServer\integrations\runtime\log` folder.
3. Open the `fcmdb.push.all.log` file in a text editor.

1.6.2.3. The query is not configured in smPushConf.xml when using ServiceManagerEnhancedAdapter9.41

The push job fails with a “Failed” status. From both the log file and the detail error message of the failed job in the Universal CMDB studio, you receive an error like the following:

```
java.lang.RuntimeException: No mapping is found for TQL: "SM Business
Service Push 2.0", or Cannot retrieve the mapping from SM side by
QueryNodeName [bizservice", please configure in smPushConf.xml or SM
configuration
        at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.SmGenericPusher.p
ush(SmGenericPusher.java:119)
        ... 35 more
--- End of probe-side exception ---
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adhoctasks.AdHocProbeRequestO
peration.convertThrowableToStringSafeException(AdHocProbeRequestOperation.
java:86)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adhoctasks.AdHocProbeRequestO
peration.performAction(AdHocProbeRequestOperation.java:77)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskdispatcher.AdHocTaskDispa
tcher.dispatchTask(AdHocTaskDispatcher.java:70)
        at sun.reflect.GeneratedMethodAccessor59.invoke(Unknown Source)
        at
sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorIm
pl.java:43)
        at java.lang.reflect.Method.invoke(Method.java:601)
        at
com.sun.jmx.mbeanserver.StandardMBeanIntrospector.invokeM2(StandardMBeanIn
trospector.java:111)
        at
com.sun.jmx.mbeanserver.StandardMBeanIntrospector.invokeM2(StandardMBeanIn
trospector.java:45)
        at
com.sun.jmx.mbeanserver.MBeanIntrospector.invokeM(MBeanIntrospector.java:2
35)
        at
com.sun.jmx.mbeanserver.PerInterface.invoke(PerInterface.java:138)
        at
com.sun.jmx.mbeanserver.MBeanSupport.invoke(MBeanSupport.java:252)
        at javax.management.StandardMBean.invoke(StandardMBean.java:405)
        at
com.sun.jmx.interceptor.DefaultMBeanServerInterceptor.invoke(DefaultMBeanS
erverInterceptor.java:819)
        at
com.sun.jmx.mbeanserver.JmxMBeanServer.invoke(JmxMBeanServer.java:792)
        at
javax.management.MBeanServerInvocationHandler.invoke(MBeanServerInvocation
```

```
Handler.java:305)
        at
org.springframework.jmx.access.MBeanClientInterceptor.doInvoke(MBeanClient
Interceptor.java:405)
        at
org.springframework.jmx.access.MBeanClientInterceptor.invoke(MBeanClientIn
terceptor.java:353)
        at
org.springframework.aop.framework.ReflectiveMethodInvocation.proceed(Refle
ctiveMethodInvocation.java:172)
        at
org.springframework.aop.framework.JdkDynamicAopProxy.invoke(JdkDynamicAopP
roxy.java:202)
        at com.sun.proxy.$Proxy57.dispatchTask(Unknown Source)
        at
com.hp.ucmdb.discovery.probe.agents.probegw.managementtasks.adhoctasks.Adh
ocThread.run(AdhocThread.java:54)
        ... 3 more
```

Cause

The TQL queries used for pushing relationships is not configured in the smPushConf.xml file.

Solution

The TQL queries used for pushing relationships must be configured in the smPushConf.xml file.

```
<config>
        <mapping>
                <!--
                   Wildcard is supported for TQL names while you configure
the mapping now,
                   you can add the '*' at the end of the TQL name while
configuring the mapping.
                   We use the wildcard for mapping in OOTB, so each time
you do save as to a TQL,
                   you could still push it to SM without manual changing
the mapping.
                   For e.g., if you have saved the <TQL_name> query to
<TQL_name>_1, and <TQL_name>_2,
                   the TQL name is specified as <TQL_name>* in this
configuration file, and the
                   integration will automatically use this mapping entry
on all of the three TQLs.
                   However, using the exact TQL name to configure the
mapping is still supported.
                -->
<tql name="SM Business Service Relations Push 2.0"
resourceCollectionName="Relationships" resourceName="Relationship"/>
<tql name="SM CRG Relations Push 2.0"
resourceCollectionName="Relationships" resourceName="Relationship"/>
<tql name="SM Node Relations Push 2.0"
resourceCollectionName="Relationships" resourceName="Relationship"/>
<tql name="SM Layer2 Topology Relations Push 2.0"
resourceCollectionName="Relationships" resourceName="Relationship"/>
        </mapping>
</config>
```

In this case, search for text string `No mapping is found for TQL` to find the name of the query that's not yet configured, and then add a mapping entry for the query in the `smPushConf.xml` file.

Contents

- 1 Cause
- 2 Solution

1.6.2.4. The mapping file is not well-formed when using ServiceManagerEnhancedAdapter9.41

The data push job fails with a “Failed” status. From both the log file and the detailed error message of the failed job in the Universal CMDB studio, you receive an error like the following:

```
java.lang.RuntimeException: No mapping is found for TQL: "SM Business
Service Push 2.0", or Cannot retrieve the mapping from SM side by
QueryNodeName [businessservice", please configure in smPushConf.xml or SM
configuration
        at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.SmGenericPusher.p
ush(SmGenericPusher.java:119)
        ... 35 more
--- End of probe-side exception ---
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adhoctasks.AdHocProbeRequestO
peration.convertThrowableToStringSafeException(AdHocProbeRequestOperation.
java:86)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adhoctasks.AdHocProbeRequestO
peration.performAction(AdHocProbeRequestOperation.java:77)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskdispatcher.AdHocTaskDispa
tcher.dispatchTask(AdHocTaskDispatcher.java:70)
        at sun.reflect.GeneratedMethodAccessor59.invoke(Unknown Source)
        at
sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorIm
pl.java:43)
        at java.lang.reflect.Method.invoke(Method.java:601)
        at
com.sun.jmx.mbeanserver.StandardMBeanIntrospector.invokeM2(StandardMBeanIn
trospector.java:111)
        at
com.sun.jmx.mbeanserver.StandardMBeanIntrospector.invokeM2(StandardMBeanIn
trospector.java:45)
        at
com.sun.jmx.mbeanserver.MBeanIntrospector.invokeM(MBeanIntrospector.java:2
35)
        at
com.sun.jmx.mbeanserver.PerInterface.invoke(PerInterface.java:138)
        at
com.sun.jmx.mbeanserver.MBeanSupport.invoke(MBeanSupport.java:252)
        at javax.management.StandardMBean.invoke(StandardMBean.java:405)
        at
com.sun.jmx.interceptor.DefaultMBeanServerInterceptor.invoke(DefaultMBeanS
erverInterceptor.java:819)
        at
com.sun.jmx.mbeanserver.JmxMBeanServer.invoke(JmxMBeanServer.java:792)
        at
javax.management.MBeanServerInvocationHandler.invoke(MBeanServerInvocation
```

```
Handler.java:305)
        at
org.springframework.jmx.access.MBeanClientInterceptor.doInvoke(MBeanClient
Interceptor.java:405)
        at
org.springframework.jmx.access.MBeanClientInterceptor.invoke(MBeanClientIn
terceptor.java:353)
        at
org.springframework.aop.framework.ReflectiveMethodInvocation.proceed(Refle
ctiveMethodInvocation.java:172)
        at
org.springframework.aop.framework.JdkDynamicAopProxy.invoke(JdkDynamicAopP
roxy.java:202)
        at com.sun.proxy.$Proxy57.dispatchTask(Unknown Source)
        at
com.hp.ucmdb.discovery.probe.agents.probegw.managementtasks.adhoctasks.Adh
ocThread.run(AdhocThread.java:54)
        ... 3 more
```

Cause

The "target_entity name" should be **bizservice** (which is the CI type display name defined in Service Manager, and also the external class model name displayed in the UCMDB Visual Mapping tool interface); however you have configured a wrong name **businessservice**.

```
<?xml version="1.0" encoding="UTF-8"?>
<integration xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="../mappings_schema.xsd">
<info>
<source name="UCMDB"         version="10.20" vendor="HP"/>
<target name="SM"    version="9.40"  vendor="HP"/>
</info>
<import>
<scriptFile path="mappings.scripts.SMPushFunctions"/>
</import>
<!--
        Push uCMDB CIT to SM Business Service.
-->
<target_entities>
<source_instance query-name="SM Business Service Push 2.0" root-element-
name="Root" >
<target_entity name="businessservice">
<target_mapping name="UCMDBId" datatype="STRING"
value="Root['global_id']"/>
<target_mapping name="CustomerId" datatype="STRING"
value="SMPushFunctions.getCustomerId(CustomerInformation)"/>
<target_mapping name="Type" datatype="STRING" value="'bizservice'"/>
<target_mapping name="Subtype" datatype="STRING"
value="SMPushFunctions.getSMSubType(Root['element_type'],ClassModel,'BizSe
rvice')"/>
<target_mapping name="ServiceProvider" datatype="STRING"
value="Root['provider']"/>
<target_mapping name="ServiceName" datatype="STRING"
value="Root['display_label']"/>
<target_mapping name="CIIdentifier" datatype="STRING"
value="Root['display_label']"/>
</target_entity>
</source_instance>
</target_entities>
</integration>
```

Solution

In this case, search for the relevant CI type in Service Manager to find the correct display name, and then modify the mapping file with the correct name.



Tip

You can easily fix such kind of validation issues by using the UCMDB Visual Mapping tool to generate the mapping file.

Contents

- 1 Cause
- 2 Solution

1.6.2.5. Data population fails when using ServiceManagerEnhancedAdapter9.41

Data population fails when using **ServiceManagerEnhancedAdapter9.41**.

Cause

When a population job has failed, the job status becomes **Failed**. When population errors or problems occur, you can check the error messages and the population log file to identify the root causes and then solve the problems.

Solution

Troubleshoot the failed job as follows:

- Check the error message of the failed job in the Universal CMDB studio.

  See *How to check the error message of a failed population job* below.

- Check the log file for more details.

  See *How to check the population log file* below.

How to check the error message of a failed population job

While a population job fails, you can check the detailed error messages in the Universal CMDB studio.

To check the error message of a failed population job:

1. Sign in to UCMDB as an administrator.
2. Navigate to **Data Flow Management** > **Integration Studio**.
3. Select the integration point for this integration.
4. Click the **Population** tab.
5. Select the failed job from Integration Jobs, and click the **Job Errors** tab.
6. Double-click an error message from the list.

   A pop-up window opens to display the error details.

How to check the population log file

You can set the Development adapter log level to `DEBUG`. Hence, you can check the population result tree nodes of UCMDB, and send messages to and receive messages from Service Manager. Or, you can set the log level to `TRACE` so as to check the conversion period based on a mapping file.

Note

To troubleshoot push, population, and federation issues, you need to set the Development adapter log level to `DEBUG` or `TRACE` in the `fcmdb.properties` and `fcmdb.push.properties` files, which are located in the `\conf\log` folder of your Data Flow Probe or Integration Service installation. Additionally, you can view push, population and federation log information in the **fcmdb.push.all.log** file, which resides in the `\runtime\log` folder of your Data Flow Probe or Integration Service installation. You may install Data Flow Probe or Integration Service on the same host as your UCMDB Server or on a separate host.

To set the Development adapter log level and view the population log file, follow the steps described in How to check the push log file.

1.6.2.6. The TQL query is not configured in smPopConf.xml when using ServiceManagerEnhancedAdapter9.41

If you have not yet added a TQL query to your job, you can't select this query from the list while you create or update your job.

Cause

You have already added this TQL query to your job, but may forget to add the configuration for this query in smPopConf.xml. You will get a “Failed” status while you run this population job. In addition, in the Universal CMDB studio, you will get an error message like the following (see How to Check the Error Message of a Failed Population Job).

```
running population. Destination ID: sm, Failed during query: SM Business
Application Population 2.0, all queries: [SM Business Application
Population 2.0, SM Business Service Population 2.0],  finished queries:
[].
ERROR: com.mercury.topaz.cmdb.shared.base.CmdbException: [ErrorCode [802]
General Integration Error{sm}]
appilog.framework.shared.manage.impl.MamResponseException: [ErrorCode
[802] General Integration Error{sm}]
CMDB Operation Internal Error: class
com.mercury.topaz.cmdb.shared.fcmdb.dataAccess.exception.AdapterAccessGene
ralException : Unsupported Query [SM Business Application Population 2.0],
only population TQL is supported : operation Data Access Adapter Query:
Retrieve Changed Data
        at
com.mercury.topaz.cmdb.shared.manage.operation.impl.AbstractCommonOperatio
n.execute(AbstractCommonOperation.java:160)
        at
com.hp.ucmdb.dataAccess.manager.DataAccessAdapterManagerProbeImpl.executeO
peration(DataAccessAdapterManagerProbeImpl.java:50)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adapters.DataAccessAdaptersFa
cade.invokeOperation(DataAccessAdaptersFacade.java:406)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runChang
esOnPopulateChangesAdapter(AdapterService.java:1262)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runQueri
esOnPopulateChangesAdapter(AdapterService.java:1102)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runQueri
es(AdapterService.java:354)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runDisco
very(AdapterService.java:198)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.discover
(AdapterService.java:149)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter.laun
chTask(JobExecuter.java:1188)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobE
xecuterWorker.launch(JobExecuter.java:945)
```

In the fcmdb.push.all.log file, you can find more details like the following:

```
2014-11-20 10:40:50,949 [JobExecuterWorker-0:DS_sm_SM BS pop] ERROR - sm
>> Fail to Create or Return PopulationConnectorOutput
com.hp.ucmdb.federationspi.exception.DataAccessGeneralException:
Unsupported Query [SM Business Application Population 2.0], only
population TQL is supported
        at
com.hp.ucmdb.adapter.smpush.ServiceManagerGenericAdapter.populate(ServiceM
anagerGenericAdapter.java:337)
        at
com.hp.ucmdb.adapters.GenericAdapter.getChanges(GenericAdapter.java:881)
        at
com.hp.ucmdb.dataAccess.operations.DataAccessAdapterQueryRetrieveChanges.g
etChangesResult(DataAccessAdapterQueryRetrieveChanges.java:50)
        at
com.hp.ucmdb.dataAccess.operations.DataAccessAdapterQueryRetrieveChanges.d
oDataAccessQueryExecute(DataAccessAdapterQueryRetrieveChanges.java:38)
        at
com.hp.ucmdb.dataAccess.operations.AbstractDataAccessLifeCycleAdapterQuery
.doLifeCycleExecute(AbstractDataAccessLifeCycleAdapterQuery.java:34)
        at
com.hp.ucmdb.dataAccess.operations.AbstractDataAccessLifeCycleAdapterOpera
tion.doDataAccessExecute(AbstractDataAccessLifeCycleAdapterOperation.java:
57)
        at
com.hp.ucmdb.dataAccess.operations.AbstractDataAccessAdapterOperation.data
AccessExecute(AbstractDataAccessAdapterOperation.java:59)
        at
com.hp.ucmdb.dataAccess.operations.AbstractDataAccessAdapterOperation.doEx
ecute(AbstractDataAccessAdapterOperation.java:37)
        at
com.mercury.topaz.cmdb.shared.manage.operation.impl.AbstractFrameworkOpera
tion.commonExecute(AbstractFrameworkOperation.java:17)
        at
com.mercury.topaz.cmdb.shared.manage.operation.impl.AbstractCommonOperatio
n$OperationExecuteFlowTrackingCommand.execute(AbstractCommonOperation.java
:87)
        at
com.mercury.topaz.cmdb.shared.manage.operation.impl.AbstractCommonOperatio
n$OperationExecuteFlowTrackingCommand.execute(AbstractCommonOperation.java
:60)
        at
com.mercury.topaz.cmdb.shared.manage.flowmanagement.api.FlowManager.execut
e(FlowManager.java:227)
        at
com.mercury.topaz.cmdb.shared.manage.operation.flow.OperationInFlowDefault
```

```
Executor.execute(OperationInFlowDefaultExecutor.java:23)
        at
com.mercury.topaz.cmdb.shared.manage.operation.impl.AbstractCommonOperatio
n.execute(AbstractCommonOperation.java:158)
        at
com.hp.ucmdb.dataAccess.manager.DataAccessAdapterManagerProbeImpl.executeO
peration(DataAccessAdapterManagerProbeImpl.java:50)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adapters.DataAccessAdaptersFa
cade.invokeOperation(DataAccessAdaptersFacade.java:406)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runChang
esOnPopulateChangesAdapter(AdapterService.java:1262)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runQueri
esOnPopulateChangesAdapter(AdapterService.java:1102)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runQueri
es(AdapterService.java:354)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runDisco
very(AdapterService.java:198)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.discover
(AdapterService.java:149)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter.laun
chTask(JobExecuter.java:1188)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobE
xecuterWorker.launch(JobExecuter.java:945)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobE
xecuterWorker.executeTask(JobExecuter.java:867)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobE
xecuterWorker.run(JobExecuter.java:728)
```

Solution

Search for text string “Unsupported Query by this adapter” to find out the TQL query name that has not yet been configured, and then configure it in the smPopConf.xml file.

1.6.2.7. Nonexistent mapping file name is defined for a TQL query in smPopConf.xml when using ServiceManagerEnhancedAdapter9.41

You will get a “Failed” status while you run the population job. In addition, from the Universal CMDB studio, you will get an error message like the following:

```
running population. Destination ID: sm, Failed during query: SM Business
Service Population 2.0, all queries: [SM Business Application Population
2.0, SM Business Service Population 2.0],  finished queries: [SM Business
Application Population 2.0].
ERROR: com.mercury.topaz.cmdb.shared.base.CmdbException: [ErrorCode [802]
General Integration Error{sm}]
appilog.framework.shared.manage.impl.MamResponseException: [ErrorCode
[802] General Integration Error{sm}]
CMDB Operation Internal Error: class
com.mercury.topaz.cmdb.shared.fcmdb.dataAccess.exception.AdapterAccessGene
ralException : Query Definition [SM Business Service Population 2.0] has
no matching mapping. Make sure the mapping exist exists under folder
'mapping' and contains the exact query name and root name. Available
mappings:
[QueryRoot{queryName='SM Business Application Population 2.0',
rootName='bizservice'}] : operation Data Access Adapter Query: Retrieve
Changed Data
        at
com.mercury.topaz.cmdb.shared.manage.operation.impl.AbstractCommonOperatio
n.execute(AbstractCommonOperation.java:160)
        at
com.hp.ucmdb.dataAccess.manager.DataAccessAdapterManagerProbeImpl.executeO
peration(DataAccessAdapterManagerProbeImpl.java:50)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adapters.DataAccessAdaptersFa
cade.invokeOperation(DataAccessAdaptersFacade.java:406)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runChang
esOnPopulateChangesAdapter(AdapterService.java:1262)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runQueri
esOnPopulateChangesAdapter(AdapterService.java:1102)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runQueri
es(AdapterService.java:354)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runDisco
very(AdapterService.java:198)
        at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.discover
(AdapterService.java:149)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter.laun
chTask(JobExecuter.java:1188)
        at
```

```
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobE
xecuterWorker.launch(JobExecuter.java:945)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobE
xecuterWorker.executeTask(JobExecuter.java:867)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobE
xecuterWorker.run(JobExecuter.java:728)
```

Cause

Some mapping files are missing.

Solution

Search for text “has no matching mapping” to find out the missing mapping file name, and then add the mapping file to the adapter package.

Contents

- 1 Cause
- 2 Solution

1.6.2.8. Data federation fails when using ServiceManagerEnhancedAdapter9.41

A federation query fails and a pop-up window similar to the following appears in the UCMDB studio.



Cause

When a federation error occurs, you can check the error message in the federation log file to identify the root cause and then solve the problem.

Solution

To set the Development adapter log level and view the detailed error message of a failed federation query, follow the steps described in How to check the push log file.

Note

To troubleshoot push, population, and federation issues, you need to set the Development adapter log level to DEBUG or TRACE in the **fcmdb.properties** and **fcmdb.push.properties** files, which are located in the \conf\log folder of your Data Flow Probe or Integration Service installation. Additionally, you can view push, population and federation log information in the **fcmdb.push.all.log** file, which is located in the \runtime\log folder of your Data Flow Probe or Integration Service installation. Your Data Flow Probe or Integration Service may be installed on the same host as your UCMDB Server or on a separate host.

1.6.2.9. The mapping file for the federation TQL query is not well-formed when using ServiceManagerEnhancedAdapter9.41

The mapping file for the federation TQL query is not well-formed when using ServiceManagerEnhancedAdapter9.41.

Cause

There are cases where the mapping file for a federation TQL query is not well formed. For example, target_mapping_name “priority” in the following example federation mapping file specifies a wrong Groovy function name (SMFederationFunctions.getEnumValue1):

```
<?xml version="1.0" encoding="UTF-8"?>
<integration xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="../mappings_schema.xsd">
<info>
<source name="SM"         version="9.40"    vendor="HP"/>
<target name="UCMDB"    version="10.20"    vendor="HP"/>
</info>
<import>
<scriptFile path="mappings.scripts.SMFederationFunctions"/>
</import>
<target_entities>
<source_instance query-name="SM Incident 2.0" root-element-
name="ucmdbIncident">
<target_entity name="incident">
<target_mapping name="reference_number"     datatype="STRING"
value="ucmdbIncident['IncidentID']"/>
<target_mapping name="name"                 datatype="STRING"
value="ucmdbIncident['BriefDescription']"/>
<target_mapping name="priority"             datatype="STRING"
value="SMFederationFunctions.getEnumValue1(ucmdbIncident['PriorityCode'],'
Priority')"/>
<target_mapping name="incident_status"         datatype="STRING"
value="SMFederationFunctions.firstLetterToLowerAndReplaceSpaceWithUndersco
re(ucmdbIncident['IMTicketStatus'])"/>
<target_mapping name="category"             datatype="STRING"
value="SMFederationFunctions.replaceSpaceWithUnderscore(ucmdbIncident['Cat
egory'])"/>
<target_mapping name="closed_time"             datatype="DATE"
value="SMFederationFunctions.convertDate(ucmdbIncident['ClosedTime'])"/>
<target_mapping name="create_time"             datatype="DATE"
value="SMFederationFunctions.convertDate(ucmdbIncident['OpenTime'])"/>
<target_mapping name="last_modified_time"     datatype="DATE"
value="SMFederationFunctions.convertDate(ucmdbIncident['UpdatedTime'])"/>
<target_mapping name="impact_scope"         datatype="STRING"
value="SMFederationFunctions.getEnumValue(ucmdbIncident['ImpactScope'],'Im
pactScope')"/>
<target_mapping name="urgency"                 datatype="STRING"
value="SMFederationFunctions.getEnumValue(ucmdbIncident['Urgency'],'Urgenc
y')"/>
</target_entity>
</source_instance>
</target_entities>
</integration>
```

When you try to federate incidents, no records are retrieved from Service Manager. If you check the fcmdb.push.all.log file, you can find a detailed error message that resembles the following for the federation query:

```
2014-11-20 14:09:58,670 [AdHoc:AD_HOC_TASK_PATTERN_ID-66-1416463796366]
ERROR -  >> Failed executing value
[SMFederationFunctions.getEnumValue1(ucmdbIncident['PriorityCode'],'Priori
ty')] of mapping <target_mapping name="priority">  <target_ci_type
name="incident"> , Root cmdbId [null]
groovy.lang.MissingMethodException: No signature of method: static
mappings.scripts.SMFederationFunctions.getEnumValue1() is applicable for
argument types: (java.lang.String, java.lang.String) values: [1, Priority]
Possible solutions: getEnumValue(java.lang.String, java.lang.String)
        at
groovy.lang.MetaClassImpl.invokeStaticMissingMethod(MetaClassImpl.java:135
9)
        at
groovy.lang.MetaClassImpl.invokeStaticMethod(MetaClassImpl.java:1345)
        at
org.codehaus.groovy.runtime.callsite.StaticMetaClassSite.call(StaticMetaCl
assSite.java:50)
        at
org.codehaus.groovy.runtime.callsite.CallSiteArray.defaultCall(CallSiteArr
ay.java:42)
        at
org.codehaus.groovy.runtime.callsite.AbstractCallSite.call(AbstractCallSit
e.java:108)
        at
org.codehaus.groovy.runtime.callsite.AbstractCallSite.call(AbstractCallSit
e.java:120)
        at
Mapping_4a8761e7adc009e8a7d268c2f1349ab4.run(Mapping_4a8761e7adc009e8a7d26
8c2f1349ab4.groovy:1)
        at
com.hp.ucmdb.adapters.instance.mapping.AbstractResultTreeNodeMapper.calcul
ateProperties(AbstractResultTreeNodeMapper.java:231)
        at
com.hp.ucmdb.adapters.instance.mapping.AbstractResultTreeNodeMapper.proces
sTargetEntity(AbstractResultTreeNodeMapper.java:301)
        at
com.hp.ucmdb.adapters.instance.mapping.RtnToRtnMapper.processTargetEntitie
s(RtnToRtnMapper.java:214)
        at
com.hp.ucmdb.adapters.instance.mapping.RtnToRtnMapper.processSourceInstanc
eWrapper(RtnToRtnMapper.java:101)
        at
com.hp.ucmdb.adapters.instance.mapping.RtnToRtnMapper.buildResultTreeNode(
RtnToRtnMapper.java:76)
        at
```

```
com.hp.ucmdb.adapters.GenericAdapter$PopulationFederationChunkProcessor.in
voke(GenericAdapter.java:1213)
        at
com.hp.ucmdb.adapters.GenericAdapter.getDataResult(GenericAdapter.java:743
)
        at
com.hp.ucmdb.discovery.probe.processor.FederationTopologyGetDataResultProb
eRequestProcessor.processRequest(FederationTopologyGetDataResultProbeReque
stProcessor.java:40)
        at
com.hp.ucmdb.discovery.probe.processor.FederationTopologyGetDataResultProb
eRequestProcessor.processRequest(FederationTopologyGetDataResultProbeReque
stProcessor.java:26)
        at
com.hp.ucmdb.discovery.probe.processor.AbstractProbeProcessor.process(Abst
ractProbeProcessor.java:56)
        at
com.hp.ucmdb.discovery.probe.processor.AbstractProbeProcessor.process(Abst
ractProbeProcessor.java:19)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.adhoctasks.AdHocProbeRequestO
peration.performAction(AdHocProbeRequestOperation.java:63)
        at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskdispatcher.AdHocTaskDispa
tcher.dispatchTask(AdHocTaskDispatcher.java:70)
        at sun.reflect.GeneratedMethodAccessor75.invoke(Unknown Source)
        at
sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorIm
pl.java:43)
        at java.lang.reflect.Method.invoke(Method.java:601)
        at
com.sun.jmx.mbeanserver.StandardMBeanIntrospector.invokeM2(StandardMBeanIn
trospector.java:111)
        at
com.sun.jmx.mbeanserver.StandardMBeanIntrospector.invokeM2(StandardMBeanIn
trospector.java:45)
        at
com.sun.jmx.mbeanserver.MBeanIntrospector.invokeM(MBeanIntrospector.java:2
35)
        at
com.sun.jmx.mbeanserver.PerInterface.invoke(PerInterface.java:138)
        at
com.sun.jmx.mbeanserver.MBeanSupport.invoke(MBeanSupport.java:252)
        at javax.management.StandardMBean.invoke(StandardMBean.java:405)
        at
com.sun.jmx.interceptor.DefaultMBeanServerInterceptor.invoke(DefaultMBeanS
```

```
erverInterceptor.java:819)
        at
com.sun.jmx.mbeanserver.JmxMBeanServer.invoke(JmxMBeanServer.java:792)
        at
javax.management.MBeanServerInvocationHandler.invoke(MBeanServerInvocation
Handler.java:305)
        at
org.springframework.jmx.access.MBeanClientInterceptor.doInvoke(MBeanClient
Interceptor.java:405)
        at
org.springframework.jmx.access.MBeanClientInterceptor.invoke(MBeanClientIn
terceptor.java:353)
        at
org.springframework.aop.framework.ReflectiveMethodInvocation.proceed(Refle
ctiveMethodInvocation.java:172)
        at
org.springframework.aop.framework.JdkDynamicAopProxy.invoke(JdkDynamicAopP
roxy.java:202)
        at com.sun.proxy.$Proxy51.dispatchTask(Unknown Source)
        at
com.hp.ucmdb.discovery.probe.agents.probegw.managementtasks.adhoctasks.Adh
ocThread.run(AdhocThread.java:54)
        at
org.eclipse.jetty.util.thread.QueuedThreadPool.runJob(QueuedThreadPool.jav
a:599)
        at
org.eclipse.jetty.util.thread.QueuedThreadPool$3.run(QueuedThreadPool.java
:534)
        at java.lang.Thread.run(Thread.java:722)
```

Solution

Search for word "Exception" in the fcmdb.push.all.log file to find the wrong method name, and then correct it in the corresponding federation mapping file.

Contents

- 1 Cause
- 2 Solution

1.6.2.10. Data push fails when using ServiceManagerAdapter9.41

Data push fails when using ServiceManagerAdapter9.41.

Cause

When data push errors or problems occur, you can check the error messages and the log file to figure out the root causes and then fix the errors.

This integration uses the following error codes for data push.

Data push error codes

Error code	Description
-1	Unspecified error.
0	Success.
3	Resource unavailable.
28	Not authorized.
51	Record modified since last retrieved.
70	Invalid SOAP action / unrecognized application action.
71	Validation failed.
881	CI doesn't exist in Service Manager.
882	Unable to remove the relationship because at least one of CIs involved in the relationship doesn't exist in Service Manager.

Solution

When a data push job has failed, the job status becomes **Failed**. Troubleshoot the failed job as follows:

- Check the error messages of the failed job in the Universal CMDB studio.

  See *How to check the error message of a failed push job* below.

- Check the log file for more details.

  See *How to check the push log file* below.

When a data push job was completed, but with partial records failed, the job status becomes **Passed with failures**. Troubleshoot the failed records as follows:

- Check the error messages of failed CIs in the Universal CMDB studio.

  See *How to check the error messages of failed CIs/CI relationships in a push job* below.

- Check the log file for more details.

  See *How to check the push log file* below.

Once you have fixed the issues with the failed records, you can re-push them one by one or in batches. For details, see *How to re-push failed CI/CI relationship records* below.

How to check the error message of a failed push job

To check the error message of a failed job:

1. Sign in to UCMDB as an administrator.
2. Navigate to **Data Flow Management** > **Integration Studio**.
3. Select the integration point for this integration from Integration Point list.
4. Click the **Data Push** tab.
5. Select the job from Integration Jobs.
6. Click the **Job Errors** sub-tab, and double-click the Severity of a message from the list.

   A popup window displays the detailed error message of this failed job. Following is an excerpt of a sample error message indicating that an XSLT file was not found.

```
java.io.FileNotFoundException: Resource:
business_service_push_file_not_exist.xslt, was not found

at
com.mercury.topaz.cmdb.server.fcmdb.adapterstate.AdapterStateResourceServerFacade
.openResourceForReading(AdapterStateResourceServerFacade.java:51)

at
com.mercury.topaz.cmdb.server.fcmdb.spi.adapter.environment.DataAdapterEnvironmen
tImpl.openResourceForReading(DataAdapterEnvironmentImpl.java:119)

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.TransformUtils.readFromFile(T
ransformUtils.java:113)

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.XsltTransformer.createTe
mplates(XsltTransformer.java:93)

... 36 more

& &
```

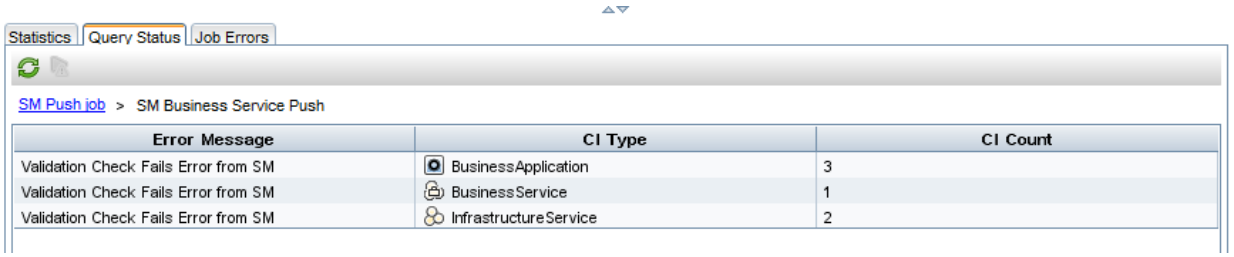
How to check the error messages of failed CIs/CI relationships in a push job

When a data push job is completed with partial records failed, in the Universal CMDB studio, you can check the source CI XML of Universal CMDB, the XSLT-transformed XML and the response XML from Service Manager to see if the failure is caused by data issues.

To check the error messages of failed records in a data push job:

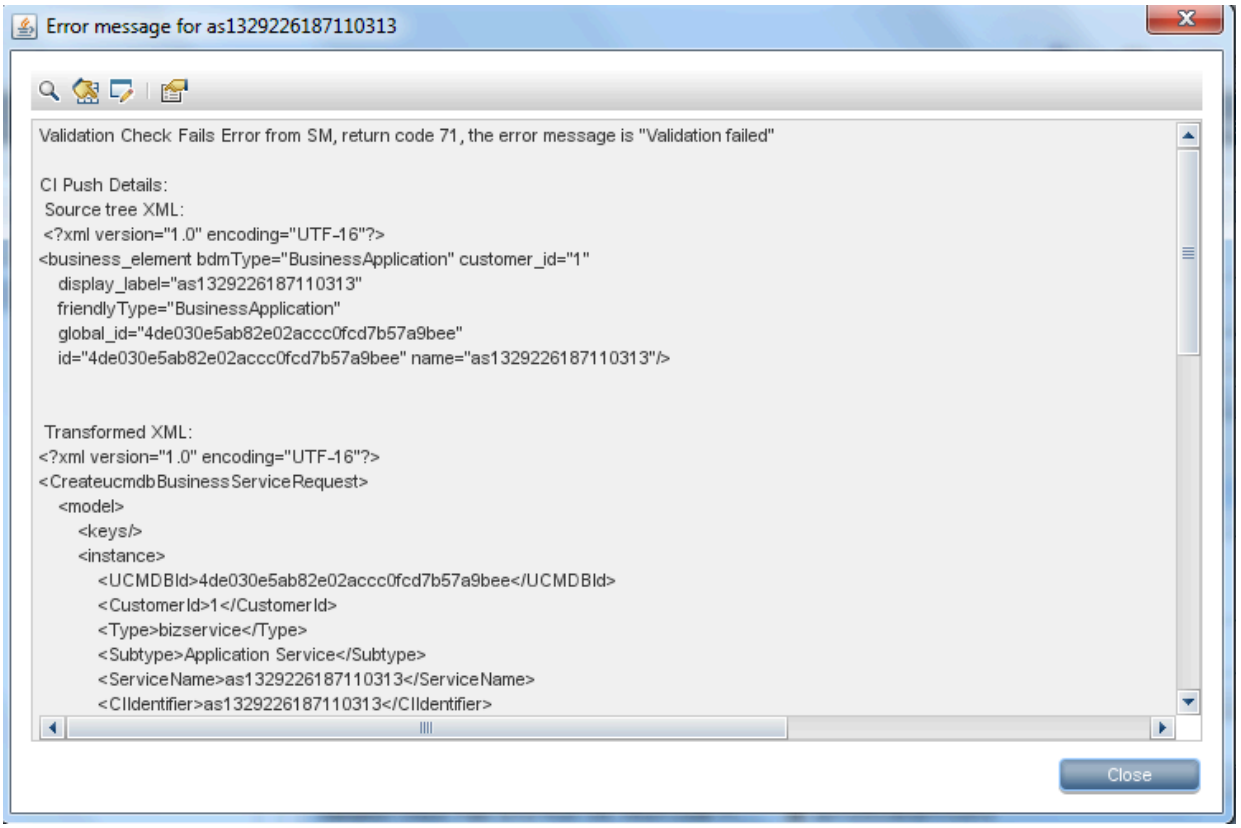
1. Sign in to UCMDB as an administrator.

2. Navigate to **Data Flow Management** > **Integration Studio**.
3. Select the integration point for this integration from Integration Point list.
4. Click the **Data Push** tab.
5. Select the job from Integration Jobs.
6. Click the **Query Status** sub-tab.
7. Double-click a query with failures. The Error Message and CI Count for each failed CI Type are displayed.



8. Double-click an error message. A list of failed records is displayed.
9. Double-click a failed record.

   The detailed error message of the record is displayed.



   The following is a sample error message that indicates a validation failure, and the root cause is that a not-null key (key definition #18 of the device table) contains an empty value.

```
Validation Check Fails Error from SM, return code 71, the error message is
"Validation failed",
```

CI Push Details:

Source tree XML:

```
<node bdmType="Node" customer_id="1"
display_label="XMA16.asiapacific.hpqcorp.net" friendly-Type="Windows"
global_id="fd21b94ddf26dc9ee054e21adca3d4c0"
id="fd21b94ddf26dc9ee054e21adca3d4c0" os_vendor="Microsoft"
primary_dns_name="XMA16.asiapacific.hpqcorp.net">
    <ip_addresss direction="outgoing" linkLabel="Containment" linkName="32"
linkType="Containment">
        <ip_address bdmType="IpAddress" customer_id="1"
friendlyType="IpAddress" id="bd03e34189c713898712dcac500730ba"
name="16.158.154.152" realRelationType="Containment"/>
    </ip_addresss>
    <interfaces direction="outgoing" linkLabel="Composition" linkName="33"
linkType="Composition">
       <interface bdmType="Interface" customer_id="1" friendlyType="Interface"
id="5897567531c55a7bd196b42d5a6aa485" mac_address="B499BAE9AC7A"
realRelationType="Composition"
root_container="UCMDB%0Ant%0A1%0Ainternal_id%3DSTRING%3Dfd21b94ddf26dc9ee054e2
1adca3d4c0%0A"/>
    </interfaces>
</node>
```

Transformed XML:

```
<CreateucmdbNodeRequest>
  <model>
    <keys/>
    <instance>
      <file.device>
        <UCMDBId>fd21b94ddf26dc9ee054e21adca3d4c0</UCMDBId>
        <CustomerId>1</CustomerId>
        <Type>computer</Type>
        <Subtype>Server</Subtype>

           <addlIPAddr>
             <AddlIPAddress>16.158.154.152</AddlIPAddress>
             <AddlSubnet/>
           </addlIPAddr>
     </addlIPAddr>
```

```
 <CIIdentifier>XMA16.asiapacific.hpqcorp.net</CIIdentifier>
        <DNSName>XMA16.asiapacific.hpqcorp.net</DNSName>
      </file.device>
      <file.node>
         <OSVendor>Microsoft</OSVendor>
        <addlIPAddr>
<AddlMacAddress>
          <AddlMacAddress>B499BAE9AC7A</AddlMacAddress>
        </AddlMacAddress>
      </file.node>
    </instance>
  </model>
</CreateucmdbNodeRequest>
```

Response message from SM:

```
  <CreateucmdbNodeResponse message="Validation failed" returnCode="71"
schemaRevision-Date="2011-09-18" schemaRevisionLevel="5" status="FAILURE"
xmlns="http://servicecenter.peregrine.com/PWS"
xmlns:cmn="http://servicecenter.peregrine.com/PWS/Common"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://servicecenter.peregrine.com/PWS
http://XMA16.asiapacific.hpqcorp.net:13080/sc62server/ucmdbNode.xsd">
      <model>
        <keys>
          <CIIdentifier
type="String">XMA16.asiapacific.hpqcorp.net</CIIdentifier>
        </keys>
        <instance
uniquequery="file.device,logical.name=&amp;quot;XMA16.asiapacific.hpqcorp.net&amp;quot
;">
          <file.device type="Structure">
            <CIIdentifier
type="String">XMA16.asiapacific.hpqcorp.net</CIIdentifier>
            <DNSName type="String">XMA16.asiapacific.hpqcorp.net</DNSName>
            <Type type="String">computer</Type>
            <CIName type="String">CI10869</CIName>
            <Subtype type="String">Server</Subtype>
            <UCMDBId type="String">fd21b94ddf26dc9ee054e21adca3d4c0</UCMDBId>
          </file.device>
          <file.node type="Structure">
            <addlIPAddr type="Array">
              <addlIPAddr type="StructureType">
```

```
    </addlIPAddr>
            </addlIPAddr>
            <AddlMacAddress type="Array">
              <AddlMacAddress type="String">B499BAE9AC7A</AddlMacAddress>
            </AddlMacAddress>
          </file.node>
        </instance>
      </model>
      <messages>
        <cmn:message type="String">7=====add</cmn:message>
        <cmn:message type="String">Key #18 is empty.
(se.base.method,add.record.radd)</cmn:message>
        <cmn:message type="String">file:(device)
key:(logical.name=XMA16.asiapacific.hpqcorp.net)
(se.base.method,add.record.radd)</cmn:message>
        <cmn:message type="String">The record being added contains a NULL key
(se.base.method,add.record.radd)</cmn:message>
        <cmn:message type="String">file:(joinnode)
key:(file.device,logical.name=XMA16.asiapacific.hpqcorp.net)
(se.base.method,add.record.radd)</cmn:message>
        <cmn:message type="String">The record being added contains a NULL key
(se.base.method,add.record.radd)</cmn:message>
        <cmn:message type="String">This record contains an invalid null
key.</cmn:message>
      </messages>
    </CreateucmdbNodeResponse>
```

How to check the push log file

You need to set the Development adapter log level to DEBUG so that you can check the source tree XML file of UCMDB, the XSLT-transformed XML file, and the response XML file from Service Manager.



Tip

You are recommended to enable the Development Mode for the integration point so that the above-mentioned three XML files are in a good format. See the following for the steps.

To set the Development adapter log level to DEBUG:

1. Sign in to the UCMDB server host as an administrator.
2. Navigate to the `<UCMDB installation folder>\UCMDBServer\conf\log\fcmdb.properties` file. For example: `C:\UCMDB\UCMDBServer\conf\log\fcmdb.properties`.
3. Open the `fcmdb.properties` file in a text editor.
4. Update the `log4j.category.fcmdb.adapters` log level to **DEBUG**.

```
log4j.category.com.mercury.topaz.cmdb.server.fcmdb.dataAccess=${loglevel},fcmdb.
dataAccess.appender
log4j.category.com.mercury.topaz.cmdb.shared.fcmdb.dataAccess=${loglevel},fcmdb.
dataAccess.appender
log4j.category.fcmdb.ftql.calc=${loglevel},fcmdb.ftql.calc.appender
log4j.category.fcmdb.config.audit=info,fcmdb.config.audit.appender
log4j.category.fcmdb.lifecycle=info,fcmdb.lifecycle.appender
log4j.category.fcmdb.mapping.engine=${loglevel},fcmdb.mapping.engine.appender

#every change in fcmdb.adapters will affect fcmdb.adapters.<integration_point_na
me>.log
log4j.category.fcmdb.adapters=DEBUG,fcmdb.adapters
log4j.category.fcmdb.statistics=${loglevel},fcmdb.statistics.appender
log4j.category.com.hp.ucmdb.dataacquisition=${loglevel},dataAcquisition.appender
,daWithApi.appender
log4j.category.com.hp.ucmdb.api.server.datastoremgmt=${loglevel},daWithApi.appen
der
log4j.category.population=${loglevel},population.appender
```

5. Save the file.

To enable the Development Mode for the integration point:

1. Sign in to UCMDB as an administrator.
2. Navigate to **Data Flow Management** > **Integration Studio**.
3. Select the integration point for this integration.
4. Click the **Edit Integration Point** button.
5. For Development Mode, select **True** from the list.
6. Click the **OK** button to save the integration point.

To check the push log file:

1. Sign in to the UCMDB server host as an administrator.
2. Navigate to the `<UCMDB installation folder>\UCMDBServer\runtime\log\fcmdb.adapters.<integration_point_name>.log` file. For example: `C:\UCMDB\UCMDBServer\runtime\log\fcmdb.adapters.SM Integration.log`
3. Open the log file in a text editor.
4. Search for text strings `Source tree XML:`, `Transformed XML:`, and `Response message from SM:`.

   The following is a sample log file.

```
2011-11-30 14:22:03,325 [1751920131@Default-7] DEBUG - SM Integration >> Source tree XML:
<node bdmType="Node" customer_id="1" display_label="XMA16.asiapacific.hpqcorp.net" friendlyType="Windows" global_id="fd21b94ddf26dc9ee054e21adca3d4c0" id="fd21b94ddf26dc9ee054e21adca3d4c0" os_vendor="Microsoft" primary_dns_name="XMA16.asiapacific.hpqcorp.net">
    <ip_addresss direction="outgoing" linkLabel="Containment" linkName="32" linkType="Containment">
        <ip_address bdmType="IpAddress" customer_id="1" friendlyType="IpAddress" id="bd03e34189c713898712dcac500730ba" name="16.158.154.152" realRelationType="Containment"/>
    </ip_addresss>
    <interfaces direction="outgoing" linkLabel="Composition" linkName="33" linkType="Composition">
       <interface bdmType="Interface" customer_id="1" friendlyType="Interface" id="5897567531c55a7bd196b42d5a6aa485" mac_address="B499BAE9AC7A" realRelationType="Composition" root_container="UCMDB%0Ant%0A1%0Ainternal_id%3DSTRING%3Dfd21b94ddf26dc9ee054e21adca3d4c0%0A"/>
    </interfaces>
</node>
```

```
2011-11-30 14:22:03,423 [1751920131@Default-7] DEBUG - SM Integration >> Transformed XML:
<CreateucmdbNodeRequest>
  <model>
    <keys/>
    <instance>
      <file.device>
        <UCMDBId>fd21b94ddf26dc9ee054e21adca3d4c0</UCMDBId>
        <CustomerId>1</CustomerId>
        <Type>computer</Type>
        <Subtype>Server</Subtype>
        <CIIdentifier>XMA16.asiapacific.hpqcorp.net</CIIdentifier>
        <DNSName>XMA16.asiapacific.hpqcorp.net</DNSName>
      </file.device>
      <file.node>
        <OSVendor>Microsoft</OSVendor>
        <addlIPAddr>
          <addlIPAddr>
            <AddlIPAddress>16.158.154.152</AddlIPAddress>
            <AddlSubnet/>
          </addlIPAddr>
        </addlIPAddr>
        <AddlMacAddress>
          <AddlMacAddress>B499BAE9AC7A</AddlMacAddress>
        </AddlMacAddress>
      </file.node>
    </instance>
  </model>
</CreateucmdbNodeRequest>
```

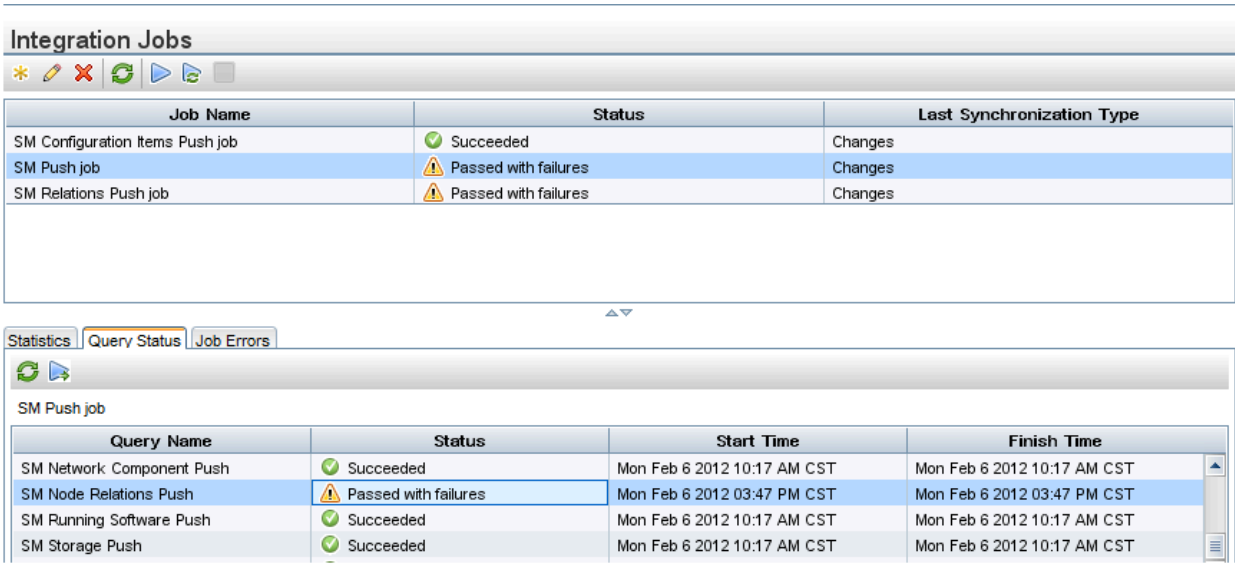
```
2011-11-30 14:22:07,615 [1751920131@Default-7] DEBUG - SM Integration >>
Response message from SM:
    <CreateucmdbNodeResponse message="Success" returnCode="0"
schemaRevisionDate="2011-09-18" schemaRevisionLevel="5" status="SUCCESS"
xmlns="http://servicecenter.peregrine.com/PWS"
xmlns:cmn="http://servicecenter.peregrine.com/PWS/Common"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://servicecenter.peregrine.com/PWS
http://XMA16.asiapacific.hpqcorp.net:13080/sc62server/ucmdbNode.xsd">
      <model>
        <keys>
          <CIIdentifier
type="String">XMA16.asiapacific.hpqcorp.net</CIIdentifier>
        </keys>
        <instance
unique-query="file.device,logical.name=&amp;quot;XMA16.asiapacific.hpqcorp.net&amp;quo
t;">
          <file.device type="Structure">
            <CIIdentifier
type="String">XMA16.asiapacific.hpqcorp.net</CIIdentifier>
            <DNSName type="String">XMA16.asiapacific.hpqcorp.net</DNSName>
            <Type type="String">computer</Type>
            <Subtype type="String">Server</Subtype>
         <UCMDBId type="String">fd21b94ddf26dc9ee054e21adca3d4c0</UCMDBId>
</file.device>
          <file.node type="Structure">
            <addlIPAddr type="Array">
              <addlIPAddr type="StructureType">
                <AddlIPAddress type="String">16.158.154.152</AddlIPAddress>
              </addlIPAddr>
            </addlIPAddr>
            <AddlMacAddress type="Array">
              <AddlMacAddress type="String">B499BAE9AC7A</AddlMacAddress>
            </AddlMacAddress>
          </file.node>
        </instance>
      </model>
      <messages>
         <cmn:message type="String">0=====update</cmn:message>
       </messages>
```

How to re-push failed CI/CI relationship records

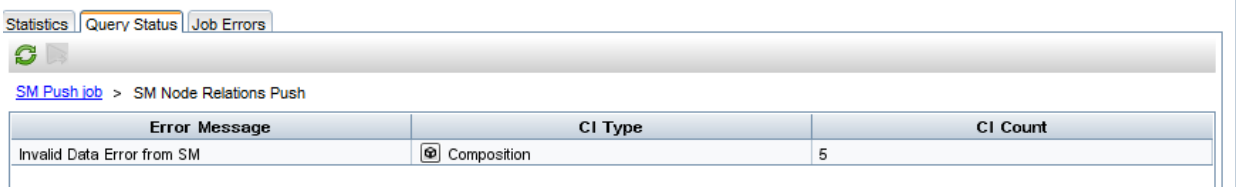
The push error handling mechanism allows you to re-push failed CI/CI Relationship records either one by one or in batches.

To re-push failed data in a query of a data push job:

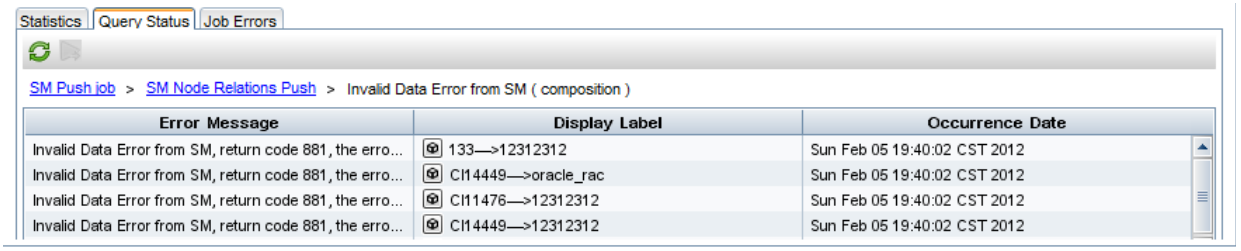
1. Select the data push job.
2. On the Query Status tab, double-click the failed query.



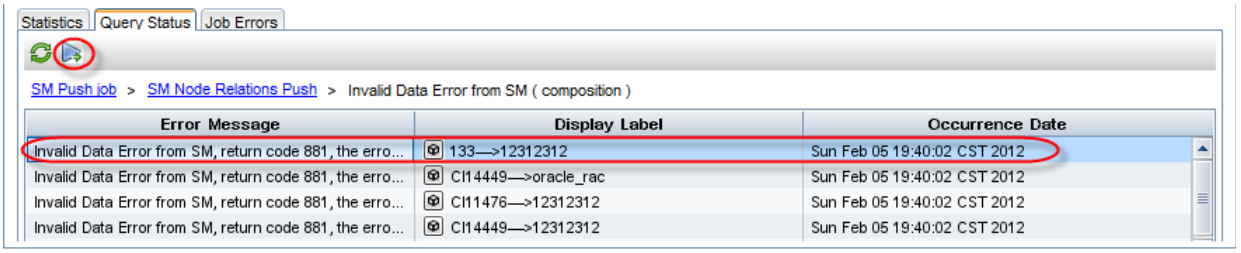
The query failure details (Error Message, CI Type, and CI Count) are displayed.



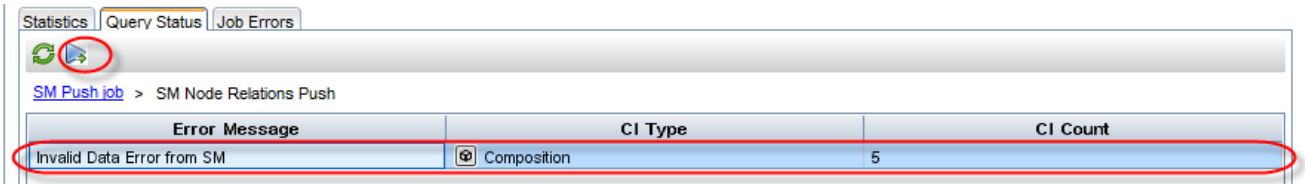
3. Double-click the Error Message, to see more details of the failed records.



4. Double-click each failed record to see the detailed error message of the record.
5. Fix the issues with each failed record according to the error information.
6. Select a failed record, and then click the **Push selected failed data** button to re-synchronize the record.



Alternatively, click the query name link (in this example, **SM Node Relations Push**), select the error message of the query, and then click the **Push selected failed data** button to re-push all failed records in the query.


Statistics
Query Status
Job Errors
SM Push job > SM Node Relations Push
Error Message
CI Type
CI Count
Invalid Data Error from SM
Composition
5

1.6.2.11. Query is not configured in smSyncConfFile.xml when using ServiceManagerAdapter9.41

The push job fails with a “Failed” status. From both the log file and the detail error message of the failed job in the Universal CMDB studio (see How to check the error message of a failed push job), you receive an error like the following:

```
java.lang.RuntimeException: No mapping is found for TQL: "SM Business Service Push",
please configure in smSyncConfFile.xml

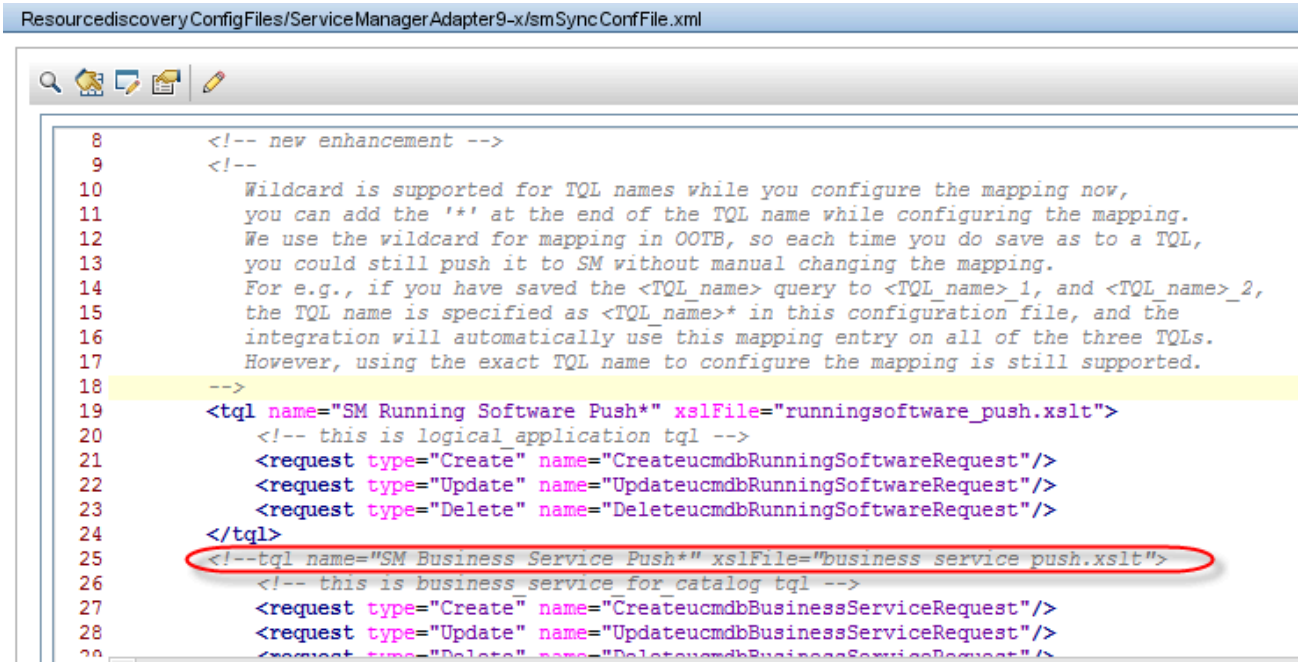
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.SmPusher.pushObjects(SmPush
er.java:273)

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.SmPusher.push(SmPusher.java
:93)

  ... 31 more
```

Cause

The query used for populating business service CIs is named `SM Business Service Push`, however you have not configured it (or have commented it out) in the `smSyncConfFile.xml` file:



Solution

Search for text `No mapping is found for TQL` to find the query name that's not yet configured, and then configure the query name in the `smSyncConfFile.xml` file.

For instructions on how to add a mapping for a query, see How to map the CI Type’s query to an XSL transformation file.

1.6.2.12. Nonexistent XSLT file name is defined for a query in smSyncConfFile.xml when using ServiceManagerAdapter9.41

The data push job fails with a “Failed” status. From both the log file and the detail error message of the failed job in the Universal CMDB studio (see How to check the error message of a failed push job), you receive an error like the following:

```
java.io.FileNotFoundException: Resource: business_service_push_wrong_name.xslt, was
not found

at
com.mercury.topaz.cmdb.server.fcmdb.adapterstate.AdapterStateResourceServerFacade.op
enResourceForReading(AdapterStateResourceServerFacade.java:51)

at
com.mercury.topaz.cmdb.server.fcmdb.spi.adapter.environment.DataAdapterEnvironmentIm
pl.openResourceForReading(DataAdapterEnvironmentImpl.java:119)

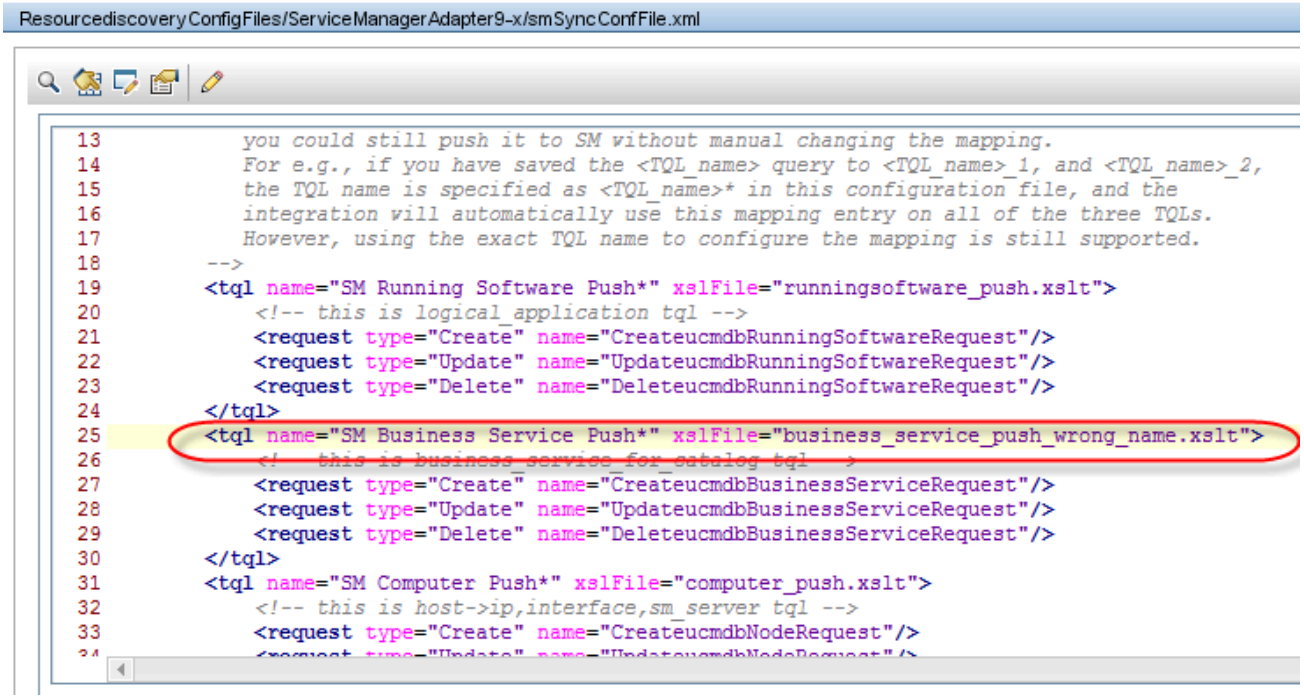
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.TransformUtils.readFromFile(Tran
sformUtils.java:113)

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.XsltTransformer.createTempl
ates(XsltTransformer.java:93)

... 36 more
```

Cause

The XSLT file for populating business service CIs is named `business_service_push.xslt`, however you have configured a wrong name `business_service_push_wrong_name.xslt` in `smSyncConfFile.xml`.



Solution

Search for text `java.io.FileNotFoundException: Resource:` to find the wrong XSLT file name, and then correct the XSLT file name in the `smSyncConfFile.xml` file.

For instructions on how to configure an XSLT file name for a query, see How to map the CI Type's query to an XSL transformation file.

1.6.2.13. Request name is not found for a query in smSyncConfFile.xml when using ServiceManagerAdapter9.41

The data push job fails with a “Failed” status. From both the log file and the detail error message of the failed job in the Universal CMDB studio (see How to check the error message of a failed push iob), you receive an error like the following:

```
java.lang.RuntimeException: No request name was found for operation "Create" of TQL
name "SM Business Service Push", please check in smSyncConfFile.xml file.

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.Mapping.createTransformer(M
apping.java:89)

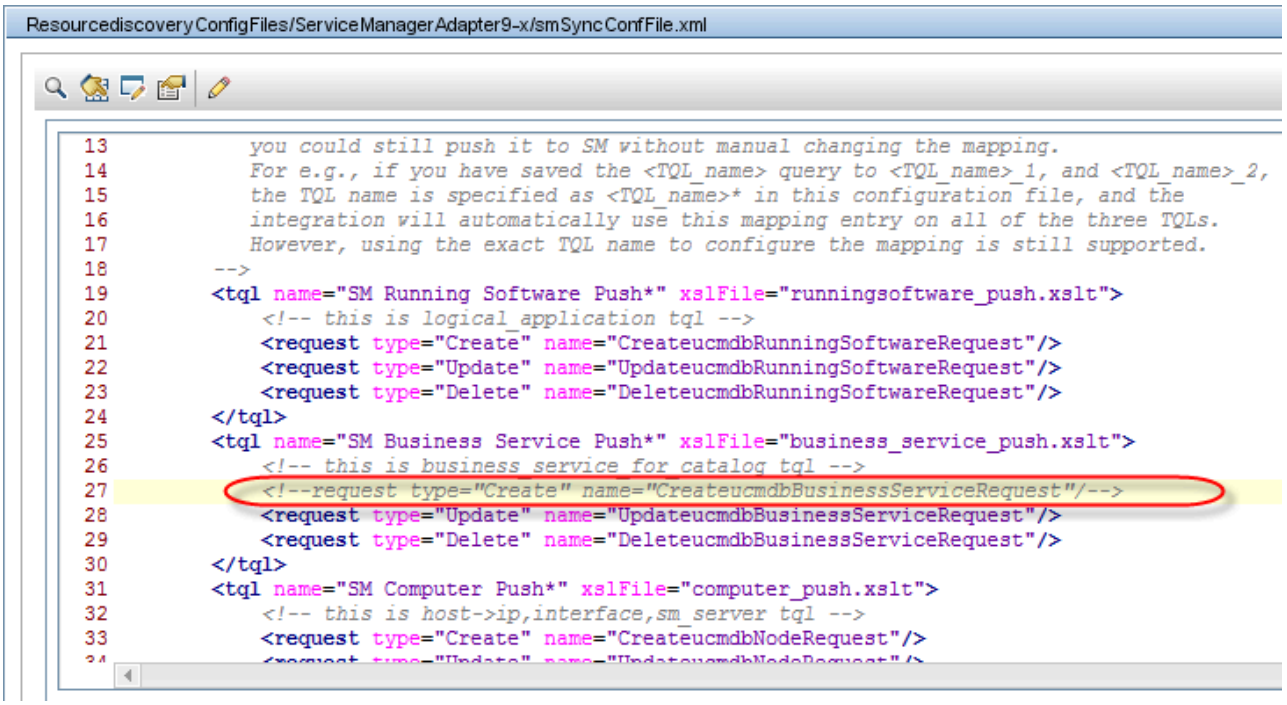
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.SmPusher.pushObjects(SmPush
er.java:276)

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.SmPusher.push(SmPusher.java
:95)

... 31 more & &
```

Cause

The Service Manager web service request of the Create type is named `CreateucmdbBusinessServiceRequest`, however you have not configured it (or have commented it out) in the `smSyncConfFile.xml` file.



Solution

In the error message find the query name for which a request name was not found, and then in the `smSyncConfFile.xml` file add the request name for the query name.

For instructions on how to configure a request for a query, see How to map the CI Type's query to an XSL transformation file.

1.6.2.14. Wrong Service Manager WS request name is defined in smSyncConfFile.xml when using ServiceManagerAdapter9.41

The data push job is completed with a “Passed with failures” status. From both the log file and the detailed error messages of the failed CIs in the Universal CMDB studio (see How to Check the Error Messages of Failed CIs/CI Relationships in a Push Job), you receive an error like the following:

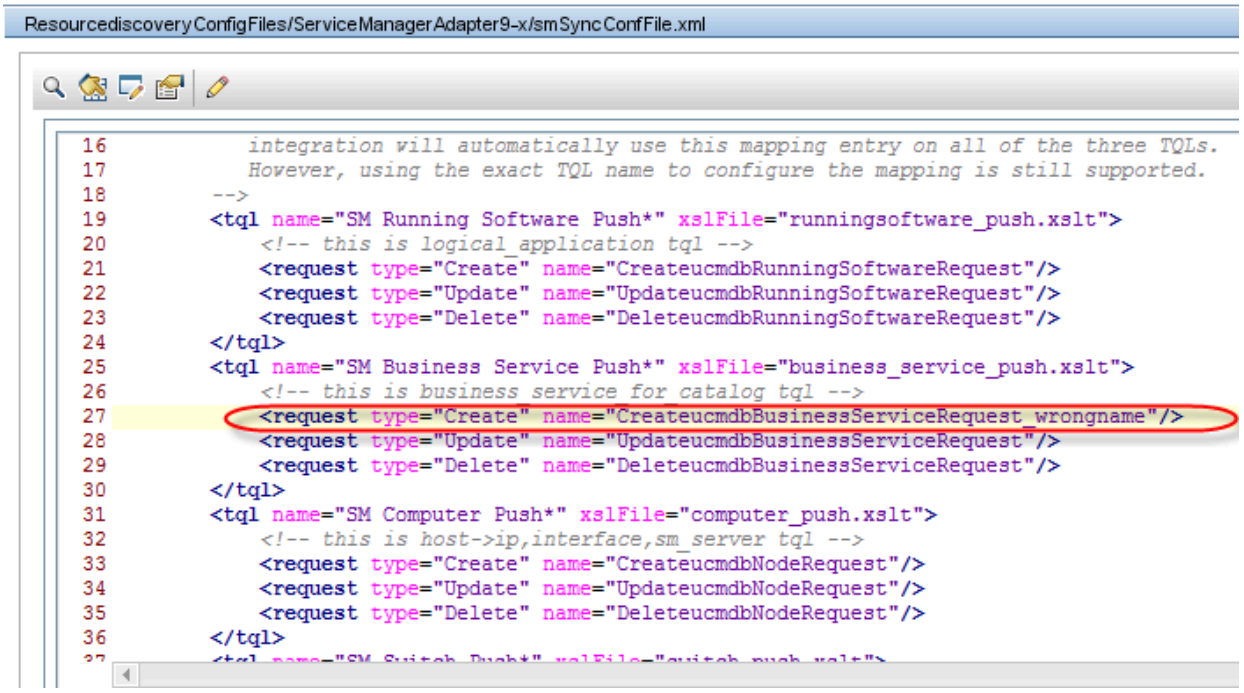
```
Other exception while sending message to SM, the exception message is "SOAP fault
received: A CXmlApiException was raised in native code : error 16 : scxmlapi(16) -
Invalid or missing file name in XML request"


& &
 Source tree XML:
 <?xml version="1.0" encoding="UTF-16"?>
<business_element bdmType="BusinessService" customer_id="1"
    display_label="test_bizservice_push_1"
    friendlyType="BusinessService" id="dc57d623182d759df82c7bccd2448630"
    name="test_bizservice_push_1" provider="provider1"/>

 Transformed XML:
<?xml version="1.0" encoding="UTF-16"?>
<CreateucmdbBusinessServiceRequest_wrongname>
    <model>
        <keys/>
        <instance>
            <UCMDBId>dc57d623182d759df82c7bccd2448630</UCMDBId>
            <CustomerId>1</CustomerId>
            <Type>bizservice</Type>
            <Subtype>Business Service</Subtype>
            <ServiceProvider>provider1</ServiceProvider>
            <ServiceName>test_bizservice_push_1</ServiceName>
            <CIIdentifier>test_bizservice_push_1</CIIdentifier>
        </instance>
    </model>
</CreateucmdbBusinessServiceRequest_wrongname>
```

Cause

The Service Manager web service request of the Create type is named `CreateucmdbBusinessServiceRequest`, however you have configured a wrong request name `CreateucmdbBusinessServiceRequest_wrongname` in the `smSyncConfFile.xml` file.



Solution

In the `Transformed XML` section of the error message, check the Request name (which is configured as `CreateucmdbBusinessServiceRequest_wrongname` in this sample), and make sure that the request name specified in `smSyncConfFile.xml` is the exact name defined in the WSDL.

For instructions on how to configure a Service Manager web service request for a query, see How to map the CI type's query to an XSL transformation file.

1.6.2.15. The XSLT file is not well-formed when using ServiceManagerAdapter9.41

The data push job fails with a “Failed” status. From both the log file and the detailed error message of the failed job in the Universal CMDB studio (see How to Check the Error Message of a Failed Push Job), you receive an error like the following:

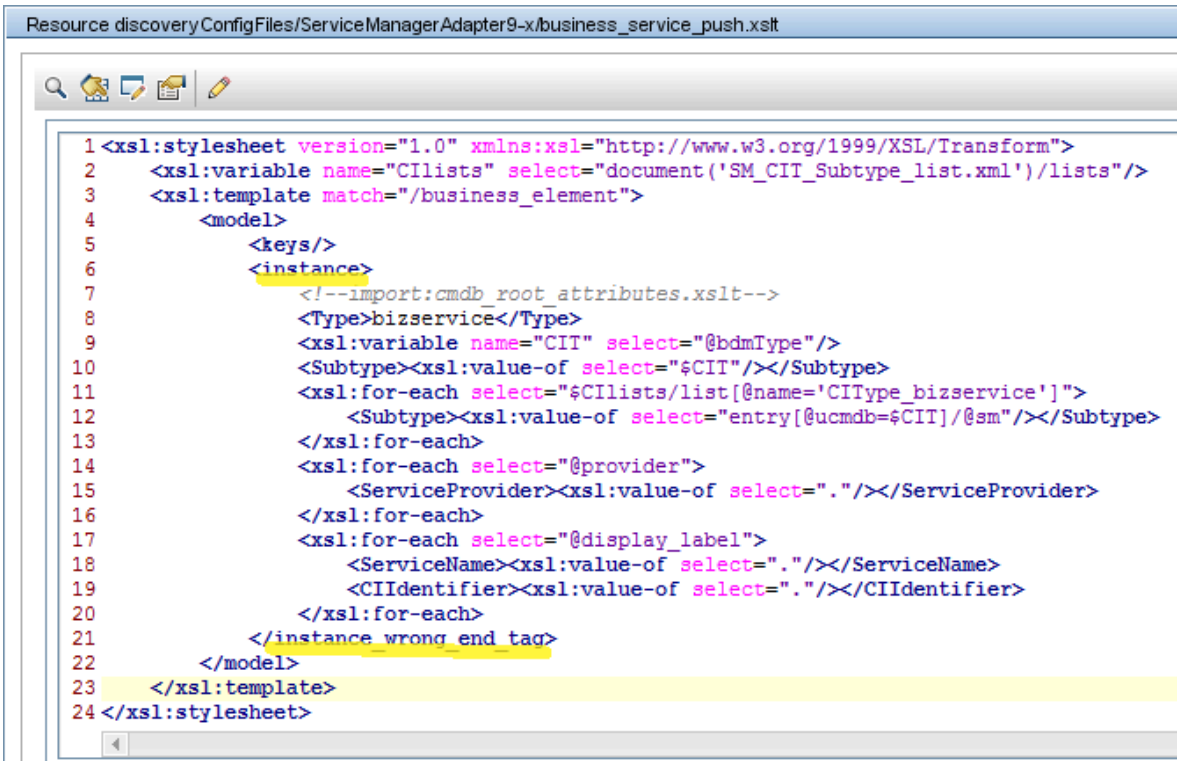
```
javax.xml.transform.TransformerConfigurationException: Failed to compile stylesheet
at
com.sun.org.apache.xalan.internal.xsltc.trax.TransformerFactoryImpl.newTemplates(Unk
nown Source)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.XsltTransformer.createTempl
ates(XsltTransformer.java:100)
... 36 more
```

In addition, you can find more detailed error message in the log file to see which XSLT file is not well formed. You should see something like following:

```
2011-12-12 14:54:58,108 [170025072@Default-10] ERROR - SM Integration >> Got
DataAccessException while updateData
com.hp.ucmdb.federationspi.exception.DataAccessGeneralException: Not valid XSLT file
"busi-ness_service_push.xslt", please check this XSLT file
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.XsltTransformer.createTempl
ates(XsltTransformer.java:103)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.push.Mapping.parseConfig(Mapping
.java:54)
```

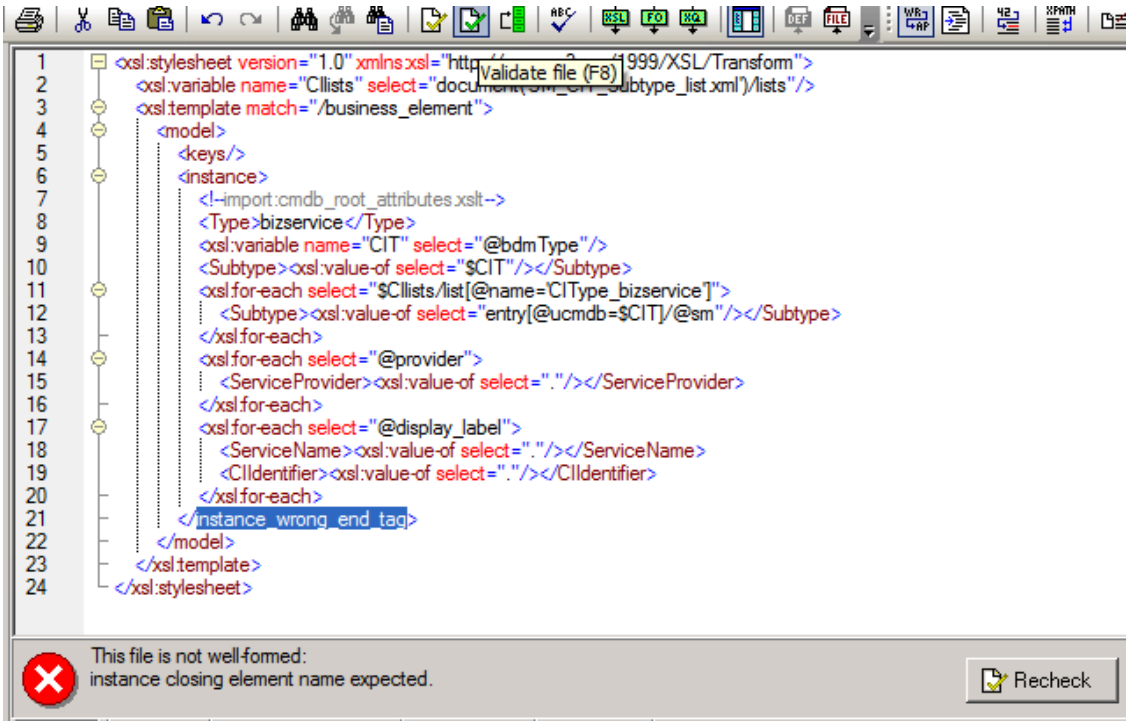
Cause

The end tag of “instance” should be `</instance>`, however you have configured a wrong end tag `</instance_wrong_end_tag>`.



Solution

Search for text `Not valid XSLT file` to find the XSLT file name, and then validate the XSLT file in an XML editor (for example, XMLSpy). You can easily find and fix any validation issues.

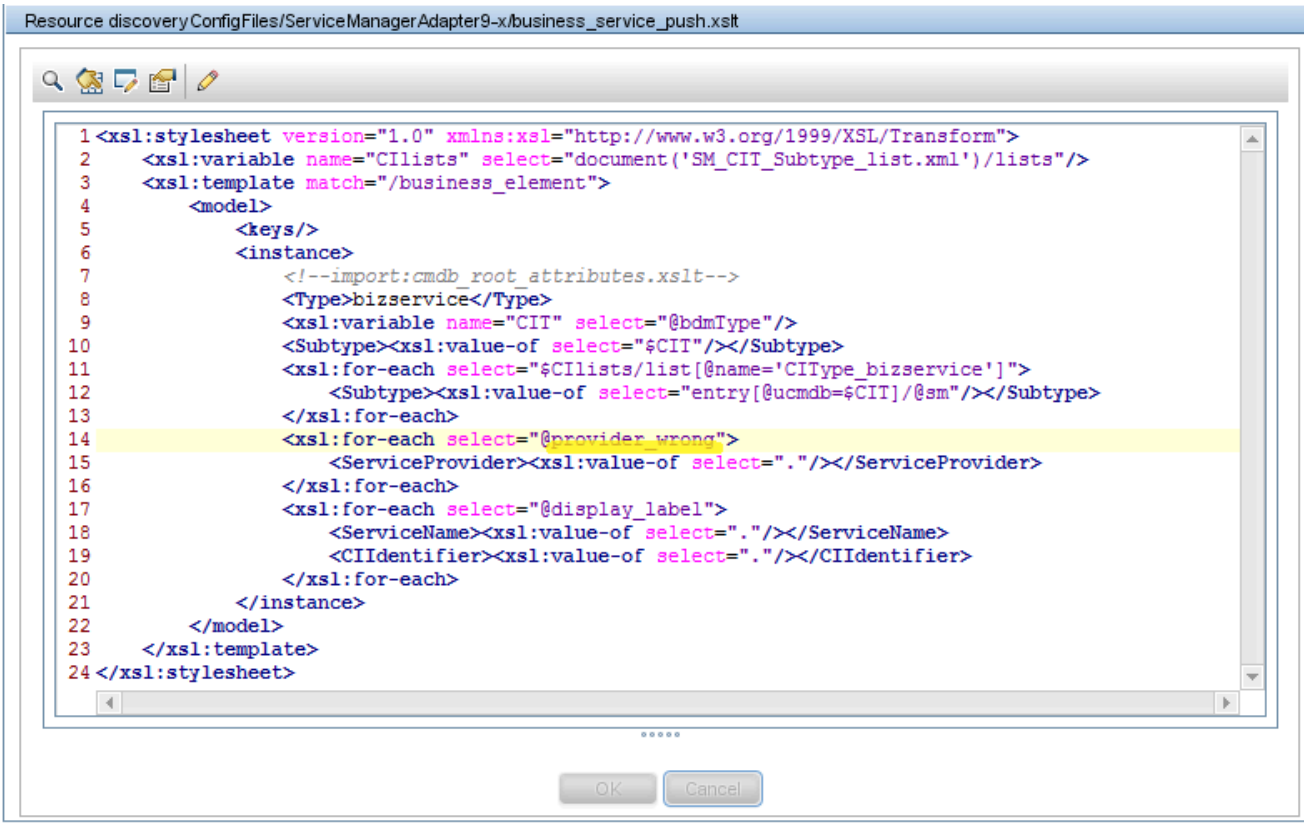


1.6.2.16. The UCMDB attribute name is wrong in the XSLT file when using ServiceManagerAdapter9.41

You will NOT get any error messages either in the log file or in the Universal CMDB studio, however the Universal CMDB attribute value will not be pushed to Service Manager.

Cause

The Universal CMCB attribute name is `provider`, however you have configured a wrong attribute named `provider_wrong`.



Solution

If you find a UCMDB attribute value that can't be pushed to Service Manager, double-check the UCDMB attribute name and Service Manager field name of the mapping in the XSLT file.

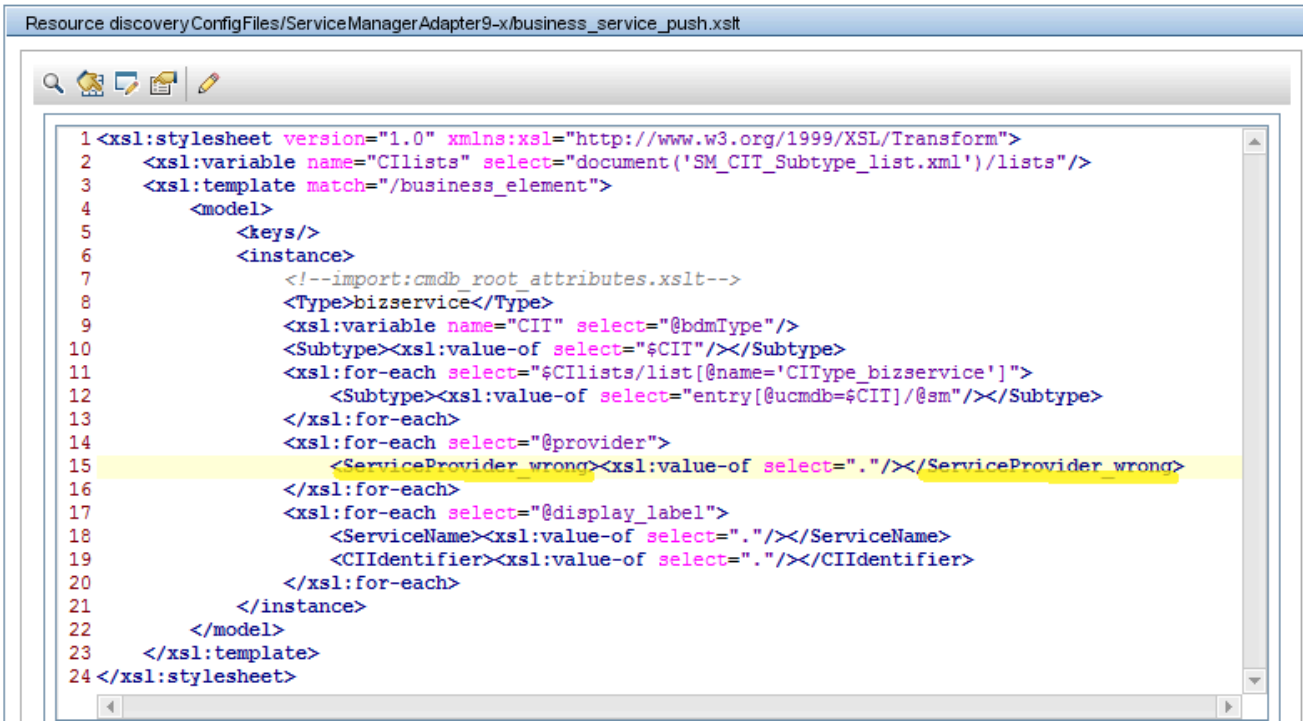
For instructions on how to configure an attribute mapping, see How to map the CI Attribute to a Web Service field.

1.6.2.17. The Service Manager field name is wrong in the XSLT File when using ServiceManagerAdapter9.41

You will NOT get any error messages either in the log file or in the Universal CMDB studio, however, the Universal CMDB attribute value will not be pushed to Service Manager.

Cause

The Service Manager field name is `ServiceProvider`, however you have configured a wrong attribute named `ServiceProvider_wrong` in the XSLT file. This error also commonly occurs since the attribute is case-sensitive.



Solution

If you find a Universal CMDB attribute value that can't be pushed to Service Manager, check both the Universal CDMB attribute name and Service Manager field name of the mapping in the XSLT file.

For instructions on how to configure an attribute mapping, see How to map the CI Attribute to a Web Service field.

1.6.2.18. The data push job passes with failures when using ServiceManagerAdapter9.41

The data push job completes with a `Passed with failures` status. From both the log file and the detailed error messages of the failed CIs in the Universal CMDB studio, you receive an error like the following.

```
Validation Check Fails Error from SM, return code 71, the error message is "Validation
failed"


CI Push Details:
 Source tree XML:
 &

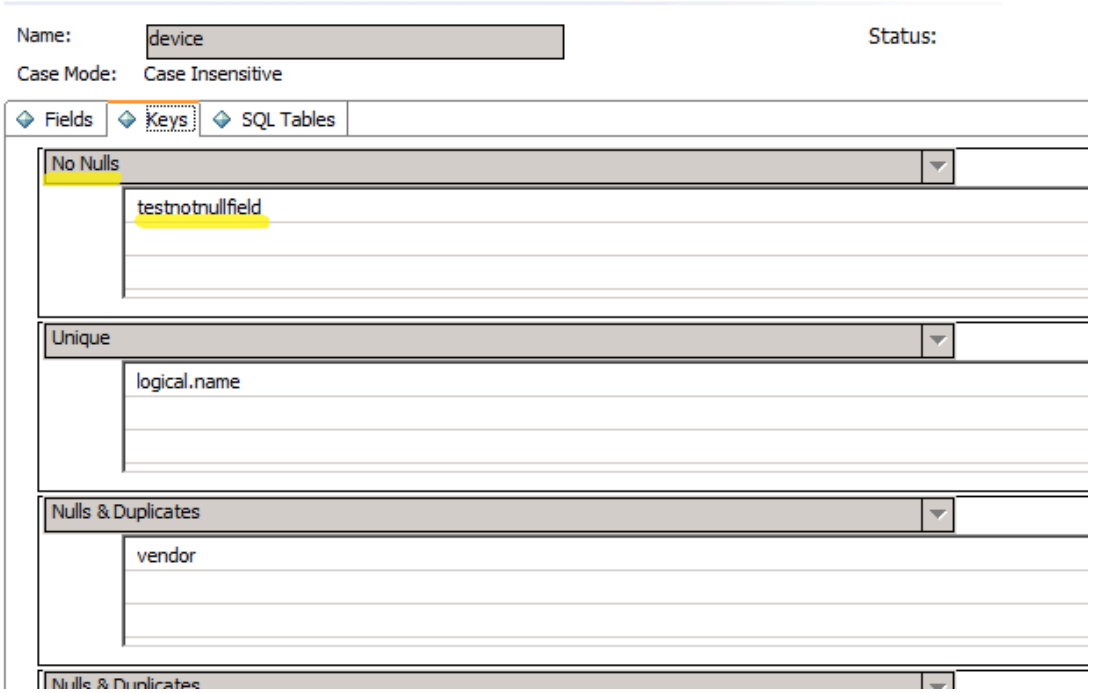

 Transformed XML:
&


 Response message from SM:
<?xml version="1.0" encoding="UTF-16"?>
<CreateucmdbBusinessServiceResponse message="Validation failed"
    returnCode="71" schemaRevisionDate="2011-09-18"
    schemaRevisionLevel="5" status="FAILURE"
    xmlns="http://servicecenter.peregrine.com/PWS"
    xmlns:cmn="http://servicecenter.peregrine.com/PWS/Common"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://servicecenter.peregrine.com/PWS
http://XMA16.asiapacific.hpqcorp.net:13080/sc62server/ucmdbBusinessService.xsd">
    <model>
        <keys>
            <CIIdentifier type="String">test_bizservice_push_1</CIIdentifier>
        </keys>
<instance uniquequery="file.device,logical.name=&amp;quot;test_bizservice_push_1&amp;quot;">
            <file.device type="Structure">
                <CIIdentifier type="String">test_bizservice_push_1</CIIdentifier>
                <ServiceProvider type="String">provider1</ServiceProvider>
                <Type type="String">bizservice</Type>
                <CIName type="String">CI10894</CIName>
                <Subtype type="String">Business Service</Subtype>
                <ServiceName type="String">test_bizservice_push_1</ServiceName>
                <UCMDBId type="String">dc57d623182d759df82c7bccd2448630</UCMDBId>
             </file.device>
</instance>
```

```
</model>
    <messages>
        <cmn:message type="String">6=====add</cmn:message>
        <cmn:message type="String">Key #1 is empty.
(se.base.method,add.record.radd)</cmn:message>
        <cmn:message type="String">file:(device)
key:(logical.name=test_bizservice_push_1)
(se.base.method,add.record.radd)</cmn:message>
        <cmn:message type="String">The record being added contains a NULL key
(se.base.method,add.record.radd)</cmn:message>
        &
    </messages>
</CreateucmdbBusinessServiceResponse>
```

Cause

In Service Manager you have configured a No Nulls key for the **testnotnullfield** field of the device table, however you haven't mapped this field in the `XSLT` file.



Solution

Find the key number in the `<messages>` section to see which No Nulls key has a null value. For example, suppose you have found a message `Key #1 is empty`, check the first key definition of the device table to see which field(s) this key is for. Next, make sure to map the field(s) in the `XSLT` file with a value that isn't null.

For instructions on how to configure an attribute mapping, see How to map the CI Attribute to a Web Service field.

1.6.2.19. The CI display name is truncated when using ServiceManagerAdapter9.41

The CI display name is truncated when using ServiceManagerAdapter9.41.

Cause

The length of a UCMDB CI name exceeds 200 characters, which is the maximum allowed field length for `display.name` (Display Name) in Service Manager. When this CI is pushed to Service Manager, the CI name is truncated.

Solution

Update the CI type's XSL transformation file so that, before running a push, the integration can make sure that all CI names are much less than 200 characters in length.

Contents

- 1 Cause
- 2 Solution

1.6.2.20. Global ID and Customer ID are missing from XSLT when using ServiceManagerAdapter9.41

No error message occurs, however when you update or delete a CI record in UCMDB, the update or deletion will not be pushed to Service Manager.

Cause

You create an XSL transformation file for push without the following element:

```
<!--import:cmdb_root_attributes.xslt-->
```

The `cmdb_root_attributes.xslt` file contains Global ID and Customer ID, which are required for data push:

```
<UCMDBId><xsl:value-of select="@id"/></UCMDBId>
```

```
<CustomerId><xsl:value-of select="@customer_id"/></CustomerId>
```

Solution

In the XSL transformation file, include the missing element:

```
<!--import:cmdb_root_attributes.xslt-->
```

Contents

- 1 Cause
- 2 Solution

1.6.2.21. Data population fails when using ServiceManagerAdapter9.41

Data population fails when using **ServiceManagerAdapter9.41**.

Cause

When population errors or problems occur, you can check the error messages and the population log file to identify the root causes and then solve the problems.

Solution

When a population job has failed, the job status becomes **Failed**. Troubleshoot the failed job as follows:

- Check the error message of the failed job in the Universal CMDB studio.

  See *How to check the error message of a failed population job* below.

- Check the log file for more details.

  See *How to check the population log file* below.

How to check the error message of a failed population job

While a population job fails, you can check the detailed error messages in the Universal CMDB studio.

To check the error message of a failed population job:

1. Sign in to UCMDB as an administrator.
2. Navigate to **Data Flow Management** > **Integration Studio**.
3. Select the integration point for this integration.
4. Click the **Population** tab.
5. Select the failed job from Integration Jobs, and click the **Job Errors** tab.
6. Double-click an error message from the list.

   A pop-up window opens to display the error details. The following is a sample error message about an `XSLT` file that doesn't exist.

```
Failed initializing the datastore. adapterID:ServiceManagerAdapter9-x destID:SM
Integration
ER-ROR:com.mercury.topaz.cmdb.shared.fcmdb.dataAccess.exception.AdapterAccessFaile
dToStartAdapterException: [ErrorCode [850] Integration Point cannot start{SM
Integration}]

Failed to start adapter [SM Integration].java.lang.RuntimeException: No XSLT file
is found "busi-ness_service_population_wrong.xslt", please configure the right XSLT
file in smPopConfFile.xml

com.hp.ucmdb.federationspi.exception.DataAccessCommunicationException:
java.lang.RuntimeException: No XSLT file is found
"business_service_population_wrong.xslt", please configure the right XSLT file in
smPopConf-File.xml

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.ServiceDeskAdapter.start(Servi
ceDeskAdapter.java:841)

at
com.mercury.topaz.cmdb.server.fcmdb.dataAccess.manager.impl.AbstractDataAccessCont
ainerManagerImpl.startAdapter(AbstractDataAccessContainerManagerImpl.java:244)

at
com.mercury.topaz.cmdb.server.fcmdb.dataAccess.manager.impl.AbstractDataAccessCont
ainerManagerImpl.addStartAndReturnBasicDataAdapterWrapper(AbstractDataAccessContai
nerManagerImpl.java:162)

at
com.mercury.topaz.cmdb.server.fcmdb.dataAccess.manager.impl.AbstractDataAccessCont
ainerManagerImpl.getStartedBasicDataAdapterWrapper(AbstractDataAccessContainerMana
gerImpl.java:187)

at
com.hp.ucmdb.discovery.probe.agents.probemgr.adapters.DataAccessAdaptersFacade.get
StartedBasicDataAdapterWrapper(DataAccessAdaptersFacade.java:139)

at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runDiscovery(Ada
pterService.java:172)

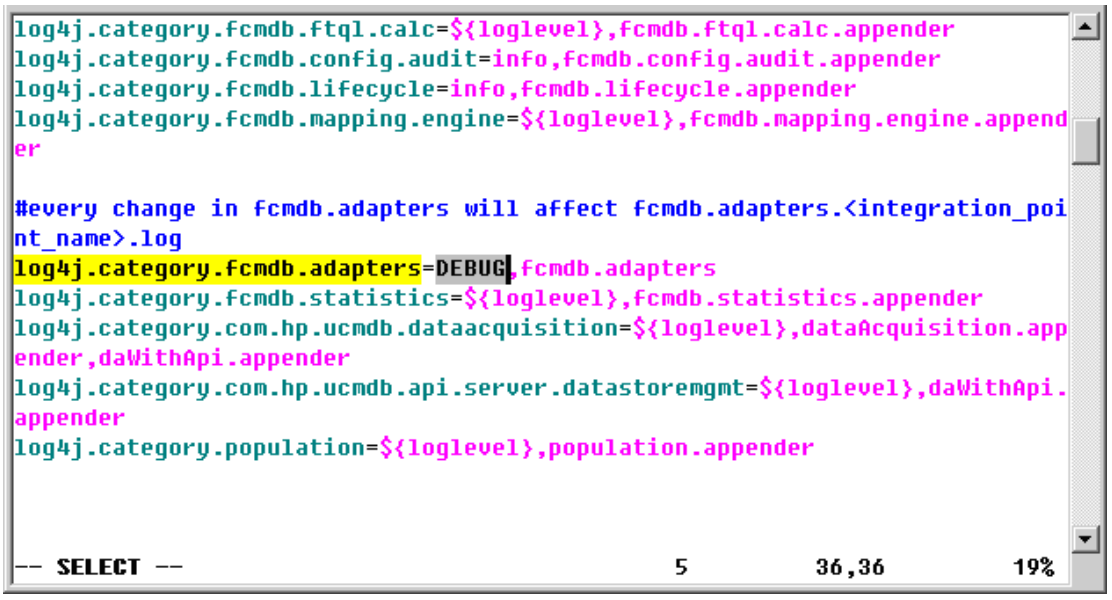
   at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.discover(Adapter
Service.java:140)& &
```

How to check the population log file

You can set the Development adapter log level to `DEBUG` to check the incoming CI XML file of Service Manager, and the XML file transformed by `XSLT`. It's advisable that you enable the Development Mode for the integration point so that these two XML files are in a good format.

To set the adapter log level to `DEBUG`:

1. Sign in to the UCMDB server host as an administrator.
2. Navigate to the `<UCMDB installation folder>\DataFlowProbe\conf\log\fcmdb.properties` file. For example, `C:\UCMDB\DataFlowProbe\conf\log\fcmdb.properties`
3. Open the `fcmdb.properties` configuration file in a text editor.
4. Update the `log4j.category.fcmdb.adapters` log level to **DEBUG**:



5. Save the file.
6. Wait a while for the change to take effect.

To enable the Development Mode of the integration point:

For detailed steps, see How to Check the Push Log File.

To check the population log file:

1. Sign in to the UCMDB server host as an administrator.
2. Navigate to the `<UCMDB installation folder>\DataFlowProbe\runtime\log\fcmdb.adapters.<integration_point_name>.log` file. For example, `C:\UCMDB\DataFlowProbe\runtime\log\fcmdb.adapters.SM Integration.log`
3. Open the log file in a text editor.
4. Search for text strings `Source SM CI XML`, `Transformed XML`, `Source tree XML for ID Pushback`, `Transformed XML for ID Pushback` and `Response message from SM For ID Pushback`. The log looks like the following:

```
2011-12-13 11:01:18,063 [JobExecuterWorker-0:DS_SM Integration_bizservice
population] DEBUG - SM Integra-tion >> Source SM CI XML:
<?xml version="1.0" encoding="UTF-16"?>
<SOAP-ENV:Envelope xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
    <SOAP-ENV:Body>
        <RetrieveucmdbBusinessServiceListResponse
            xmlns="http://servicecenter.peregrine.com/PWS"
            message="Success" returnCode="0"
            schemaRevisionDate="2011-09-18" schemaRevisionLevel="5"
            status="SUCCESS"
            xmlns:cmn="http://servicecenter.peregrine.com/PWS/Common"
            xmlns:xsd="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://servicecenter.peregrine.com/PWS
http://XMA16.asiapacific.hpqcorp.net:13080/sc62server/ucmdbBusinessService.xsd">
            <instance query=""
uniquequery="file.device,logical.name=&quot;test_biz_servcie_1&quot;">
                <file.device type="Structure">
                    <CIIdentifier type="String">test_biz_servcie_1</CIIdentifier>
                    <ServiceProvider type="String">VENDOR1</ServiceProvider>
                    <Type type="String">bizservice</Type>
                    <CIName type="String">CI10873</CIName>
                    <Subtype type="String">Business Service</Subtype>
                </file.device>
            </instance>
            <instance query=""
uniquequery="file.device,logical.name=&quot;test_biz_servcie_2&quot;">
                <file.device type="Structure">
                    <CIIdentifier type="String">test_biz_servcie_2</CIIdentifier>
                    <ServiceProvider type="String">VENDOR2</ServiceProvider>
                    <Type type="String">bizservice</Type>
                    <CIName type="String">CI10872</CIName>
                    <Subtype type="String">Business Service</Subtype>
                </file.device>
            </instance>
        </RetrieveucmdbBusinessServiceListResponse>
  </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```

```
2011-12-13 11:01:18,414 [JobExecuterWorker-0:DS_SM Integration_bizservice
population] DEBUG - SM Integra-tion >> Transformed XML:
<?xml version="1.0" encoding="UTF-16"?>
<topology>
    <ci class="business_service">
        <attribute name="name" type="String">test_biz_servcie_1</attribute>
        <attribute name="sm_id" type="String">CI10873</attribute>
        <attribute name="global_id" type="String"/>
        <attribute name="provider" type="String">VENDOR1</attribute>
    </ci>
    <ci class="business_service">
        <attribute name="name" type="String">test_biz_servcie_2</attribute>
        <attribute name="sm_id" type="String">CI10872</attribute>
        <attribute name="global_id" type="String"/>
        <attribute name="provider" type="String">VENDOR2</attribute>
    </ci>
</topology>
& &
2011-12-13 11:01:32,061 [JobExecuterWorker-0:DS_SM Integration_bizservice
population] DEBUG - SM Integra-tion >> Source tree XML for ID Pushback to update SM
CI "CI10873" with uCMDB ID "50fccdeaec49b1d81a1f54ca99942b27":
<?xml version="1.0" encoding="UTF-16"?>
<ucmdbIDPushBack>
    <ci id="CI10873" ucmdbid="50fccdeaec49b1d81a1f54ca99942b27"/>
</ucmdbIDPushBack>

2011-12-13 11:01:32,084 [JobExecuterWorker-0:DS_SM Integration_bizservice
population] DEBUG - SM Integra-tion >> Transformed XML for ID Pushback:
<?xml version="1.0" encoding="UTF-16"?>
<UpdateucmdbIDPushBackRequest>
    <model xmlns:ns="http://schemas.hp.com/SM/7">
        <keys/>
        <instance>
            <ConfigurationItem>CI10873</ConfigurationItem>
            <UcmdbID>50fccdeaec49b1d81a1f54ca99942b27</UcmdbID>
        </instance>
    </model>
</UpdateucmdbIDPushBackRequest>
```

```
& &
2011-12-13 11:01:32,240 [JobExecuterWorker-0:DS_SM Integration_bizservice
population] DEBUG - SM Integra-tion >> Response message from SM For ID Pushback:
<?xml version="1.0" encoding="UTF-16"?>
<SOAP-ENV:Envelope xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
    <SOAP-ENV:Body>
        <UpdateucmdbIDPushBackResponse
            xmlns="http://servicecenter.peregrine.com/PWS"
            message="Success" returnCode="0"
            schemaRevisionDate="2011-09-18" schemaRevisionLevel="5"
            status="SUCCESS"
            xmlns:cmn="http://servicecenter.peregrine.com/PWS/Common"
            xmlns:xsd="http://www.w3.org/2001/XMLSchema"
            xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://servicecenter.peregrine.com/PWS
http://XMA16.asiapacific.hpqcorp.net:13080/sc62server/ucmdbIDPushBack.xsd">
            <model>
                <keys>
                    <ConfigurationItem type="String">CI10873</ConfigurationItem>
                </keys>
                <instance recordid="CI10873 -  - "
uniquequery="logical.name=&amp;quot;CI10873&amp;quot;">
                    <ConfigurationItem type="String">CI10873</ConfigurationItem>
                    <UcmdbID
type="String">50fccdeaec49b1d81a1f54ca99942b27</UcmdbID>
                </instance>
            </model>
            <messages>
                <cmn:message type="String">==Foundabcdefg</cmn:message>
                <cmn:message type="String">Found</cmn:message>
                <cmn:message type="String">going to update ucmdb_id to
abcdefg</cmn:message>
                <cmn:message type="String">Update done!</cmn:message>
                <cmn:message type="String">Foundabcdefg</cmn:message>
            </messages>
        </UpdateucmdbIDPushBackResponse>
    </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```

1.6.2.22. Query is not configured in smPopConfFile.xml when using ServiceManagerAdapter9.41

Query isn't configured in `smPopConfFile.xml` when using **ServiceManagerAdapter9.41**.

Cause

If you haven't yet added a query to your job, you can't select this query from the list while you create or update your job.

If you have already added this query to your job before removing it from `smPopConfFile.xml`, you will get a `Failed` status while you run this population job. In addition, in the Universal CMDB studio, you will get an error message like the following (see How to Check the Error Message of a Failed Population Job):

```
Failed running population. destID:SM Integration, Failed during query: , all
queries:[SM Business Service Popula-tion],  finished queries:[]
ERROR:com.hp.ucmdb.federationspi.exception.DataAccessGeneralException: Query [SM
Business Service Population] is not supported by this adapter.

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.ServiceDeskAdapter.getQueryByN
ame(ServiceDeskAdapter.java:990)

at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.retrieveQueryDef
initionByRootType(AdapterService.java:247)

at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.runDiscovery(Ada
pterService.java:183)

at
com.hp.ucmdb.discovery.probe.services.dynamic.core.AdapterService.discover(Adapter
Service.java:140)

at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter.launchTask(J
obExecuter.java:1194)

at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobExecuterW
orker.launch(JobExecuter.java:963)

at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobExecuterW
orker.executeTask(JobExecuter.java:908)

at
com.hp.ucmdb.discovery.probe.agents.probemgr.taskexecuter.JobExecuter$JobExecuterW
orker.run(JobExecuter.java:813)
```

Solution

Search for text `is unsupported by this adapter` to find the query name that you haven't configured, and then configure it in the `smPopConfFile.xml` file.

For instructions on how to add a mapping to a query, see How to map the CI type's query to an `XSL` transformation file.

1.6.2.23. Nonexistent XSLT file name is defined for a query in smPopConfFile.xml when using ServiceManagerAdapter9.41

You will get a “Failed” status while you run the population job. In addition, from both the population log file (see How to Check the Population Log File) and the Universal CMDB studio (see How to Check the Error Message of a Failed Population Job), you will get an error message like the following:

```
java.lang.RuntimeException: No XSLT file is found
"business_service_population_wrong.xslt", please configure the right XSLT file in
smPopConfFile.xml

com.hp.ucmdb.federationspi.exception.DataAccessCommunicationException:
java.lang.RuntimeException: No XSLT file is found
"business_service_population_wrong.xslt", please configure the right XSLT file in
smPopConf-File.xml

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.ServiceDeskAdapter.start(Servi
ceDeskAdapter.java:841)

at
com.mercury.topaz.cmdb.server.fcmdb.dataAccess.manager.impl.AbstractDataAccessCont
ainerManagerImpl.startAdapter(AbstractDataAccessContainerManagerImpl.java:244) & &
```

Cause

The XSLT file name is not correct.

Solution

Search for text `No XSLT file is found` to find the wrong XSLT file name, and then correct the name in the `smPopConfFile.xml` file.

For instructions on how to configure an XSLT file name for a query, see How to map the CI type's query to an XSL transformation file.

1.6.2.24. The "Retrieve" type request is undefined in smPopConfFile.xml when using ServiceManagerAdapter9.41

You will get a “Failed” status while you run the population job. In addition, from both the population log file and the Universal CMDB studio, you will get an error message like the following:

```
java.lang.RuntimeException: No request was configured for operation "Retrieve" of
TQL name "SM Business Ser-vice Population", please check in smPopConfFile.xml file.
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.PopChunkGetter.init(
PopChunkGetter.java:134)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.PopChunkGetter.<init
>(PopChunkGetter.java:104)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.ServiceDeskAdapter.getChanges(S
erviceDeskAdapter.java:1108)
at
com.mercury.topaz.cmdb.server.fcmdb.dataAccess.operation.query.impl.DataAccessAdapt
erQueryRetrieveChanges.getChangesResult(DataAccessAdapterQueryRetrieveChanges.java:
48)
```

Cause

The "Retrieve" type request is undefined in smPopConfFile.xml.

Solution

Search for text `No request was configured for operation` to find the query name for which a retrieve type request is missing, and then add the missing request name in the `smPopConfFile.xml` file.

For instructions on how to configure a request for a query, see How to map the CI type's query to an XSL transformation file.

1.6.2.25. Wrong request name of retrieveKeysQueryName is configured in smPopConfFile.xml when using ServiceManagerAdapter9.41

You will get a “Failed” status while you run the population job. In addition, from both the population log file (see How to Check the Population Log File) and the Universal CMDB studio (see How to Check the Error Message of a Failed Population Job), you will get an error message like the following:

```
2011-12-13 17:18:04,394 [JobExecuterWorker-0:DS_SM Integration_bizservice
population] ERROR - SM Integra-tion >> Populate ci data failed for element

com.hp.ucmdb.federationspi.exception.DataAccessGeneralException: SOAP fault received
for retrieving SM CI Keys: A CXmlApiException was raised in native code : error 16 :
scxmlapi(16) - Invalid or missing file name in XML request

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.SmPopulater.response
ExceptionHandle(SmPopulater.java:322)

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.SmPopulater.queryKey
List(SmPopulater.java:116)

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.PopChunkGetter.init(
PopChunkGetter.java:139)

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.PopChunkGetter.<init
>(PopChunkGetter.java:104)

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.ServiceDeskAdapter.getChanges(S
erviceDeskAdapter.java:1111)

& &
```

Further more, you can find more detailed error message in the log file that indicates which request of retrieving CI keys is wrong. The following is an example:

```
2011-12-13 17:18:02,832 [JobExecuterWorker-0:DS_SM Integration_bizservice
population] DEBUG - SM Integra-tion >> Web Service Request XMl For getting SM CI
keys:
<?xml version="1.0" encoding="UTF-16"?>
<soapenv:Envelope xmlns:enc="http://schemas.xmlsoap.org/soap/encoding/"
    xmlns:env="http://schemas.xmlsoap.org/soap/envelop/"
    xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
    <soapenv:Body>
        <axis2ns1:RetrieveucmdbBusinessServiceKeysListRequest_wrong>
            <axis2ns2:model>
                <axis2ns3:keys query="type#&quot;bizservice&quot;"/>
                <axis2ns4:instance>
                    <axis2ns5:file.device/>
                </axis2ns4:instance>
                <axis2ns6:messages/>
            </axis2ns2:model>
        </axis2ns1:RetrieveucmdbBusinessServiceKeysListRequest_wrong>
    </soapenv:Body>
</soapenv:Envelope>
```

Cause

Wrong request name of retrieveKeysQueryName is configured in smPopConfFile.xml.

Solution

Search for text `Web Service Request XMl For getting SM CI keys` to find the wrong request name, and then specify the right request name in the `smPopConfFile.xml` file for the query name.

For instructions on how to configure a request for a query, see How to map the CI type's query to an XSL transformation file.

1.6.2.26. Wrong request name of retrieveListQueryName is configured in smPopConfFile.xml when using ServiceManagerAdapter9.41

You will get a “Failed” status while you run the population job. In addition, from both the population log file (see How to Check the Population Log File) and the Universal CMDB studio (see How to Check the Error Message of a Failed Population Job), you will get an error message like the following:

```
com.hp.ucmdb.federationspi.exception.DataAccessGeneralException: SOAP fault received
for retrieving SM CI List: A CXmlApiException was raised in native code : error 16 :
scxmlapi(16) - Invalid or missing file name in XML request
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.SmPopulater.response
ExceptionHandle(SmPopulater.java:324)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.SmPopulater.queryRec
ordList(SmPopulater.java:188)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.PopChunkGetter.addOr
DelCIsToTopology(PopChunkGetter.java:177)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.PopChunkGetter.getNe
xtResultChunk(PopChunkGetter.java:164)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.ServiceDeskAdapter.getChanges(S
erviceDeskAdapter.java:1149)
& &
```

Further more, you can find more detailed error message in the log file that indicates which request of retrieving a CI list is wrong. The following is an example.

```
2011-12-13 17:30:23,424 [JobExecuterWorker-0:DS_SM Integration_bizservice
population] DEBUG - SM Integra-tion >> Web Service Request XMl For getting SM CI
List:
<?xml version="1.0" encoding="UTF-16"?>
<soapenv:Envelope xmlns:enc="http://schemas.xmlsoap.org/soap/encoding/"
    xmlns:env="http://schemas.xmlsoap.org/soap/envelop/"
    xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
    <soapenv:Body>
        <axis2ns13:RetrieveucmdbBusinessServiceListRequest_wrong>
            <axis2ns14:keys>
                <axis2ns15:CIIdentifier>test_biz_servcie_1</axis2ns15:CIIdentifier>
            </axis2ns14:keys>
            <axis2ns16:keys>
                <axis2ns17:CIIdentifier>test_biz_servcie_2</axis2ns17:CIIdentifier>
            </axis2ns16:keys>
        </axis2ns13:RetrieveucmdbBusinessServiceListRequest_wrong>
    </soapenv:Body>
</soapenv:Envelope>
```

Cause

Wrong request name of retrieveListQueryName is configured in smPopConfFile.xml.

Solution

Search for text `Web Service Request XMl For getting SM CI List` to find the wrong request name, and then specify the right request name in the `smPopConfFile.xml` file for the query name.

For instructions on how to configure a request for a query, see How to map the CI type's query to an XSL transformation file.

1.6.2.27. The XSLT file is not well-formed when using ServiceManagerAdapter9.41

Suppose you haven't yet created your integration point. When you create it an error message similar to the following example will occur in the Universal CMDB studio or the log file, causing the creation to fail.

Suppose you have already created your integration point, but haven't yet activated it, you can no longer activate it. When you try to activate it you will get a detailed error message in the probe error log file (`probe-error.log`), similar to the following example.

If you have created and activated your integration point, you will get a `Failed` status when running the population job. In addition, from both the population log file and the Universal CMDB studio, you will get an error message similar to the following example.

```
java.lang.RuntimeException: Got SAXException while parsing the XSLT file
"business_service_population.xslt"!

com.hp.ucmdb.federationspi.exception.DataAccessCommunicationException:
java.lang.RuntimeException: Got SAXException while parsing the XSLT file
"business_service_population.xslt"!

at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.ServiceDeskAdapter.start(Servic
eDeskAdapter.java:841)

at
com.mercury.topaz.cmdb.server.fcmdb.dataAccess.manager.impl.AbstractDataAccessConta
inerManagerImpl.startAdapter(AbstractDataAccessContainerManagerImpl.java:244)

at
com.mercury.topaz.cmdb.server.fcmdb.dataAccess.manager.impl.AbstractDataAccessConta
inerManagerImpl.addStartAndReturnBasicDataAdapterWrapper(AbstractDataAccessContaine
rManagerImpl.java:162)

& &

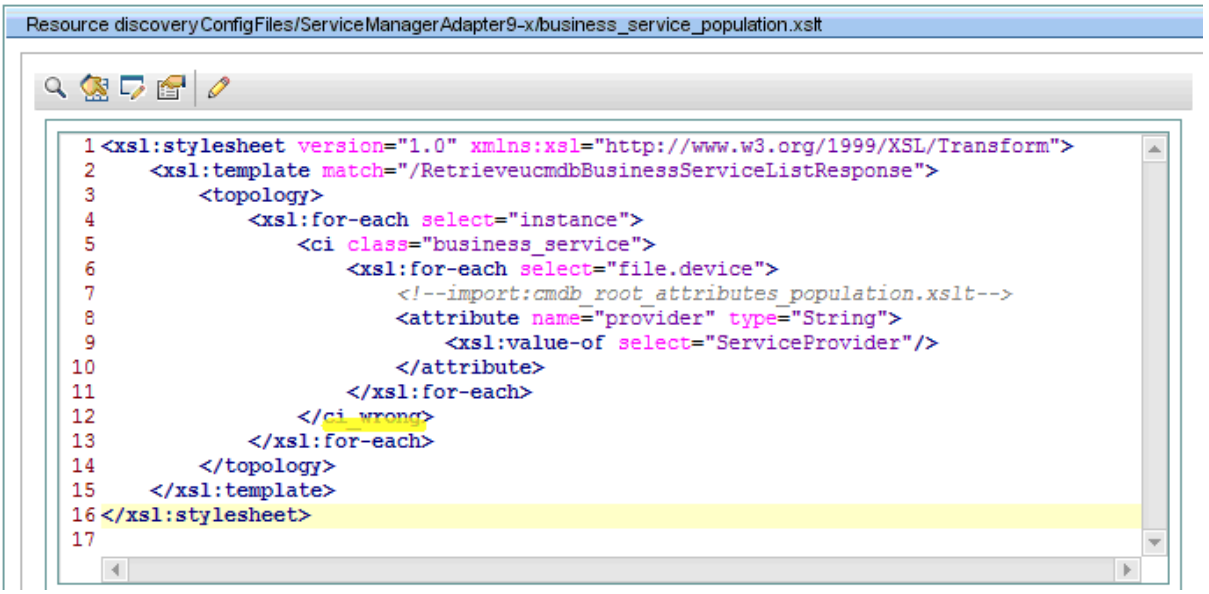
Caused by: org.xml.sax.SAXParseException: The end-tag for element type "ci" must end
with a '>' delimiter.

at org.apache.xerces.parsers.DOMParser.parse(Unknown Source)

at org.apache.xerces.jaxp.DocumentBuilderImpl.parse(Unknown Source)& &
```

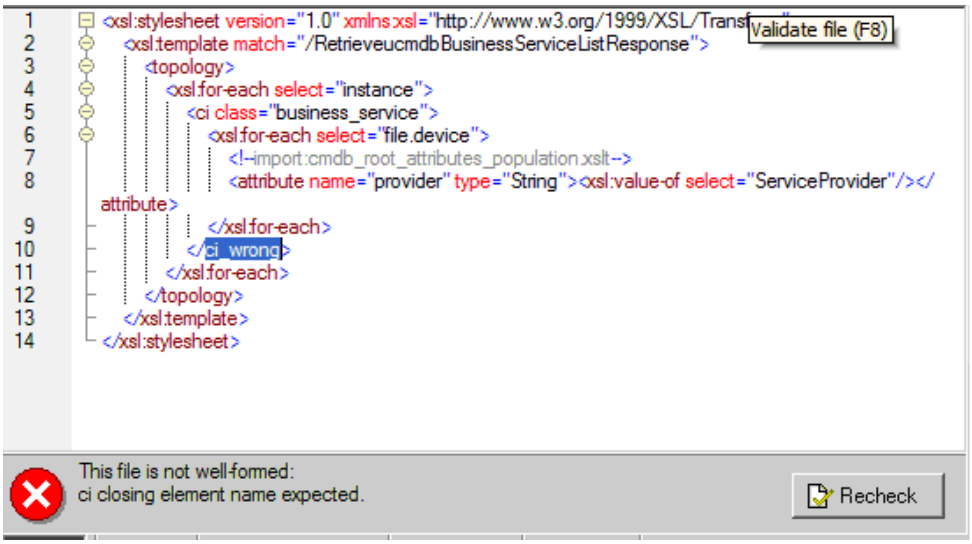
Cause

The end tag of `ci` should be `</ci>` , however you configured a wrong end tag `</ci_wrong>` .



Solution

Search for text `Got SAXException while parsing the XSLT file` to find the name of the problematic `XSLT` file, and then validate the file in an XML editor (for example, **XMLSpy**). You can find and fix any validation issues at ease.

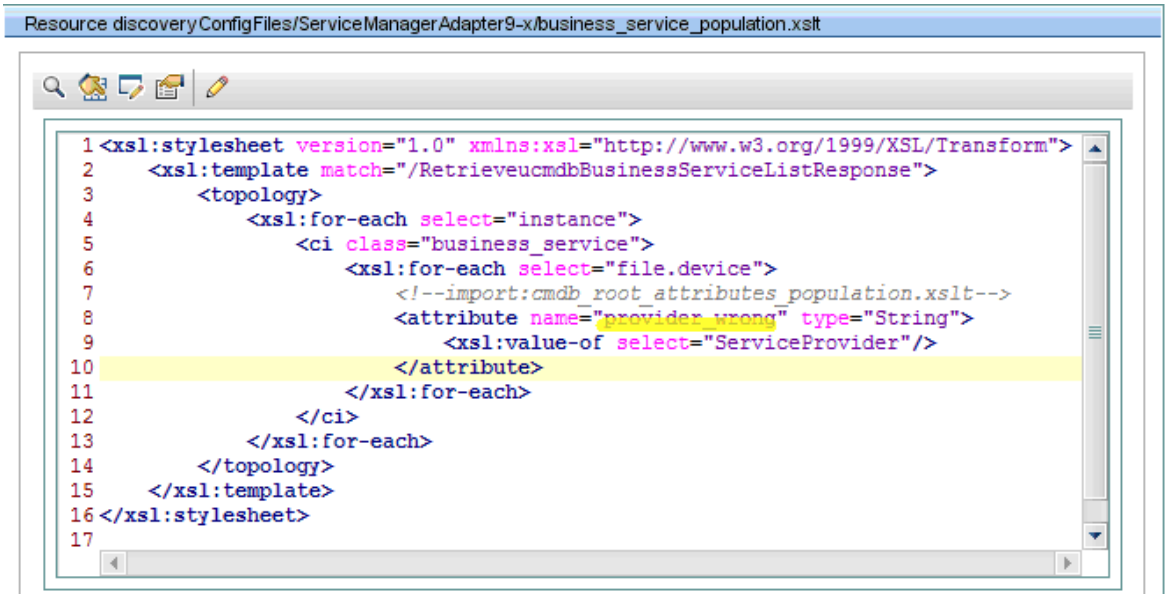


1.6.2.28. The UCMDB attribute name is wrong in the XSLT File when using ServiceManagerAdapter9.41

You will NOT get any error messages either in the log file or the Universal CMDB studio, however the Service Manager field value will not be populated to Universal CMDB.

Cause

The UCMDB attribute name is `provider`, however you configured a wrong attribute `provider_wrong`. This error also occurs if you configure an attribute name in the wrong case because attribute names are case-sensitive.



Solution

If you find that a Service Manager field value can't be populated to Universal CMDB, check both the UCDMB attribute name and the Service Manager field name of the mapping in the XSLT file.

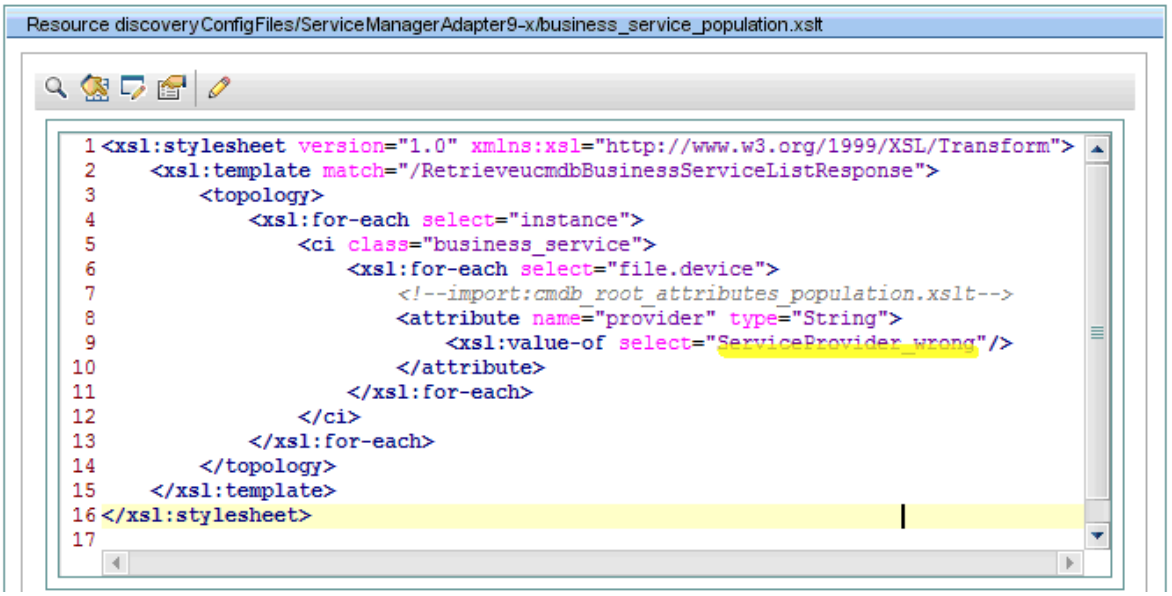
For instructions on how to configure an attribute mapping, see How to map the CI attribute to the Web Service field.

1.6.2.29. The Service Manager field name is wrong in the XSLT file when using ServiceManagerAdapter9.41

You will NOT get any error messages either in the log file or the Universal CMDB studio, however the Service Manager field value will not be populated to Universal CMDB.

Cause

The Service Manager field name is `ServiceProvider`, however you configured a wrong attribute `ServiceProvider_wrong`:



Solution

If you find that a Service Manager field value can't be populated to Universal CMDB, check both the UCDMB attribute name and the Service Manager field name of the mapping in the XSLT file.

For instructions on how to configure an attribute mapping, see How to map the CI attribute to the Web Service field.

1.6.2.30. The Universal CMDB attribute data type is wrong in the XSLT file when using ServiceManagerAdapter9.41

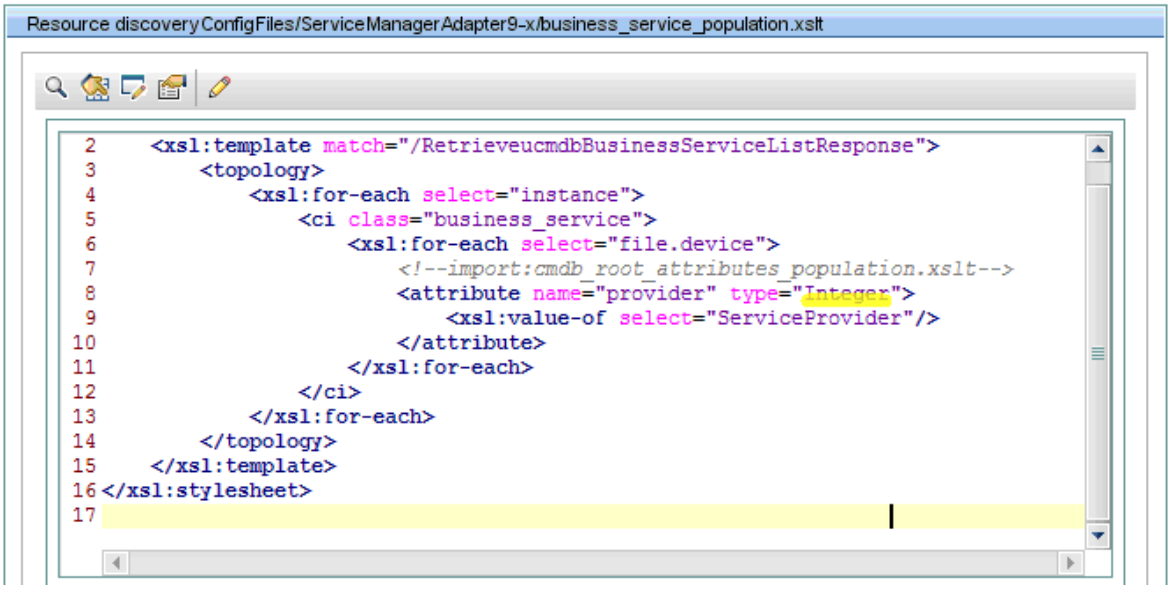
The CIs will be populated to Universal CMDB, but the field value with a wrong data type configuration will not be populated to Universal CMDB.

When you run the population job, you will get a “Failed” status. In addition, from both the population log file (see How to check the population log file) and the Universal CMDB studio (see How to check the error message of a failed population job), you will get an error message similar to the following:

```
General datastore error: SM Integration >> The value "VENDOR1" of field "provider"
is not a integer number, ig-nore this field value!
 java.lang.NumberFormatException: For input string: "VENDOR1"
at java.lang.NumberFormatException.forInputString(NumberFormatException.java:48)
at java.lang.Integer.parseInt(Integer.java:449)
```

Cause

The data type of Universal CMCB attribute “provider” is `String`, however you configured a wrong type `Integer`:



Solution

Search for text `The value "xxx" of field "yyy" is not` to find the name of the attribute with a wrong data type, and then specify the right data type in the XSLT file.

For instructions on how to configure an attribute data type, see How to map the CI attribute to the Web Service field.

1.6.2.31. sm_id is not mapped to the right Service Manager field in XSLT when using ServiceManagerAdapter9.41

When you run the population job, you will get a “Failed” status. In addition, from both the population log file (see How to Check the Population Log File) and the Universal CMDB studio (see How to Check the Error Message of a Failed Population Job), you will get an error message similar to the following:

```
java.lang.RuntimeException: The Universal CMDB attribute "sm_id" of CI type
"business_service" is mapped to an empty value, please check this mapping in XSLT
file.
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.PopChunkGetter.addOr
DelCIToTopology(PopChunkGetter.java:198)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.PopChunkGetter.addOr
DelCIsToTopology(PopChunkGetter.java:184)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.population.PopChunkGetter.getNe
xtResultChunk(PopChunkGetter.java:164)
at
com.mercury.topaz.fcmdb.adapters.serviceDeskAdapter.ServiceDeskAdapter.getChanges(S
erviceDeskAdapter.java:1149)
at
& &
```

Cause

The Service Manager CI ID must be mapped to Universal CMDB CI attribute `sm_id`, because it is used to push the Universal CMDB CI ID back to Service Manager.

Out-of-the-box, this attribute mapping is configured in XSLT file `cmdb_root_attributes_population.xslt`, which is imported by the other XSLT files as a common field mapping, and the Service Manager CI ID field is exposed as the caption `CIName`.

If you configured the mapping for `sm_id` in one of the following ways:

- You did not configure the mapping for `sm_id`;
- You did not expose the Service Manager CI ID field in the Service Manager web service;
- You exposed the Service Manager CI ID field with a caption other than `CIName`;
- The Service Manager CI ID field was exposed in the web service as the caption `CIName`, but you configured a wrong name (for example, `CIName_wrong`) in the XSLT file.

Solution

Search for text `The Universal CMDB attribute "sm_id" of CI type` to find the CI type, and then configure the attribute mapping for this CI Type in the XSLT file.

For instructions on how to configure an attribute mapping, see How to map the CI attribute to the Web Service field.

1.6.2.32. The CI Visualization page displays Fail to render widget

The CI Visualization page displays "Fail to render widget".

Cause

The port set in the UCMDB Gateway URL in the SM integration instance is different from the UCMDB Gateway port in `config.properties`.

Solution

Make sure to use the same port. After the configuration change, you may also need to clear the browser cache before you access the SM web tier to verify.

1.6.3. Troubleshoot SAP integration

This section helps you identify and resolve some of the issues that you might come across while using the SAP integration. Some of these issues include:

- checker.bat and encryptPasswords.bat fail
- Cannot send Incidents to SAP AGS
- java.lang.OutOfMemoryError
- Record in EventIn is not executed
- Incident Update or Process Action fails
- Information is not updated in SAP Solution Manager

1.6.3.1. checker.bat and encryptPasswords.bat fail

The exception “Class not found” appears in the console when running `checker.bat` or `encryptPasswords.bat`.

Cause

The library files that checker requires were not extracted to the required Tomcat.

Solution

1. Run `setup startup`. Tomcat extracts `ovictex.war` and a copies the required jar files.
2. Restart Tomcat.

Contents

- 1 Cause
- 2 Solution

1.6.3.2. Cannot send Incident to SAP AGS

Incident is not sent to SAP AGS when using a newly configured priority in Solution Manager.

Cause

Incidents that have set new priorities in Solution Manager can't be sent to SAP AGS (only default priorities can be sent).

Solution

`fieldMapping.xml` maps to default priorities.

Contents

- 1 Cause
- 2 Solution

1.6.3.3. java.lang.OutOfMemoryError

The java.lang.OutOfMemoryError occurs.

Cause

Too many incidents with big attachments are exchanged simultaneously.

Solution

Increase the Java Virtual Machine heap size in `catalina.bat` (Tomcat).

set JAVA_OPTS=-Xms512m -Xmx1024m

Contents

- 1 Cause
- 2 Solution

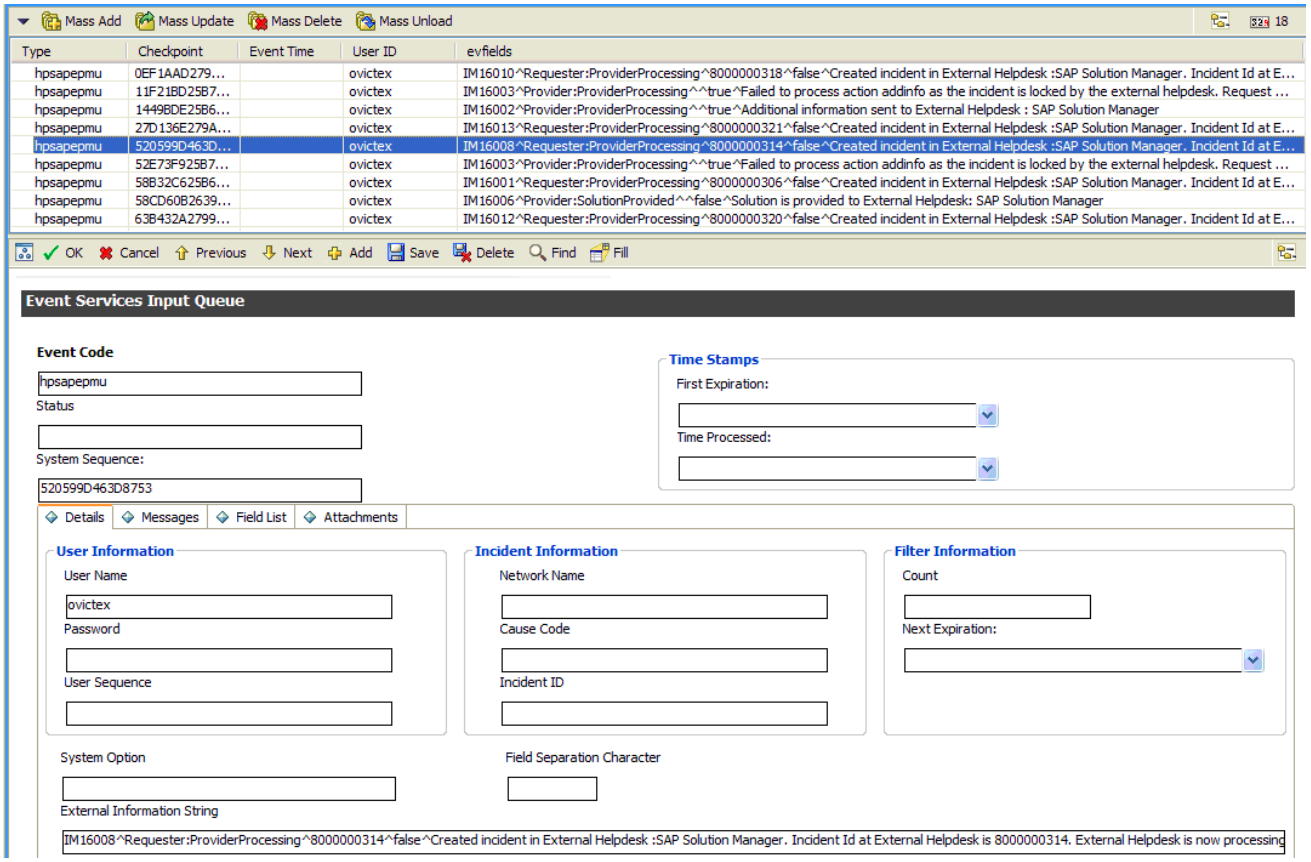
1.6.3.4. Record in EventIn is not executed

The record in the `EvenIn` table is not executed. After Service Manager sends the incident to Solution Manager, the process is finished, but the following problems occur:

- Integration buttons for the incident aren't shown correctly.
- Field `hidden.meta.info` is not updated.

Cause

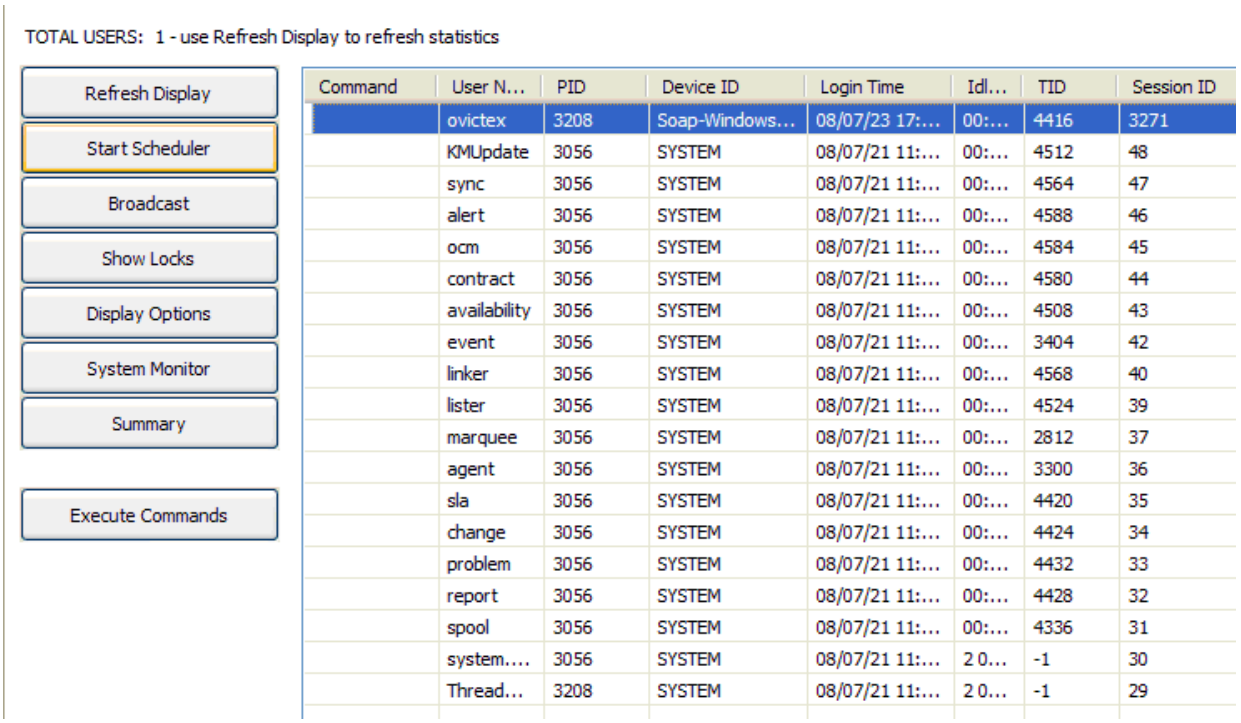
The Event In process threads aren't started when the Service Manager server starts, so in the Input Events window (**Tailoring** > **Event Services** > **Input Events**) the input events aren't handled (as shown in the following diagram).



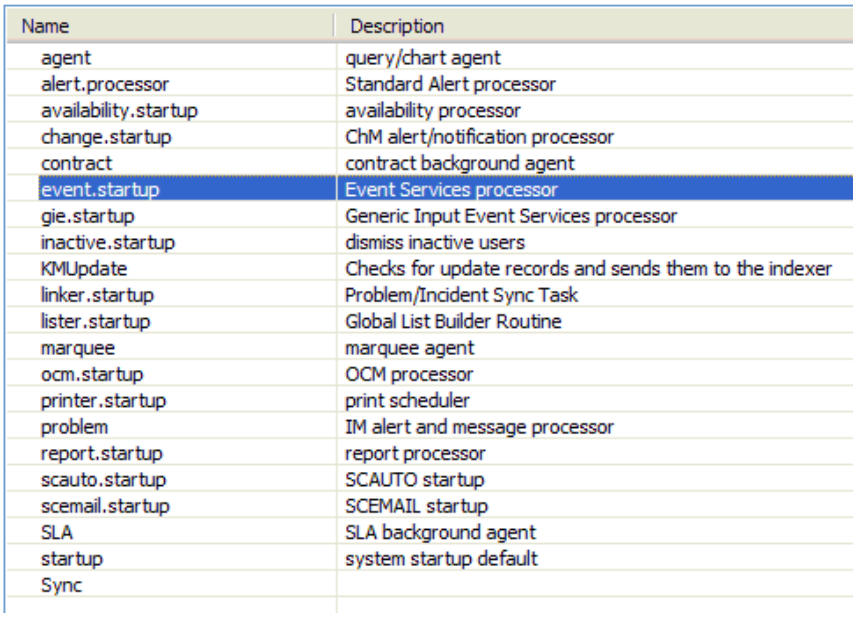
Solution

To handle the input events, start the event in process threads.

1. Go to **System Status**



2. Click **Start Scheduler**.



3. Start `event.startup` and dependent process threads.

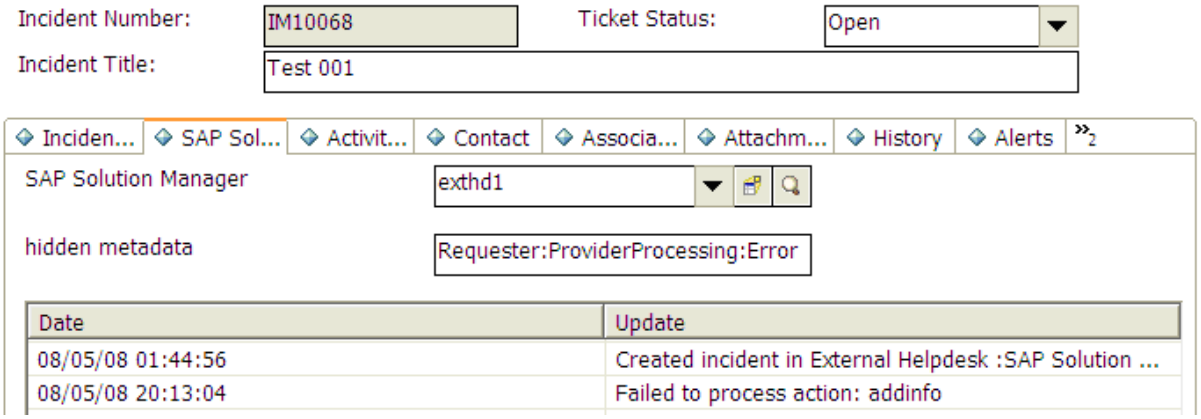
1.6.3.5. Incident Update or Process Action fails

Some incidents aren't exchanged between Service Manager and SAP Solution Manager. The log file or console message of SMSSMEX displays WARN or FATAL level information as described below.

1. Service Manager sends an incident to SAP Solution Manager:

   WARN com.hp.ov.ictex - Failed to process action addinfo as the incident is locked by the external helpdesk. Request will be sent again later.

   The following alert may appear in Service Manager:



2. SAP Solution Manager updates the incident to Service Manager.

   DEBUG com.hp.ov.ictex - Failed to update incident. id:IM10068
   DEBUG com.hp.ov.ictex - Response code = 3. Probably an Incident: IM10068 is locked.
   FATAL com.hp.ov.ictex - Saving of incident failed. Received Message from ServiceCenter: Resource Unavailable
   null
   FATAL com.hp.ov.ictex - An error occured while processing incident ID IM10068. Message: Resource Unavailable
   null
   DEBUG com.hp.ov.ictex - An error occured while processing incident ID IM10068. Message: Resource Unavailable
   null
   com.hp.ov.ictex.ovhdaccess.OvHDException: Resource Unavailable
   null
   at com.hp.ov.ictex.ovhdaccess.servicecenter.Incident.save(Unknown Source)
   at com.hp.ov.ictex.exthdrequesthandler.OvictexServer.updateIncident(Unknown Source)
   ...

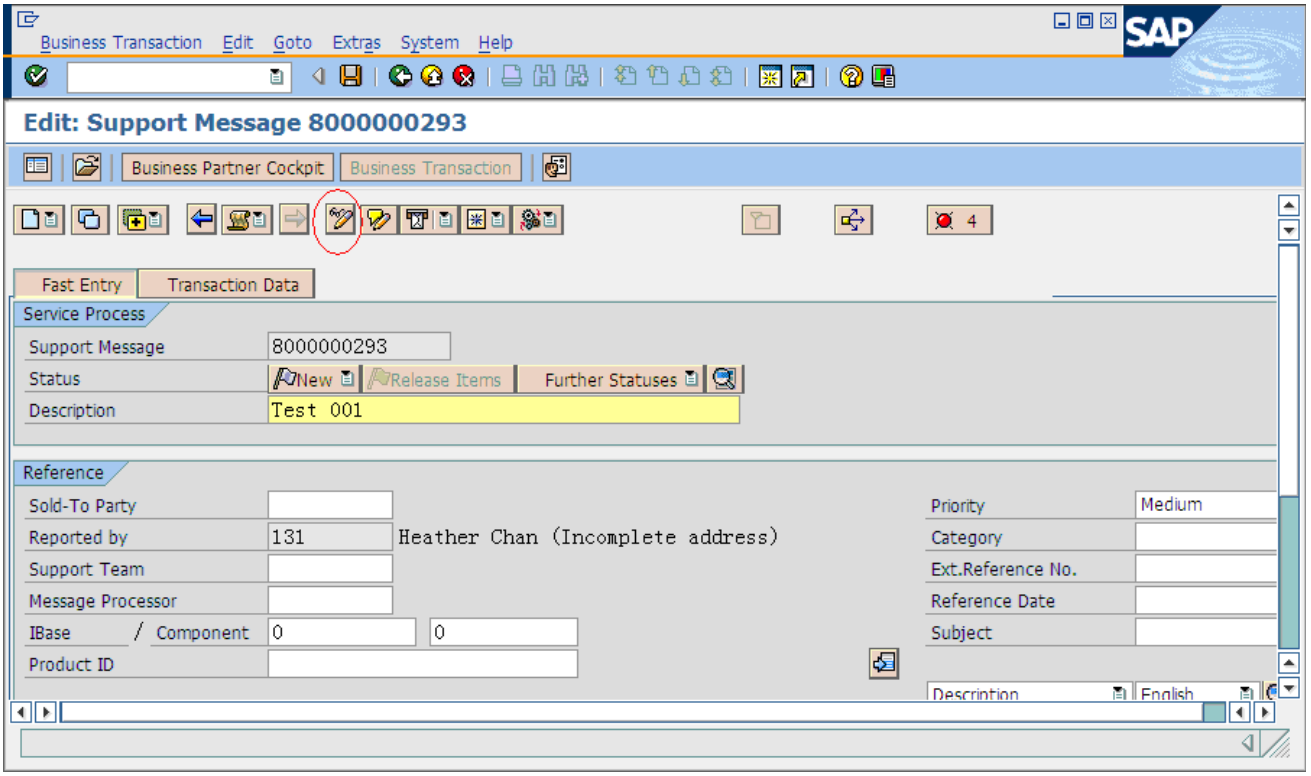
Cause

The incident in Service Manager or SAP Solution Manager is locked:

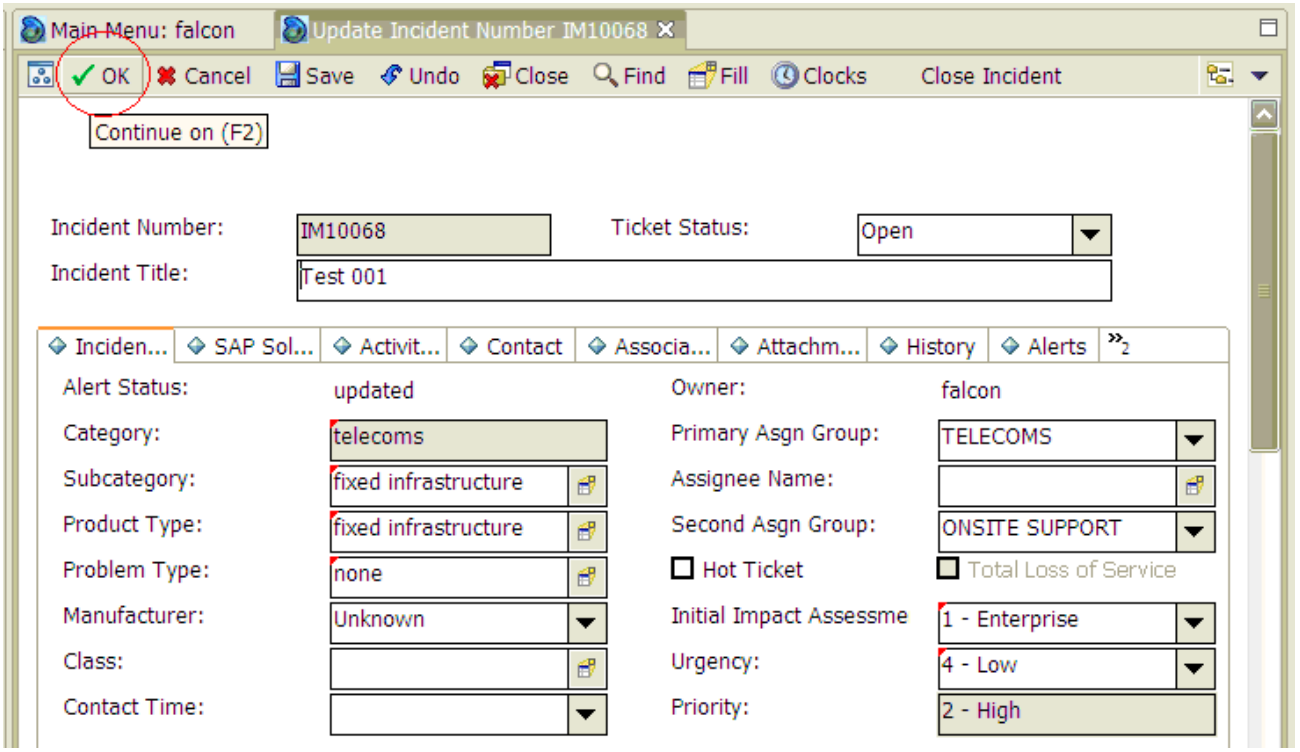
1. In SAP Solution Manager, if the user doesn't click the button **Display/Change Trans** to release an incident write lock, Service Manager can't update or send a message to SAP.
2. In Service Manager, if the user doesn't click **OK** to release an incident write lock in time, the incident maintains the “Updating” status and no message from SAP can be accepted (until the status changes).

Solution

In SAP Solution Manager always click **Display/Change Trans** after finishing or updating an activity.



In Service Manager, click **OK** after finishing or updating an activity.



Contents

- 1 Cause
- 2 Solution

1.6.3.6. Information is not updated in SAP Solution Manager

An open support message is not changed after synchronization from Service Manager to SAP.

Cause

SAP solution manager doesn't refresh the support message automatically.

Solution

In SAP GUI, exit from the current transaction and execute transaction `crmd_order`.

Contents

- 1 Cause
- 2 Solution

1.6.4. Troubleshoot PPM integration

This section helps you identify and resolve some of the issues that you might come across while using the PPM integration. Some of these issues include:

- Fails to submit a Project Proposal Catalog Item to PPM Center
- Cannot access a SM record from the URL on a PPM Center web page
- No Proposal Request is submitted to PPM Center

1.6.4.1. Fails to submit a Project Proposal Catalog Item to PPM Center

Fails to submit a Project Proposal Catalog Item to PPM Center.

Possible causes

- Cause 1: The **CallID** field is not added to the PPM Request Type field
- Cause 2: The `linker.startup` scheduler is not started
- Cause 3: The **SMCatalog-PPMProjectProposal** integration instance is not properly configured
- Cause 4: The PPM Server is not running

Solutions

Solution for Cause 1: The CallID field is not added to the PPM Request Type field

Add the **CallID** field to the PPM Request Type field (for example, PFM-Proposal).

Solution for Cause 2: The `linker.startup` scheduler is not started

Start the `linker.startup` scheduler in Service Manager.

Solution for Cause 3: The SMCatalog-PPMProjectProposal integration instance is not properly configured

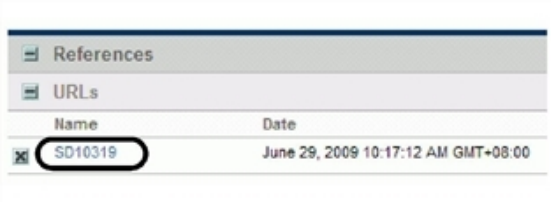
Configure the **SMCatalog-PPMProjectProposal** integration instance properly in the Integration Manager of Service Manager, and then enable the instance.

Solution for Cause 4: The PPM Server is not running

Start the PPM Server, and then connect the Service Manager server host to the PPM Server host.

1.6.4.2. Cannot access a SM record from the URL on a PPM Center web page

Cannot access a SM record from the URL on a PPM Center web page.



Possible causes

- Cause 1: The configuration item querySecurity is not set correctly.
- Cause 2: The value for the server host is not set correctly.

Solutions

Solution for Cause 1: The configuration item querySecurity is not set correctly

Set the configuration item querySecurity to `false` in `web.xml` under Service Manager web application's `WEB-INF` folder.

```
<!-- Enables secure encoding of queries -->
<init-param>
        <param-name>querySecurity</param-name>
        <param-value>false</param-value>
</init-param>
```

Solution for Cause 2: The value for the server host is not set correctly

Value for the server host is the machine name.

```
<init-param>
<!-- Specify the Service Manager server host and port location -->
        <param-name>serverHost</param-name>
        <param-value>smcfrd78</param-value>
</init-param>
<init-param>
        <param-name>serverPort</param-name>
        <param-value>13080</param-value>
</init-param>
```

1.6.4.3. No Proposal Request is submitted to PPM Center

There is no Proposal Request submitted to PPM Center after the specified interval time.

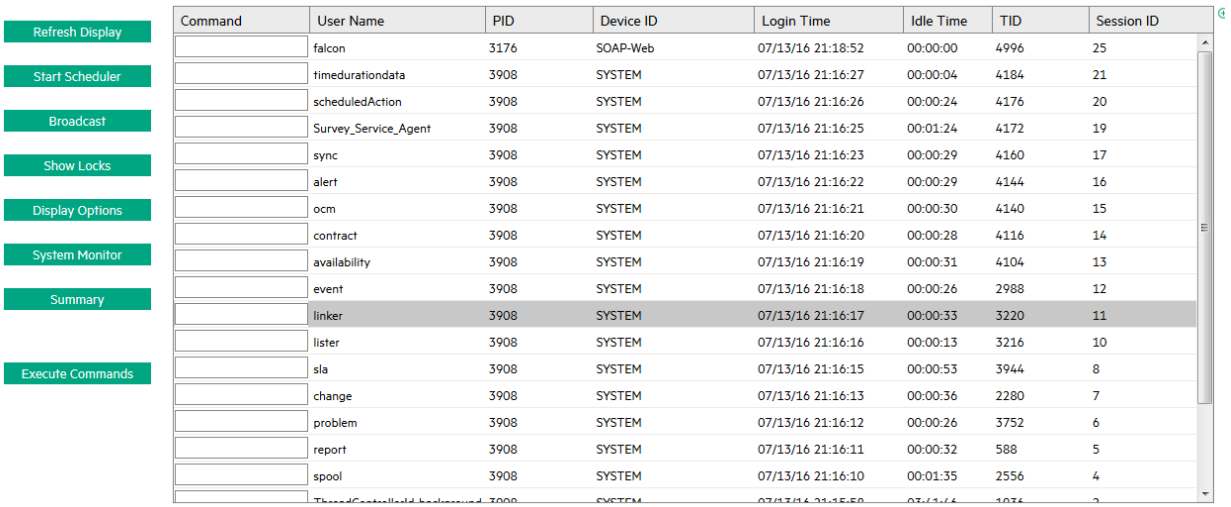
Cause

The `linker` scheduler is running in Service Manager.

Solution

Make sure that the scheduler `linker` is running in Service Manager.

1. Go to **System Status**, and make sure that there is an entry with a User Name of `linker`.

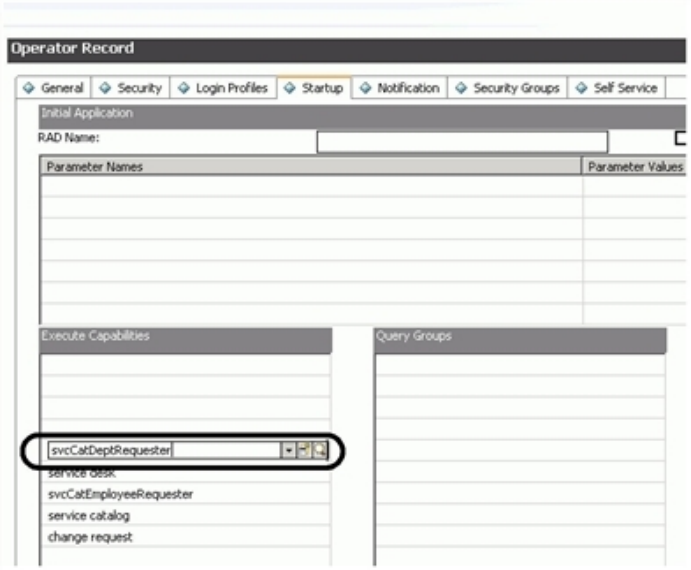


2. If the entry doesn't exist, click **Start Scheduler** and then double-click **linker.startup** to start the `linker` scheduler.

End | More

Name	Description
agent	query/chart agent
alert.processor	Standard Alert processor
AutomationTaskUpdate	Refresh the status of the Automation Task executed in OO
availability.startup	availability processor
change.startup	ChM alert/notification processor
contract	contract background agent
dmt.unload.load	background schedulers for Delta Migration Tool
emailout.startup	emailout startup
event.startup	Event Services processor
gie.startup	Generic Input Event Services processor
KMAttachment	Index attachment for KM into IDOL
KMUpdate	Checks for update records and sends them to the indexer
linker.startup	Problem/Incident Sync Task
lister.startup	Global List Builder Routine
marquee	marquee agent
ocm.startup	OCM processor
printer.startup	print scheduler
problem	IM alert and message processor
propel	Propel Catalog Item Aggregation
refcheck.startup	reference missing check scheduler
report.startup	report processor
scauto.startup	SCAUTO startup

3. If an employee self-service (ESS) user needs to access the Project Proposal option in Service Catalog, add the **svcCatDeptRequester** capability word in the Startup tab of the operator record for this user.



1.6.5. Troubleshoot CIT integration

This section helps you identify and resolve some of the issues that you might come across while using the CIT integration. Some of these issues include:

- Unable to download SM attachments

1.6.5.1. Unable to download SM attachments

When Service Manager (SM) is down but the SM database is still up and running, you can get SM records from the SM database by using CIT, but you can't get the SM attachments from the database.

Cause

You can't retrieve attachments records when the Service Manager server is down.

Solution

To solve the issue, see the sample CIT scenario and use the SM utility included in the following knowledge article: https://softwaresupport.softwaregrp.com/doc/KM03470125.

1.7. Troubleshoot applications

This section helps you identify and resolve some issues that you might come across while using Service Manager applications. Some of these issues include:

- The task planner only displays one value in Process Designer
- Cannot modify the option ID of workflow-based actions
- The User Option condition doesn't work correctly
- The WSDL2JS utility doesn't support the duplicated type names in different namespaces
- The Activity Viewer displays No data found
- Journal Update is saved whereas CI creation failed

1.7.1. The task planner only displays one value in Process Designer

When you select a value for a task category in the task planner two times, only one value is available the second time.

Cause

There is a technical limitation for the drop-down list display of this widget in task planner.

Solution

In this case, clean the combo field and click the drop-down button again to see all list items.

1.7.2. Cannot modify the option ID of workflow-based actions

You can't modify the option ID of workflow-based actions, and an option ID that duplicates an ID in display options may be generated.

Cause

The option ID of workflow-based actions is automatically created by the codes. You can't update the option ID of workflow-based actions by design.

Solution

In this case, manually change the display option ID.

1.7.3. The User Option condition doesn't work correctly when the option value is an array type

The User Option condition doesn't work correctly when the option value is an array type.

Cause

The value of a User Option has an incorrect format.

Solution

In this case, set the value of User Option UO1 = {"1"} instead of User Option UO1 = "1" In Condition Editor.

1.7.4. The WSDL2JS utility doesn't support the duplicated type names in different namespaces

Customers can generate a new JavaScript record in the Service Manager ScriptLibrary table by using the WSDL2JS utility. However, in some special cases, there are types with the duplicated names but in different namespaces.

Cause

Currently, we don't support this use case.

Solution

Try the following solutions one by one.

Solution 1

If there are existing WSDL files that have duplicated types, you can rename the type names to unique values in the XML file.

Solution 2

To create a new WSDL file to support the new business use case, you can leverage the existing RTE methods. See JavaScript global method: doHTTPRequest and JavaScript global method: doSOAPRequest for more information.

1.7.5. The Activity Viewer displays No data found

The Activity Viewer displays the `No data found` message.

Possible causes

1. Both the binaries and applications versions are 9.80. In Forms Designer, the value for the **Input** property of the Activity Viewer doesn't match the virtual join field of the activity table.
2. Both the binaries and applications versions are 9.80. In Forms Designer, the value for the **Input** property of the Activity Viewer is correct. However, there is no data in the activity table.
3. Both the binaries and applications versions are 9.80. In Forms Designer, the value for the **Input** property of the Activity Viewer is correct. However, there is no data after filtering the Activity Log data.
4. The binaries version is 9.80, whereas the applications version is earlier than 9.80.

Solutions

Solution for cause 1:

Set the value of the **Input** property to the same as that in the virtual join field of the activity table.

Solution for cause 2 and cause 3:

This is the expected result by design and it requires no user actions.

Solution for cause 4:

Upgrade the applications to version 9.80.

1.7.6. Journal Update is saved whereas CI creation failed

Suppose you have redesigned the old forms and adopted the latest changes for the activities feature. Later, when saving a record, Service Manager displays the message that a certain field validation failed. However, the Journal Updates field still displays the activity update information whereas the actual CI creation is unsuccessful.

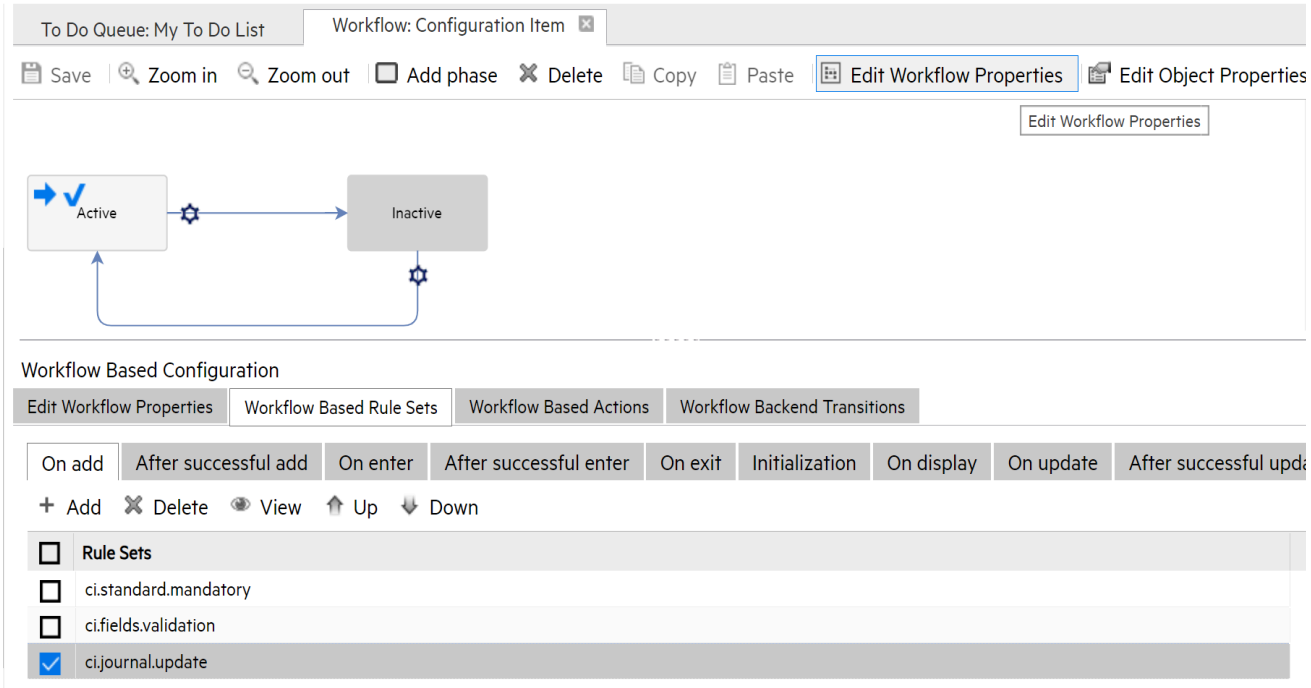
Cause

The field validation information is in the Format Control validation, while the logic for saving journal updates is in the rule sets. Service Manager runs this logic before the Format Control validation.

Solution

Move the content of Format Control validation to the rule sets and ensure that Service Manager runs this validation before saving journal updates. You may add **ci.standard.mandatory** on top of **ci.journal.update**.

See the following screenshot as an example:



1.7.7. Fails to set up read-only conditions in form designer for an assigned user only to upload attachments.

Read-Only Condition is not evaluated for relational operators in the attachment tab when validating assigned user.

Cause

The Forms Designer read-only condition evaluation does not correctly resolve runtime user context variables. While static values (specific usernames, null checks) are honored, dynamic references to the logged-in operator (e.g., current user) are not evaluated as expected. This prevents the condition from differentiating between the assignee and other users.

Solution

To resolve this issue follow the following:

1. Create a rule set with the following details:
    - **Rule Type**: Run JavaScript Validation
    - **Condition**: Always
    - **Script as:**

```
system.vars.$abc = record.assignee_name;

system.vars.$curLogin = system.functions.operator();

if (system.vars.$abc === system.vars.$curLogin) {

  system.vars.$isReadOnly = false;

} else {

  system.vars.$isReadOnly = true;

}
```

2. In form designer for the subform: `im.incident.subform.attachments.with.interaction`, in attachment tab set the read-only condition as `[$isReadOnly]`.
3. Invoke the Rule Set in your custom workflows, and add a new rule set in the workflow properties.

1.8. Troubleshoot Service Manager Reports

This section helps you identify and resolve some of the issues that you might come across while using Service Manager Reports. Some of these issues include:

- Cannot set the remote path and mapping path in the File Server Base Path
- Cannot print reports with long legend texts
- Service Manager server CPU utilization is higher than expected
- The exported and on-screen reports differ
- The JVM heap size for the report.export thread needs to be at least 1024MB

1.8.1. Cannot set the remote path and mapping path in the File Server Base Path

The remote path and mapping path can't be set in the File Server Base Path.

Cause

You have not start Service Manager with administrative privileges.

Solution

In this case, start Service Manager as an administrator.

1.8.2. Cannot print reports with long legend texts

Reports that have long legend text don't print as expected.

Cause

Cannot print reports with long legend texts Cause: There are technical limitations to print reports with long legend texts.

Solution

In this case, shorten the legend label.

1.8.3. Service Manager server CPU utilization is higher than expected

Service Manager server CPU utilization is higher than expected when the ToDo queue is set as the landing page and the inbox size exceeds 50K.

Cause

The curent thread is hogging the CPU and keeping the logical CPU cores busy for loading the inbox data. Furthermore, with huge amount of inbox data to load the application requires memory and it's getting low on what it's configured to use the garbage collector will run often which consume a lot of CPU cycles. If it can't collect anything, your memory will stay low so it will run again and again.

Solution

In this case, check the inbox size periodically and control the size manually.

1.8.4. The exported and on-screen reports differ

When you export a report from the webtier, the exported and on-screen reports may differ.

Cause

When you export a report from the webtier, the report data and definition are fetched from server. Therefore, the exported and on-screen reports may differ if you have modified the report data but not refreshed the screen.

Solution

In this case, refresh the dashboard before exporting a report.

1.8.5. Incorrect JVM heap size for the report.export thread

The JVM heap size for the report.export thread is incorrect.

Cause

The JVM heap size for the report.export thread needs to be at least 1024MB when the export threads are configured to 10.

Solution

In this case, set a dedicated Service Manager instance for the reporting schedule.

1.9. Troubleshoot web tier

This section helps you identify and resolve some of the issues that you might come across while using Service Manager web tier. Some of these issues include:

- CSRF errors occur when working with reverse proxy

1.9.1. CSRF errors occur when working with reverse proxy

When working with a reverse proxy, you may encounter the **Access is denied** error page after login. Meanwhile, the Service Manager web tier log file has the following CSRF errors:

```
[com.hp.ov.web.csrf.CSRFTokenManager] Protocol/Host/Port don't fully
matches so we block the request!
[com.hp.ov.sm.client.webtier.SCLogging] Failed to validate Login CSRF!
```

Cause

You need to set the *webtierFullUrl* parameter in the webtier.properties file, or set the `X-Forwarded-Host` and `X-Forwarded-Proto` headers in your reverse proxy.

Solution

For detailed instructions, see Configure a web server to redirect requests to the web tier.

1.10. Troubleshoot Smart Analytics

This section helps you identify and resolve some of the issues that you might come across while using Smart Analytics. Some of these issues include:

- Failed to start the Smart Analytics server
- Failed to connect the Smart Analytics Server, Media Server, or CFS server
- Unable to see the Multiple Company tab in the Smart Ticket configuration form
- Unable to train and test a Smart Ticket task after you enable the multi-company mode
- Unable to train and test a Smart Ticket task after installing or upgrading to Service Manager 9.60
- Train, Index, and Test buttons are disabled
- Indexing or training failure
- The full reindexing process hangs for a knowledgebase library
- Test Connection in Smart Analytics fails
- The upgraded content server is not running in Linux
- A removed task is still displayed in the connector task list
- Content servers aren't running after installation
- Hot Topic Analytics has no result after indexing
- Hot Topic Analytics is slow
- Unable to launch Hot Topic Analytics in the multi-company mode
- OCR consumes too much memory or is too slow
- Error occurs when running the sqlplus command
- Error occurs when running the autpassword.exe command
- Error occurs when performing a full reindex for an okmlib library on Windows
- The Document ID field in the Contact lib configuration is editable
- The IDOL content service doesn't start on machines that have 24 CPU cores configured
- Smart Search doesn't work

1.10.1. High CPU usage on the Smart Analytics Content server

The CPU usage on the Smart Analytics Content server is high.

Cause

Some indexes aren't properly used.

Solution

Add certain parameters to the **content.cfg** file of the Smart Analytics Content server. If there are multiple content servers, you need to make the changes to all of them.

1. Log in to the Smart Analysis Content server, and then add the following parameters to the **content.cfg** file of the content server.
   `[Server]`

   `RegenerateNumericDateIndex=true`

   `[FieldProcessing]`

   `23=SetNumericDateFields`

   `[SetNumericDateFields]`

   `,*/IMPLEMENTATIONEND_DATE`

   `[SetMatchFields]`

   `,*/ID`
2. Restart the Smart Analytics Content service, and then open the following two URLs from your browser to make the changes of the configuration file to take effect:
   `http://<Smart Analytics Server>:20011/DREREGENERATE?FieldProcessingSection=SetMatchFields`
   `http://<Smart Analytics Server>:20011/DREREGENERATE?FieldProcessingSection=SetNumericDateFields`
   You will then get two IDs of the two jobs, for example:
   `INDEXID=9955XXX`

   `INDEXID=9955XXX`
3. Log in to the Service Manager as system administrator.
4. Type saa in the Service Manager command line, and then press Enter. The Smart Analytics Assistant interface is launched.
5. Use the View Index Status action to make sure that the jobs are executed successfully.
   `http://<Smart Analytics`

Server>:20010/action=indexerGetStatus&MaxResults=20&INDEX=9955XXX
http://<smart Analytics
Server>:20010/action=indexerGetStatus&MaxResults=20&INDEX=9955XXX

1.10.2. Search results in Japanese aren't accurate

When you enter Japanese keywords as search conditions, the search results aren't accurate.

Cause

By default, Smart Search automatically detects the language to use, in which case the search results might be inaccurate if your primary language is Japanese.

Solution

To resolve this issue, follow these steps:

1. On the **Smart Search Settings** page (**Smart Analytics** > **Smart Search** > **More** > **Tailoring**), uncheck the **Automatically detect language in search terms** option.
2. Modify the **DefaultLanguageType** parameter in the **[LanguageTypes]** section of all **content.cfg** files.

   ```
   [LanguageTypes]

   DefaultLanguageType=japaneseUTF8
   ```

3. Update the following language configurations in the **[LanguageTypes]** section of all **content.cfg** files:
   - Set the **Transliteration** parameter to **False**.
   - Add the following lines:

     ```
     Ngram=2

     Ngramorientalonly=true

     Propernames=0

     Augmentseparators=-‘.

     Stoplists=international.dat
     ```

4. Restart all the content servers.
5. Perform a reindex.

1.10.3. Failed to start the Smart Analytics server

Failed to start the Smart Analytics server. The system displays an error message that indicates the msvcr100.dll file is missing.

Cause

The required packages are missing.

Solution

Install .Net Framework 4.0 and Visual C++ Redistributable X64 packages. You can download these two packages from Microsoft website.

1.10.4. Failed to connect the Smart Analytics Server, Media Server, or CFS server

You are unable to connect the Smart Analytics Server, Media Server, or CFS server.

Cause

The cause of this issue is unknown.

Solution

If your Service Manager failed to connect the Smart Analytics Server, check the `application.log` file in the `<SM Smart Analytics>\IDOL\logs\` directory to make sure that the Smart Analytics IDOL Server is started. You also need to make sure that the following six components are started: agent, community, category, DIH, DAH, and view at the main level. Smart Search level has two components: DIH and DAH. If any component is not running normally, restart the Smart Analytics Server.



Note

If the category and community components aren't running, check if all your content servers and Smart Search proxy server are running. If not, start all of them and make sure they are available by the proxy server. And then restart the Smart Analytics main proxy server.

The following is a sample message in the log file that indicates your Smart Analytics Server and the five components are started successfully:

```
19/08/2015 12:47:26 [0] 00-Always: Determining child engine status...
19/08/2015 12:47:26 [0] 00-Always: Engine [dah] state : RUNNING
19/08/2015 12:47:26 [0] 00-Always: Engine [dih] state : RUNNING
19/08/2015 12:47:26 [0] 00-Always: Engine [community] state : RUNNING
19/08/2015 12:47:26 [0] 00-Always: Engine [category] state : RUNNING
19/08/2015 12:47:26 [0] 00-Always: Engine [agentstore] state : RUNNING
19/08/2015 12:47:26 [0] 00-Always: Engine [view] state : RUNNING
19/08/2015 12:47:26 [0] 00-Always: All 6 components started successfully.
19/08/2015 12:47:26 [0] 30-Normal: ACI Server validated operations key.
19/08/2015 12:47:26 [0] 30-Normal: ACI Server has no QPS limit.
19/08/2015 12:47:26 [0] 30-Normal: ACI Server is licensed for SSL
encryption.
19/08/2015 12:47:26 [0] 30-Normal: Performed hostname lookup and converted
USERCLIENTS from
[16.187.190.64,127.0.0.1,127.0.0.1,SGDLITVM0581.hpswlabs.adapps.hp.com,fe8
0::d802:aa38:e68c:435] ->
[16.187.190.64,127.0.0.1,::1,127.0.0.1,::1,16.187.190.81,fe80::d802:aa38:e
68c:435].
19/08/2015 12:47:26 [0] 30-Normal: Performed hostname lookup and converted
ADMINCLIENTS from
[16.187.190.64,127.0.0.1,127.0.0.1,SGDLITVM0581.hpswlabs.adapps.hp.com,fe8
0::d802:aa38:e68c:435] ->
[16.187.190.64,127.0.0.1,::1,127.0.0.1,::1,16.187.190.81,fe80::d802:aa38:e
68c:435].
19/08/2015 12:47:26 [0] 30-Normal: This ACI Server will not accept
unencrypted communications from ACI clients.
19/08/2015 12:47:26 [0] 30-Normal: ACI Server setting MaxInputString to
64000.
19/08/2015 12:47:27 [0] 30-Normal: ACI Server successfully loaded online
help.
19/08/2015 12:47:27 [0] 30-Normal: ACI Server successfully loaded admin
UI.
19/08/2015 12:47:27 [100] 30-Normal: ACI thread 100 attached to port 9000
19/08/2015 12:47:27 [101] 30-Normal: ACI thread 101 attached to port 9000
19/08/2015 12:47:27 [102] 30-Normal: ACI thread 102 attached to port 9000
```

If your Service Manager failed to connect the Media Server, check the `application.log` file in the `<Smart Analytics>\Mediaserver\logs` directory to make sure that the Media Server is started.

The following is a sample message in the log file that indicates your Media Server is started successfully:

```
19/08/2015 10:55:43 [1] 00-Always: ACI Server starting at
xxx.xxx.xxx.xxx:18000
19/08/2015 10:55:43 [54] 00-Always: ACI Server thread 1 initialized
19/08/2015 10:55:43 [55] 00-Always: ACI Server thread 2 initialized
19/08/2015 10:55:43 [57] 00-Always: ACI Server thread 3 initialized
19/08/2015 10:55:43 [58] 00-Always: ACI Server thread 4 initialized
```

1.10.5. Unable to see the Multiple Company tab in the Smart Ticket configuration form

You are unable to see the Multiple Company tab in the Smart Ticket configuration form.

Cause

The multi-company mode is disabled in Service Manager.

Solution

Make sure that the multi-company mode is enabled in Service Manager. To enable the multi-company mode, follow these steps:

1. Click System **Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. On the **General** tab, select the **Run in Multi-Company Mode** option.
3. Click **Save**.
4. Log out and log back in Service Manage for the change to take effect.

1.10.6. Unable to train and test a Smart Ticket task after you enable the multi-company mode

You are unable to train and test a Smart Ticket task after you enable the multi-company mode.

Cause

Suppose you are working with Smart Analytics in the single company mode. After you upgrade to the latest Service Manager, you must clear the data in Service Manager and IDOL before you enable the multi-company mode.

Solution

Follow these steps:

1. Clear data in the cate2idol2 table.
    1. Click **Tailoring** > **Database Manager**.
    2. Type cate2idol2 in the Table field, select the **Administration mode** checkbox, and then click **Search**.
    3. Select the **cate2idol2.g** format, and then click **Search**.
    4. Click **Mass Delete**.
2. Clear data in IDOL.
    1. Type saa in the Service Manager command line, and then press Enter. The Smart Analytics Assistant interface is launched.
    2. Run the **http://*<Host>*:*<MainProxyPort>*/action=CategoryDelete&Category=0** command in Smart Analytics Assistant. The default port number is 9000.

1.10.7. Train, Index, and Test buttons are disabled

Train, Index, and Test buttons are disabled.

Cause

The cause of this issue is unknown.

Solution

If the **Train**, **Index**, and **Test** buttons are disabled in the configuration pages for Smart and Hot Topic Analytics, follow these steps:

1. Make sure that you have enabled the **Smart Analytics Server Enabled** option on the Smart Analytics Configuration form. Also, make sure to click the **Save** button to save this configuration.
2. If the issue still exists, make sure that the upgrade to Service Manager `9.8x` is successful. Check that all the libraries are of the Service Manager `9.8x` version. The default folder is `C:\Program Files\OpenText\Service Manager 9.8x\Server\RUN\lib`.
3. If the issue still exists, you might use an old license. In this case, launch the Service Manager server with the new license file, which includes the Smart Analytics module license.

   For Service Manager trial version installation, follow these steps:

   a. Move your old license file out of the folder. The default folder is `C:\Program Files\OpenText\Service Manager 9.8x\Server\RUN\`.

   b. Run **sm -instanton** to generate the instant on license.

   c. Restart the Service Manager server to load the instant on license.

1.10.8. Indexing or training failure

You failed to do indexing or training.

Cause

The cause of this issue is unknown.

Solution

Check the SMIS task log first, and then check the information in `sm.log`.

You need to modify the configuration files of the content server if you receive the following error message during indexing or find the error message in `sm.log`:

`2004( 3236) 02/26/2016 17:49:46 JS E IDOLKMIncrementalIndex Index database xxx is read only in IDOL server, incremental index can't proceed`

To fix the problem, follow these steps:

1. Open content server configuration files (for example, content1.cfg)
2. Locate the `[databaseN]` section in each .cfg file of the content server.
3. Set the `readonly` parameter to `false` for all readonly databases:

   `[databaseN]`

   `DatabaseReadOnly=false`
4. Restart corresponding content services.

1.10.9. The full reindexing process hangs for a knowledgebase library

If the full reindexing process for a knowledgebase library is not making progress for a long time according to your scheduled plan, the back-end processor for the knowledgebase library may already hang.

Cause

Check the "idle time" for the back-end processor (worker) from the **System Status** page. If the idle time is too long (for example, 30 minutes), the processor may hang.

Solution

In this case, you can kill and then restart the corresponding processor.

1.10.10. Test Connection in Smart Analytics fails

Test Connection in Smart Analytics fails.

Cause

The cause of this issue is unknown.

Solution

Enter the Smart Analytics Assistance by typing **SAA** in the command box. Check the status of all components that are listed in the table and see if any component is in the **Offline** status. The components and their corresponding processes are listed as below.

Services and corresponding processes

- IDOL Smart Analytics Main Server:

  `<SmartAnalytics>\IDOL\IDOLServer.exe`

  `<SmartAnalytics>\IDOL\community\community.exe`

  `<SmartAnalytics>\IDOL\view\view.exe`

  `<SmartAnalytics>\IDOL\category\category.exe`

  `<SmartAnalytics>\IDOL\agentstore\agentstore.exe`

  `<SmartAnalytics>\IDOL\dih\dih.exe`

  `<SmartAnalytics>\IDOL\dah\dah.exe`

- IDOL CFS Connector:

  `<SmartAnalytics>\IDOL\CFS\CFS.exe`

- IDOL Image Server1

  `<SmartAnalytics>\IDOL\ImageServer1\ImageServer1.exe`

- IDOL Content1

  `<SmartAnalytics>\Content1\Content1.exe`

- IDOL Content2

  `<SmartAnalytics>\Content2\Content2.exe`

- IDOL Content3

  `<SmartAnalytics>\Content3\Content3.exe`

- IDOL Smart Search Proxy

  `<SmartAnalytics>\IDOL\level2proxy\IDOLServer.exe`

`<SmartAnalytics>\IDOL\level2proxy\dah\dah.exe`

`<SmartAnalytics>\IDOL\level2proxy\dih\dih.exe`

Check the log file first. All log files are in the logs folder of the corresponding service.

Check the general log file of the application.log file first to find if there is any error message showing that some process doesn't start, such as the following error message:

`Engine [category] state : FAILED (no process detected)`

Find the corresponding component log file. For example, the `category_application.log` file.

Make sure the port number is not in conflict with others. The port number is occupied by other processes when the following message is displayed:

`70-Error: Could not create TCP listener (Bad Parameter Value).`

If detailed log information is required, set the value of **logLevel** to full (in the **[Logging]** section of the corresponding component) and restart the service for troubleshooting purpose. Rmember to modify the value back to normal afterwards.

1.10.11. The upgraded content server is not running in Linux

The upgraded content server is not running in Linux.

Cause

The cause of this issue is unknown.

Solution

1. Check the application.log file of the upgraded content server first to find if there is any error message displayed as below:

   `26/08/2015 15:25:55 [1] 70-Error: Error: SecurityCode '1' is incorrectly configured, '1' was previously used with Security Type 'AUTONOMY_SECURITY_V4_NT_MAPPED' in library file '/usr/local/IDOLServer/IDOL/modules/mapped_security'`

   `26/08/2015 15:25:55 [1] 70-Error: Please see 'SecurityCode1.txt' for the correct settings for this type.`

   `26/08/2015 15:25:55 [1] 70-Error: Error: Failed to validate security settings`

   `26/08/2015 15:25:55 [1] 70-Error: Error: SecurityCode '2' is incorrectly configured, '2' was previously used with Security Type 'AUTONOMY_SECURITY_V4_NETWARE_MAPPED' in library file '/usr/local/IDOLServer/IDOL/modules/mapped_security'`

   `26/08/2015 15:25:55 [1] 70-Error: Please see 'SecurityCode2.txt' for the correct settings for this type.`

   `26/08/2015 15:25:55 [1] 70-Error: Error: Failed to validate security settings`

2. Locate the `[Security]` section in the .cfg file of the content server, and then add the // comment characters to the parameters as displayed in the following:

   `[Security]`

   `SecurityInfoKeys=123,144,564,231`

   `DebugDecrypt=true`

   `//0=NT_V4`

   `//1=SharePoint`

3. Restart the components where the .cfg file is changed.

1.10.12. A removed task is still displayed in the connector task list

After you remove a task from the web connector configuration, this removed task may still appear in the task list options.

For example, the **MYSITE** task is removed from web connector but still shows in the task options.

Cause

The cause of this issue is unknown.

Solution

To remove this task from this list, follow these steps:

1. Delete the folder of the deleted task.

   In this example, delete the *<Smart Analytics>*/WebConnector/MYSITE folder.

2. Delete the actions folder for this deleted task.

   In this example, delete the *<Smart Analytics>*/WebConnector/actions folder.

3. Restart the Web Connector service

1.10.13. Content servers aren't running after installation

If your content servers aren't running and the following error message is displayed in the application.log file for each content server, you need to modify the configuration file of the content server:

`[1] 70-Error: Error: Attempt to open *** failed. (No such file or directory). The file open mode [rb] doesn't permit creation of a new file.`

Cause

The cause of this issue is unknown.

Solution

To fix this problem, follow these steps:

1. Modify the value of the **Threads** parameters in the configuration file of the content server. Make sure the number of **x+y** is no more than 32.

   `[Server]`

   `Threads=x`

   `[AsyncActions]`

   `Threads=y`

2. Restart the content server.

1.10.14. Hot Topic Analytics has no result after indexing

Hot Topic Analytics has no result after indexing.

Cause

Smart Analytics needs some time to commit the results from the cache.

Solution

You may need to wait for a while (by default, 120 seconds) for Smart Analytics to commit the results from the cache to the disk.

1.10.15. Hot Topic Analytics is slow

Hot Topic Analytics is slow.

Cause

If you experience slow performance when using Hot Topic Analytics, one possible reason is the field type configuration in the HTA content server.

Solution

Check your action history in SAA to find out which fields are used in the `fieldtext=MATCH{<value>}:<Field_Name>` parameter. Then, you can add the field name to the `PropertyFieldCSVs` parameter in the `[SetMatchFields]` section of the IDOL configuration file (*<Smart Analytics Installation>*/IDOL/IDOLServer.cfg). Doing so may help to enhance the performance. For example, you can add "COMPANY" and "CATEGORY" as shown in the following example:

```
[SetMatchFields]
PropertyFieldCSVs=*/*_MATCH,*/EDK_*,*/MATCH_*,*/COMPANY,*/CATEGORY
```

The new added match fields will automatically take effect for new data indexing after you restart the IDOL Server.

To regenerate the modified MatchType fields for the existing indexed data without re-indexing, you can add the `RegenerateMatchIndex=true` parameter to the Content.cfg file of the HTA content server, and then restart your content server.

```
[Server]
RegenerateMatchIndex=true
```



Tip

This regeneration is executed every time when the IDOL Server restarts. To save the start-up time of the IDOL Server, disable this option after you finish your performance tuning.

1.10.16. Unable to launch Hot Topic Analytics in the multi-company mode

The following error message is displayed and you are unable to launch Hot Topic Analytics in the multi-company mode:

`ERROR uncaught exception: Error: "company" table doesn't have mandanten field defined`

Cause

The cause of this issue is unknown.

Solution

In this case, you need to configure the company information in Mandanten correctly.

To set up company in Mandanten, follow these steps.

1. Click **System Administration** > **Ongoing Maintenance** > **Mandanten** > **Mandanten Field Restrictions**.
2. Add a new definition with “probsummary” as **File Name** and “company” as **Mandant Field Name**.

1.10.17. OCR consumes too much memory or is too slow

OCR consumes too much memory or is too slow.

Cause

When a user takes a photo of the computer screen, the grid noise in the background has negative effect on the result of OCR. To improve the OCR quality, Smart Analytics uses Fast Fourier Transform (FFT) to remove the grid noise.

However, the side effect of this feature is that it consumes around 400 MB memory for each image and makes the OCR process slower. It takes about 3 seconds on Windows and about 10 seconds on Linux.

Solution

In this case, You can disable this FFT feature in the Tailoring page of Smart Ticket. For more information, see Configure Smart Ticket for OCR.

1.10.18. Error occurs when running the sqlplus command

The system may display the following error when running the **sqlplus** command:

`execute "sqlplus": error Segmentation fault(core dumped)`

Cause

The cause of this issue is unknown.

Solution

Follow these steps to troubleshoot this problem:

Check the `LD_LIBRARY_PATH` environment variable, and then update it to `LD_LIBRARY_PATH = /$ORACLE_HOME` .

1.10.19. Error occurs when running the autpassword.exe command

The system may display the following error when running the **./autpassword.exe -e -tAES -oKeyFile=./keyfile.ky username/passwords@net_service_name** command:

```
./autpassword.exe: error while loading shared libraries: libstdc++.so.5: can't open shared object file: No such file or directory.
```

Cause

The cause of this issue is unknown.

Solution

To troubleshoot this problem, you must install libstdc++.so.5, and then running autpassword.exe again.

1.10.20. Error occurs when performing a full reindex for an okmlib library on Windows

You may find the following error message in *<Oracle Connector installation directory>*/logs/synchronize.log when performing a full reindex for an okmlib library:

```
18/01/2018 08:14:20 [9] 70-Error: Failed to complete SYNCHRONIZE for task 'MYTASK1' [MTYuMTg3LjE4OC4yMDQ6NTU1NTpGRVRDSDoxNTE2MjQzNDYwMDYzMy0xNTE0MTY3MTU3] : message: , sqlstate: , code: 1717658202, var_info:
```

Cause

This error occurs because you are using the Oracle Connector with the **Instant** type of Oracle client on Windows.

Solution

To troubleshoot this problem, you need to install the **Runtime** type of Oracle client.

1.10.21. The Document ID field in the Contact lib configuration is editable

The Document ID field in the Contact lib configuration is a contact name, which can be modified by customers.

Cause

The cause of this issue is unknown.

Solution

In this case, you can set the Document ID field another non-editable and unique field.

1.10.22. The IDOL content service doesn't start on machines that have 24 CPU cores configured

The IDOL content service doesn't start on machines that have 24 CPU cores configured.

Cause

The cause of this issue is unknown.

Solution

You can apply a workaround if the following error appears in the application.log of the content server:

`[1] 70-Error: Error: Attempt to open *** failed. (No such file or directory). The file open mode [rb] doesn't permit creation of a new file.`

In this case, modify the following parameters in the configuration of the content server, and then restart the content server.

`[Server] Threads=x [AsyncActions] Threads=y let x+y<32`

1.10.23. Smart Search doesn't work

Smart Search doesn't work if you use fields that aren't in the form in the match criteria of a predefined filter.

Cause

The cause of this issue is unknown.

Solution

In this case, add the fields used in the match criteria of the predefined filter in context aware search into the form of the current working ticket.

1.11. Troubleshoot Service Manager Collaboration

This section helps you identify and resolve some of the issues that you might come across while using Service Manager Collaboration. Some of these issues include:

- Fails to start the chat service in Linux
- Fails to start the chat server in Windows
- Fails to start chat in the ESS client
- Fails to connect to the Collaboration server
- The javax.net.ssl.SSLHandshakeException error occurs when enabling Service Manager Collaboration
- The end user chat button is not displayed in the ESS client
- 500 error for the request chatservice/api/pubsub/presubscribe
- Cannot invite anybody to chat because all online invitees appear offline
- The Previous Support Ticket ID combo box is still empty after you have made some chats in the ESS web client
- Nothing happens when an ESS user clicks any of the knowledge links proposed by the virtual agent
- An IT agent can't receive the chat requests from a portal user
- Chat configuration failed error occurs
- The icon of the chat notification button rotates forever

1.11.1. Fails to start the chat service in Linux

The chat service fails to start in Linux. The system displays the `chatservice/log/nohup.out: Permission denied` error message.

Cause

The current operator has no permission to run chatmsvc.

Solution

Give permission to the current user. For example, run either the **chmod +x chatmsvc** or **chmod 777 chatmsvc** commands.

1.11.2. Fails to start the chat server in Windows

Fails to start the chat server in Windows.

Cause

Some related processes still exist.

Solution

Check the java.exe process in Windows Task Manager when restarting the chat server. If the process still exists after the chat server service is stopped, you must end these processes manually before starting the chat server.

1.11.3. Fails to start chat in the ESS client

The chat button is displayed in the ESS client and you can open the chat window, but chat is not started successfully.

This issue may occur in several scenarios as below.

- Scenario 1: The system displays the `401 (Authentication failed)` error for the api/chats/ request.
- Scenario 2: The system displays the `500 (XMPPError: remote-server-not-found)` error
- Scenario 3: The system displays the `500 (Illegal contact info in chat request entity)` error for the api/chats/ request

Possible causes

- Cause for scenario 1:

  You can see this JavaScript error in the browser console if the end user who has already logged in is not authorized to create a chat.

- Cause for scenario 2:

  You can see this JavaScript error in the browser console if you have not configured the chat server correctly.

  For example, the server name of the Openfire may not be correct. You may have configured chat server on one host and then you move the chat server to another host, but still want to use the original database. In this case, when you start the chat server in the second host, the server name of the Openfire is still the first host.

- Cause for scenario 3:

  The current user who has logged in shares the same contact with another user in Service Manager.

Solutions

Solution for Scenario 1: 401 (Authentication failed) error

Refer to Configure end user profile to add New, Update and Close rights on Service Desk security area for this user.

Solution for Scenario 2: 500 (XMPPError: remote-server-not-found) error

Check the log file or the console to see if your admin console is listening at the correct host and port. If not, you may need to re-install and re-configure the chat server from scratch.

Solution for Scenario 3: 500 (Illegal contact info in chat request entity) error

Don't share a contact for two or more users in Service Manager.

1.11.4. Fails to connect to the Collaboration server

When you sign in to the Service Manager web client by using HTTPS, the system displays the following error:



Cause

The chat server service is not started, or Service Manager Collaboration loses connection to the chat server, or Service Manager Collaboration is not properly configured. The cause of this issue is unknown.

Solution

Perform the following tasks:

Task 1: Make sure that the chat server is started

Windows:

Go to **Windows Services** to check that the chat server service (Service Manager x.xx.xxxx Chat Server) is started.

Linux:

Run the command **./openfire status** to check the status.

Task 2: Try to reconnect to the chat server

Click the **Notifications** button to try to reconnect to the chat server.

Task 3: Verify the Service Manager Collaboration configurations

Follow these steps to verify the Service Manager Collaboration configurations:

1. Sign in to Service Manager as a system administrator.
2. Click **System Administration** > **Ongoing Maintenance** > **Collaboration** > **Configuration**.
3. Ensure the following fields are configured correctly, as described in the following table.

Field	Value
Chat Service Domain and Port	http://chatserver.yourdomain.net:8088 or https://chatserver.yourdomain.net:8448
Bosh URL	/of-http-bind/
Chat Service URL	/chatservice/
Domain Name	Set it to the Openfire Domain that you set up in Task 2: Deploy and configure the chat server. In our example steps, it is chatserver.yourdomain.net.
Plugin URL	/of-plugins/

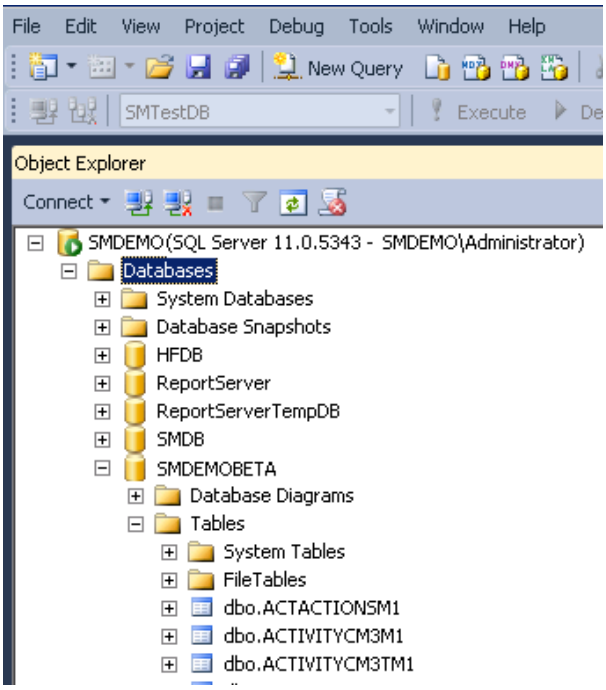
Task 4: Reinstall chat server

You may need to reinstall the chat server if you have checked all your settings but still see the following error message:



For example, follow these steps to reinstall chat server in Windows:

1. Sign out of Service Manager.
2. Stop the Windows service named **Service Manager x.xx.xxxx Chat Server**.
3. Open a DOS command prompt and change the directory to %CHAT_SERVER_HOME%\bin.
4. Run the **nssm remove SmChatServer** command to uninstall the Windows service.
5. Remove the %CHAT_SERVER_HOME% directory.
6. Sign in to your Service Manager database and list the tables.

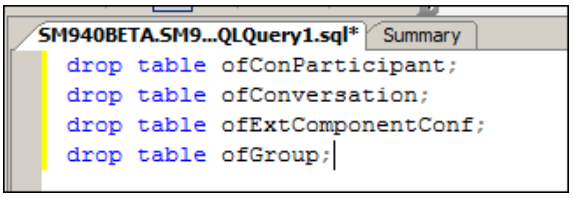


7. Locate the tables prefixed with '**of**' in the list.



   All tables prefixed with lowercase '**of**' need to be dropped.

8. Drop the tables until all tables prefixed with '**of**' are removed.



9. Reinstall the chat server. For detailed instructions, see Task 2: Deploy and configure the chat server.
10. Sign in to Service Manager again to check whether Collaboration works.

1.11.5. The javax.net.ssl.SSLHandshakeException error occurs when enabling the Service Manager Collaboration

When you enable the Service Manager Collaboration in Task 7: Enable Service Manager Collaboration, your configuration changes aren't saved and the system displays either of the following error messages. You can also find this message in the log file.

```
javax.net.ssl.SSLHandshakeException: Received fatal alert:
bad_certificate.
javax.net.ssl.SSLHandshakeException: PKIX path building failed.
```

Possible causes

- Cause 1: You may have set `ssl_reqClientAuth:2` in the sm.ini file in Prerequisite tasks > Task 3, but the certificate of the chat server is not in the trustedclients.keystore of the SM server.
- Cause 2: The CA certificate of the chat server is not the same as that of the SM server and is not imported into the cacerts of the SM server.

Solutions

Solution for Cause 1: The certificate of the chat server is not in the trustedclients.keystore of the SM server

In this case, you need to add the client certificate of the chat service to the trustedclients.keystore in the RUN folder of the SM server (smserver.yourdomain.net) and restart the SM server. Alternatively, you can start another SM process for chat by adding `ssl_reqClientAuth:0` as the startup paremeter.

You can go to the server where you generated the SSL certificates to get the updated trustedclients.keystore. Alternatively, you can copy chatserver_c_clientpubkey.cert to %SM_SERVER%/RUN and then manually import it:

cd %SM_SERVER%/RUN

keytool -import -alias chatserver_c -file chatserver_c_clientpubkey.cert -keystore trustedclients.keystore -storepass trustedclients

Solution for Cause 2: The CA certificate of the chat server is not the same as that of the SM server

Copy the CA of the chat server (for example, ca_of_chatserver.crt and alias is myca) to the %SM_SERVER%/RUN/ and manually import it to cacerts:

keytool -keystore cacerts -importcert -alias myca -file ca_of_chatserver.crt

1.11.6. The end user chat button is not displayed in the ESS client

The end user chat button is not displayed in the ESS client. This issue may occur in several scenarios as below.

- Scenario 1: The system displays no error in the browser console.
- Scenario 2: The system displays the `400 Host doesn't match SNI` error for the /api/userprofile request.
- Scenario 3: The system displays the `401 Unauthorized` error for the /api/userprofile request.
- Scenario 4: The system displays the `404` error for the /api/userprofile request.
- Scenario 5: The system displays the `404` error for the /chatui request.
- Scenario 6: The system displays the `500` error for the /api/userprofile request.
- Scenario 7: The system displays the `503 (Service Unavailable)` error for the /chat-ui request.

Possible causes

- Cause for scenario 1: You may have not correctly set the smc.ess.chat.url in webtier.properties in smwebtier.yourdomain.net.
- Cause for scenario 2: You may have configured SSL and used Apache as the reverse proxy and deployed it on a different host from where the chat server is deployed.
- Cause for scenario 3:
  - You may have set **ssl_reqClientAuth:2** in the sm.ini. The certificate of the chat server is not trusted by the SM server (smserver.yourdomain.net).
  - The SM operator you specified for **daoFactory.userName** in chat-msvc/conf/app.properties is locked out or has no rights.
- Cause for scenario 4: You may have not correctly set `chatServiceUrl` in env.js for chat UI.
- Cause for scenario 5: You may have not correctly set the reverse proxy item for chat UI.
- Cause for scenario 6: There may be some configuration errors in the app.properties file of the chat service.
- Cause for scenario 7: The server of your chat UI is not started.

Solutions

Solution for Cause 1: No error in the browser console

Set a correct value for the smc.ess.chat.url in webtier.properties in smwebtier.yourdomain.net.

Solution for Cause 2: 400 Host doesn't match SNI

Deploy the Apache on the same host as where the chat server is deployed.

Solution for Cause 3: The certificate of the chat server is not trusted by the SM server

Import client certificate of the chat server into the trustedclients.keystore in the RUN folder of the SM server (smserver.yourdomain.net) and restart the SM server.

Or, you may also set **ssl_reqClientAuth:0** in the sm.ini.

Solution for Cause 3: The SM operator is locked out or has no rights.

Either unlock the SM operator or check whether it has the **system administrator** privilege and the **RESTful API** capability.

Solution for Cause 4:

Set a correct value for the `chatServiceUrl` in env.js for chat UI.

Solution for Cause 5:

Check and set a correct value for the reverse proxy item for chat UI.

Solution for Cause 6:

Check if the below items are set correctly:

daoFactory.serviceEndPoint, daoFactory.userName and daoFactory.password

You may also perform a manual health check by accessing https://chatserver.yourdomain.net:8449/healthcheck?pretty=true. For example, if the daoFactory.userName is not set correctly, the response message of the health check will show that the dependent data service is unavailable:

```
{
  "ChatServerRESTConnection" : {
  "healthy" : true
  },
  "ChatServerXMPPConnection" : {
    "healthy" : true
  },
  "SMServerRESTConnection" : {
    "healthy" : false,
    "message" : "The dependent data service at
https://smserver.yourdomain.net:13443 is unavailable. "
  },
  "deadlocks" : {
    "healthy" : true
  }
}
```

Solution for Cause 7:

Start your chat UI server.

1.11.7. 500 error for the request chatservice/api/pubsub/presubscribe

After signing in to the Service Manager web client, you can see both the chat notification button and the end user chat request button are correctly displayed. However, there is a JavaScript error in the browser console.

```
POST
https://proxyserver.yourdomain.net/chatservice/api/pubsub/presubscribe/
500
```

Cause

See the root cause as described in Failed to start chat in the ESS client.

Solution

See the root cause as described in Failed to start chat in the ESS client.

1.11.8. Cannot invite anybody to chat because all online invitees appear offline

The system displays the chat notification button correctly after you log on to the Service Manager web client. However, when you open a Service Desk ticket to invite an online user, Service Manager displays the user as offline and you are unable to start a chat.

There’s no error in the browser console or in the log files.

Cause

You may have set up several chat instances (chat server and chat service) and have wrongly configured the reverse proxy to access another chat instance that's also running. Or if you have configured and used HTTPS before and used HTTP related proxy configuration file, the HTTPS settings may be overwritten by the HTTP settings.

Solution

Open the developer tool of your browser, and then check the response to your first `of-http-bind/` request. For example, you may see something like this:

```
<body xmlns="http://jabber.org/protocol/httpbind"
xmlns:stream="http://etherx.jabber.org/streams"
from="chatserver.yourdomain.net" authid="6zld2vum40" …><stream:features>…
</stream:features></body>
```

Ensure the value of the `from` attribute of the body element is the embedded **Openfire** in your target chat server. If not, you should set the reverse proxy to access the correct chat instance. Or you may need to comment out all the configured proxies in the HTTP related proxy configuration file so that the HTTPS settings aren't overwritten by the HTTP settings.

1.11.9. The Previous Support Ticket ID combo box is still empty after you have made some chats in the ESS web client

The Previous Support Ticket ID combo box will display all your previous support tickets. But after you have made some chats, the Previous Support Ticket ID combo box in the ESS client is still empty.

Cause

You may have not ever selected a portal for End User Chat.

Solution

Perform Task 6: Select a portal for End User Chat.

1.11.10. Nothing happens when an ESS user clicks any of the knowledge links proposed by the virtual agent

If Service Desk Chatbot is enabled, the virtual agent proposes some knowledge links when an Employee self-service (ESS) user starts a chat. But when the user clicks any of the links, nothing happens and there is a JavaScript error in the browser console:

```
"Link is empty!"
```

Cause

You may have not ever selected a portal for End User Chat.

Solution

Perform Task 6: Select a portal for End User Chat.

1.11.11. An IT agent can't receive the chat requests from a portal user

Service Manager Collaboration works between IT agents. However, if a user starts an End User Chat from a Service Portal (such as the ESS client), some IT agents can't receive the chat request from the SM web client and can't find any errors.

Cause

You may have multiple assignment groups available for an End User Chat request. In this case, the system assigns only one group to this chat based on the sequence of the records stored in the database and these agents happen to be in other groups.

Solution

You can either keep only one assignment group or setup service-based chat assignment. See Configure chat assignment settings for more information.

1.11.12. Chat configuration failed error occurs

After login, you can see a chat configuration failed error message. In the browser console, you can also see the /of-plugins request returns the 302 error and the /http-bind request returns the 404 error.

Cause

You may have not configured the reverse proxy correctly for of-http-bind in IIS. For example, you may have missed the ending backward slash (/).

Solution

Add the ending backward slash (/) to Rewrite URL in IIS.

1.11.13. The icon of the chat notification button rotates forever

When you log on to the Service Manager with the web client, the icon of the chat notification button rotates forever.



This issue may occur in several scenarios as below.

- Scenario 1: Repeated JavaScript error "reconnecting to server" in the browser console
- Scenario 2: 401 Unauthorized for the request chatservice/api/config/simplechatconfig
- Scenario 3: 502 Bad gateway for the request /of-http-bind/
- Scenario 4: 503 Service Unavailable for the request /of-http-bind/

Possible causes

- Cause for scenario 1: You may haven't correctly configured LW-SSO for the chat server, or your changes made to `lwssofmconf.xml` have been reset by the system.
- Cause for scenario 2: You may haven't correctly configured LW-SSO for the web tier or the SM server.
- Cause for scenario 3: You may haven't correctly set the reverse proxy for the /of-http-bind.
- Cause for scenario 4: You may haven't correctly set the reverse proxy.

Solutions

Solution for scenario 1: Repeated JavaScript error

Check and correctly configure LW-SSO for the chat server.

Solution for scenario 2: 401 Unauthorized error

Check and configure LW-SSO for the web tier or the SM server correctly.

Solution for scenario 3: 502 Bad gateway error

Check if your reverse proxy of /of-http-bind is correct.

For example, you may have wrongly set it to another host and the DNS lookup failed.

Solution for scenario 4: 503 Service Unavailable error

Check if you can visit the chat server specified in the reverse proxy. For example, check if you can visit `https://chatserver.yourdomain.net:7443/http-bind`.

1.11.14. Fails to redirect the Chat URL from Identity Management (IdM)

When Service Manager (SM) is configured with IDM, Chat, and a proxy server, logging in through Identity Management (IdM) does not redirect to the Chat URL as expected.

Cause

The redirect URL includes a port number, that causes the chat redirection to fail.

Solution

If you remove the port number from the URL, it allows the chat feature to function correctly.

To make sure the chat feature funtions correctly for windows you need to add the below lines in **%APACHE_HOME%\conf\extra\httpd-vhosts**.conf file (for HTTP) **or** in **%APACHE_HOME%\conf\extra\httpd-ahssl**.conf file (for HTTPS):

```
ProxyPassReverse /webtier-xxx balancer://webtierCluster/<webtier-xxx>
ProxyPassReverse /idm-service-xxx balancer://webtierCluster/<idm-service-xxx>
```

1.12. Troubleshoot upgrade

This section helps you identify and resolve some of the issues that you might come across while upgrading Service Manager. Some of these issues include:

- Fail to migrate a legacy configuration item to a new configuration item after upgrade
- Cannot run SQL Compare
- Cannot run the wizard for applying upgrade
- Some data files are missing from the Upgrade Utility package
- The Upgrade Utility appears to stop responding
- The "Session no longer valid" error occurs during an upgrade
- An unexpected error occurs during an upgrade
- Upgrade failed with a "Not enough shared memory available" error
- The database transaction log is full
- Integrations don't work after an applications upgrade
- Automatic merge fails
- Unable to check the upgrade results after upgrade
- Copying full table takes a long time
- Inbox localization information is not generated
- Invalid object name 'CM3RM2' error occurs
- INSERT INTO WORKFLOWHISTORYM1 fails due to duplicate key
- Lots of renamed records appear after upgrade

1.12.1. Fail to migrate a legacy configuration item to a new configuration item after upgrade

When migrating a legacy configuration item to a new configuration item after upgrade, the system may display the following error message:

```
RTE E Error: SQL code=12704 message=ORA-12704: character set mismatch
```

Cause

The SQL Type of the current.phase field is not VARCHAR2

Solution

Open the device record in dbdict, find the current.phase field and check if its SQL Type is not VARCHAR2. If not, perform these steps before migration:

1. Open the device record in dbdict.
2. Find the current.phase field, update its name to current.phase2 and its SQL Name to CURRENT_PHASE2.
3. Save your changes.
4. Open the device record in dbdict again.
5. Add a new field called current.phase, type is character, set the SQL Name to CURRENT_PHASE, and set the SQL Type to VARCHAR2(60).
6. Save your changes.
7. Go to database, execute the following SQL query to copy the values from the current.phase2 field to the new current.phase field:

   update DEVICE2M1 set CURRENT_PHASE=CURRENT_PHASE2;

1.12.2. Cannot run SQL Compare

When run SQL Compare, the system displays a message that the version is unsupported.

Cause

SQL Compare doesn't support the Service Manager Applications version.

Solution

SQL Compare will check whether the Service Manager Applications version is supported or not. If the version is earlier than 9.30, you need to firstly upgrade to Service Manager 9.30 or later and then upgrade to Service Manager 9.70. If the version is not a standard release, please contact Service Manager Customer Support.

1.12.3. Cannot run the wizard for applying upgrade

The wizard for applying upgrade can't continue and the system displays the message of unsupported version.

Cause

The Upgrade Utility doesn't support the Service Manager Applications version.

Solution

Upgrade Utility will check whether the Service Manager Applications version is supported or not. If the version is earlier than 9.30, you need to firstly upgrade to Service Manager 9.30 or later and then upgrade to Service Manager 9.70. If the version is not a standard release, please contact Service Manager Customer Support.

1.12.4. Some data files are missing from the Upgrade Utility package

The upgrade wizard shows that some data files are missing from the Upgrade Utility package.

Cause

The Upgrade Utility is incorrectly downloaded or extracted.

Solution

Please check whether the missed data files exist in the extracted directory of the Upgrade Utility or not. If not, please check if you have removed them manually. The best solution is to download the Upgrade Utility and extract it again.

1.12.5. The Upgrade Utility appears to stop responding

The Upgrade Utility may appear unresponsive during the upgrade process. This issue may exhibit the following symptoms:

- The Windows Task Manager indicates that the Service Manager client is not responding.
- The Upgrade Utility doesn't show the progress after you click **Next** to start the upgrade execution.
- The Upgrade Utility appears to stop at a percentage of completion.

Cause

The Upgrade Utility is executing the task to migrate mass of data.

Solution

This is normal and doesn't indicate a problem with the upgrade. Additionally, the percentage on the status screen doesn't accurately indicate the actual progress.



Tip

Refer to the following tips to check whether the Service Manager Windows Client is hang or not:

- DBA check transaction
- Check detail.log
- Check the Windows Client by using the Microsoft Process Explorer

Besides these tips, you can also try other methods as appropriate.

1.12.6. The "Session no longer valid" error occurs during an upgrade

The upgrade failed with "Session no longer valid" error when running in foreground mode. This issue is most likely to occur when you are creating or applying a custom upgrade.

Cause

The client session is terminated during an upgrade. The Service Manager client may temporarily lose heartbeat when the Upgrade Utility is running. If the server can't detect the heartbeat for a certain amount of time, it disconnects the client session.

Solution

To resolve this issue, restore the database to the latest pre-upgrade state, add `sessiontimeout:1200` and `heartbeatinterval:120` to the sm.ini file, and then restart the upgrade. See also Step 3: Perform preparatory tasks.

Note

We recommend that you set the timeout period to a length of time that's long enough for an upgrade phase to complete.

1.12.7. An unexpected error stops the upgrade

An unexpected error stops the upgrade.

Cause

The Service Manager server may be running out of memory, stack overflow, power outage, or network failure..

Solution

Fix the issue that stopped the upgrade, restore the database to the latest pre-upgrade state, and then restart the upgrade.

1.12.8. Upgrade failed with a "Not enough shared memory available" error

The upgrade process ended abruptly when you were applying an upgrade, and the following error message was logged into the sm.log file:

```
E big_alloc: Not enough shared memory available to allocate <number> bytes
```

Cause

The size of the shared memory is not enough.

Solution

Restore the database to the latest pre-upgrade state, increase the size of the shared memory (see Step 3: Perform preparatory tasks), and then restart the upgrade.

1.12.9. The database transaction log is full

The process ended abruptly when you were loading data or applying an upgrade. The sm.log file contains error messages that resemble the following:

```
[Microsoft][ODBC SQL Server Driver][SQL Server] The log file for database
'DB_Name' is full. Back up the transaction log for the database to free up
some log space. (message,add.schedule)
```

```
An error occurred while attemping to add a record (file.load,add.record.0)
```

Cause

The database transaction log is full.

Solution

Running an upgrade generates a huge number of transactions and this is likely to make the transaction log full. Restore the database to the latest pre-upgrade state, refer to the documentation for your RDBMS to increase the database transaction log size or enable auto-growth, and then restart the upgrade.

1.12.10. Integrations don't work after an applications upgrade

After you upgrade your applications to Service Manager 9.80, certain existing integrations don't work correctly.

Cause

Certain integration fields that were optional in the old applications version have become required in the new Service Manager 9.80 applications. Existing integrations that don't have those fields populated will become invalid.

Solution

In this case, you must remove and create the APM OMi integration again in the Integration Manager. For more information about your specific integration, see Integrate.

1.12.11. Automatic merge fails

The Automatic Merge task fails during the upgrade and you receive the following error message:

```
Failed to unzip <zip file path>. Auto Merge skipped.
```

Cause

This error message indicates that automatic merge was skipped because of a failure to unzip the required zip file. The Upgrade Utility skips the Automatic Merge task and continues the upgrade process.

Solution

To rerun the Automatic Merge task after running the Upgrade Utility, follow these steps:

1. In the **Upgrade\3waymerge\oob** folder, find the zip file named after the version that you selected for the **Base Version** when running the Upgrade Utility.
2. Extract the folder named after the **Base Version** that you previously selected from inside the zip file to the **Upgrade\3waymerge\oob** folder. For example, if you selected **9.3PD** for the **Base Version** previously, extract the **9.3PD** folder from inside **9.3PD.zip**, and then place the extracted folder (**9.3PD**) in the **Upgrade\3waymerge\oob** folder.
3. Open the Upgrade Results list and search for records with a type of "Renamed."
4. Select the objects that you want to merge, and click **Mass Auto Merge** from the More Actions list.

1.12.12. Unable to check the upgrade results after upgrade

Unable to open and view the upgrade results after upgrade.

Cause

The forms or codes to open and view upgrade result are conflicted.

Solution

Run the following command to check the upgrade results:

sm.-bg apm.upgrade.resolve NULL NULL showresult *<object type><object name>*

Run the following commands to revert or choose an upgrade for upgrade results:

sm -bg apm.upgrade.resolve NULL NULL revert *<object type><object name>*

sm -bg apm.upgrade.resolve NULL NULL upgrade *<object type><object name>*

1.12.13. Copying full table takes a long time

It takes a long time to copy a full table.

Cause

If there are lots of records (for example, more than 50000) in the database dictionary whose Unique/Primary key is updated by the Upgrade Utility, the system may copy the full table and take much time to update these records.

Solution

Follow these steps:

1. Before upgrade, run the SQL Compare utility and check the except.log file. You may find some information similar to the following example:

   `dbdict:cirelationship, Unique Key is {"relationship.name", "logical.name"} -- the Unique key is removed or updated and may cause full table copy.`

   This information indicates that the Unique key will be removed or updated during the upgrade process and may cause full table copy.

2. After loading the tranfer.bin file in the upgrade wizard, open upgradeKeyChanges in the ScriptLibrary, and then search for the following codes:

   `var DBDICT_IGNORE_UNIQUE_KEY = [];`

   Update it to:

   `var DBDICT_IGNORE_UNIQUE_KEY = ["cirelationship"];`

3. Save the ScriptLibrary and then continue upgrading. The Unique/Primary key of the record will not be updated.
4. After upgrade, you need to update the Unique/Primary key manually.

1.12.14. Inbox localization information is not generated

The system may display the information in except.log:

`The number of inbox is more than 5000 and Upgrade Utility will not generate inbox localization information. Please follow the upgrade guide to do it.`

Cause

There are more than 5000 inbox records and it may take much time to generate the localization information during an upgrade process.

Solution

Follow these steps to generate it manually:

1. Create a ScriptLibrary and add the following codes:

   ```
   lib.localizeTable.createAllEntries("inbox", "inbox.name");
   ```

2. Click the **Execute** button to generate localization related information.
3. Follow the detailed information as described in Step 10: Perform additional manual tasks to translate the customized inbox names.

1.12.15. Invalid object name 'CM3RM2' error occurs

You receive the following error after you upgrade Service Manager:

`Invalid object name 'CM3RM2'`

Cause

The CM3RM2 table is not created in the new Service Manager system but it is created in the old system.

Solution

Ignore this error if CM3RM2 is created after the upgrade.

1.12.16. INSERT INTO WORKFLOWHISTORYM1 fails due to duplicate key

After you upgrade from Service Manager applications versions earlier than 9.50, insert workflowHistory fails with the following error message.

`ERROR: duplicate key value violates unique constraint "workflowhistorym1" DETAIL: Key ("ID")=(10207044) already exists. (se.write.workflowHistory,add.manual)`

Cause

As of version 9.50, we have changed the generation strategy of WorkflowHistory IDs from make.sig-based to counters-based. Therefore, the newly generated IDs duplicate with the old data after the upgrade.

Solution

Go to Database Management, and then run the following SQL command to update the IDs of old data to avoid duplicating.

update WORKFLOWHISTORYM1 set ID=ID+0.5

1.12.17. Lots of renamed records appear after upgrade

When the Service Manager server is running on Linux with non-English UTF-8 locale (for example, es_ES.UTF-8), lots of records are set to renamed after upgrade.

Cause

The signatures of the records on Linux with the non-English UTF-8 locale may be calculated with different values from English UTF-8.

Solution

Change the Linux locale to English UTF-8 before upgrade.

1.13. Troubleshoot security

This section helps you identify and resolve some of the issues that you might come across while working with Service Manager security. Some of these issues include:

- The IdP login page is not displayed
- Fails to log in to IdP with an "Invalid tenant" error
- Fails to log in to IdP with an "Invalid login name/password" error
- Redirected to an "Invalid Token" page after clicking login

1.13.1. The IdP login page is not displayed

After launching the Service Manager web tier or Mobility Client URL in a browser, you are redirected to the Service Portal login page, rather than the IdP (ADFS) login page. This issue occurs when configuring SAML SSO by using the standalone IdM.

Cause

The SAML configuration in the IdM Service is incorrectly set.

Solution

If you can see the IdM login page (that is, the Service Portal login page), the IdM configuration on the Service Manager side is correct. You need to check the SAML configuration in the IdM service.

1.13.2. Fails to log in to IdP with an "Invalid tenant" error

After entering credentials in the IdP (ADFS) login page to log in, an `Invalid tenant` error is returned. This issue occurs when configuring SAML SSO by using the standalone IdM.

Cause

The IdM tenant configured on the Service Manager side is different than in the IdM service.

Solution

Check the following files:

- <idm-service>\WEB-INF\classes\seeded\samples\com.hpe.tenant1__1.3.2.1__Add_Update_Saml_Configuration.json
- /WEB-INF/webtier.properties of the SM web tier or Mobility Client

Make sure the **idm.tenant** parameter in the webtier.properties file is set to the **name** field of tenant in the .json file of the IdM service.

1.13.3. Fails to log in to IdP with an "Invalid login name/password" error

After entering credentials in the IdP (ADFS) login page to log in, the following error is returned:

`A CXmlApiException was raised in native code : error 3 : login(3): Invalid login name/password. Please try again.`

This issue occurs when configuring SAML SSO by using the standalone IdM.

Cause

The **NameID** field in ADFS and the **name** field in the SM **operator** table aren't mapped to the same LDAP attribute.

Solution

Make sure the two fields are mapped to the same LDAP attribute.

1.13.4. Redirected to an "Invalid Token" page after clicking login

You have stayed on the ADFS login page for quite a while (for example, hours) and then you click login. ADFS redirects you to an "Invalid Token" page that reads:

> The request token is invalid. It may have already been used or has expired.

This issue occurs when configuring SAML SSO by using the standalone IdM.

Cause

Token in browser is either expired or invalid.

Solution

Perform the following steps:

1. Close your browser.
2. Reopen your browser.
3. Sign in again.

1.14. Troubleshoot delta migration

This section helps you identify and resolve some of the issues that you might come across while using the delta migration tool. Some of these issues include:

- Cannot export the delta data in some tables
- The delta data in some tables is not suggested for the import operation
- The "dmt_unload_load* not started" error occurs
- The "can’t write files into output folder" error occurs
- The "empty folder" error occurs
- The "trigger disabled" error occurs
- Cannot handle skipped tables
- The delta data migration takes a long time

1.14.1. Cannot export the delta data in some tables

You can't export the delta data in some tables.

Cause

To export the delta data from a table, a field to record the last modification time is necessary. For most tables in Service Manager, this field is **sysmodtime**. However, some tables (especially custom tables) contain no such field, or the field name is not **sysmodtime**. In both cases, the tool can't find or export the delta data.

Solution

You need to migrate the data manually. For example, you can export all the data in these tables to another UNL file, and deploy this UNL file on the new system.

1.14.2. The delta data in some tables is not suggested for the import operation

The delta data in some tables is not suggested for the import operation.

Cause

There are potential risks if the table structures of the new and the original Service Manager system are different. For example, if the table for interaction in the new Service Manager system contains more fields than the original Service Manager system, the new columns will be left blank when you try to migrate data in this table by using UNL files. More seriously, if one of the new fields is inserted between the original fields instead of at the end of table, then importing the table may cause unexpected behavior.

Solution

Before the delta data is imported, the tool checks the table structure. If the structure is different, the tool will provide a risk estimation, and you can decide whether to import the data or not. If you decide not to import a table, see Cannot handle skipped tables.

1.14.3. The "dmt_unload_load* not started" error occurs

When migrating the delta data by using background tasks, the system displays the following error message:

```
background scheduler 'dmt_unload_load*' not started, …
```

Cause

The scheduler isn't started.

Solution

Follow these steps to start the schedule:

1. Sign in to Service Manager as an administrator.
2. Click **System Navigator** > **System Status** > **Start Scheduler**.
3. Click **dmt_unload_load** to start the five background schedulers.

If you have already started the schedule, return to the previous step in the wizard. Wait for one minute and then try again.

1.14.4. The can't write files into output folder error occurs

The system displays a `can't write files into output folder` error message during the export operation.

Possible causes

- Cause 1: The folder doesn't exist.
- Cause 2: Service Manager doesn't have write rights to the folder.

Solutions

Solution for Cause 1: The folder doesn't exist

Check if the specified folder exists.

Solution for Cause 2: Service Manager doesn't have write rights to the folder

Check if Service Manager has the right to add and modify files in the folder, especially when Service Manager is installed on a server with an operating system other than Windows.

1.14.5. The "empty folder" error occurs

The system displays a `selected folder is empty, or 'readme.txt' missing` error message during the import operation.

Possible causes

- Cause 1: The folder doesn't exist
- Cause 2: Service Manager doesn't have write rights to the folder

Solutions

Solution for Cause 1: The folder doesn't exist

Check if the specified folder exists, and then check if the export operation stores everything generated in the specified folder.

Solution for Cause 2: Service Manager doesn't have write rights to the folder

Check if Service Manager has the right to add and modify files in the folder, especially when Service Manager is installed on a server that has an OS other than Windows.

1.14.6. The "trigger disabled" error occurs

Receive the following error message from the background scheduler:

```
background scheduler 'dmt_unload_load*' are trying to disable triggers, …
```

Cause

All triggers will be disabled to avoid unexpected behavior during the import operation. It may take no more than one minute for each background session to disable the triggers.

Solution

Wait for a while, and then return to the previous step to try again.

1.14.7. Cannot handle skipped tables

Cannot handle skipped tables.

Cause

All the information in the original Service Manager system should be migrated to the new Service Manager system, except the information that belongs to a feature or function that's no longer valid in the new Service Manager system.

Solution

- If a table is skipped during the export operation because it contains no delta data, then no additional work is needed.
- If a table is skipped during the export operation because the **sysmodtime** field missing, migrate all the data in the table to the new system by using the Service Manager unload script.

1.14.8. The delta data migration takes a long time

The delta data migration takes a long time.

Cause

If there is a lot of delta data to migrate, the operation may take several hours. This leads to several hours' downtime, and new delta data is generated during this downtime.

Solution

You can run the tool multiple times to resolve this issue. For example, you have a delta time range from 00:00:00 on Jan 1st to 16:00:00 on Feb 2nd (the present). If you set the delta data time range from 00:00:00 on Jan 1st to 00:00:00 on Feb 2nd, it may take several hours to migrate the data. You can then run the tool again and set the delta data time range from 00:00:00 on Feb 2nd to the present time. This time, the operation may take several minutes only. These several minutes are the only downtime and after that, the new Service Manager system will function as a production system.

1.15. Troubleshoot Service Manager Wizard

This section helps you identify and resolve some of the issues that you might come across while using the Service Manager Wizard tool. Some of these issues include:

- Button options aren't displayed properly
- File variables aren't passed to wizard flow
- The "Format not found" error occurs
- JavaScript syntax errors occur
- A query fails to provide correct results
- A query returns a list of records rather than a single record
- RAD expression syntax errors occur
- Records aren't sorted properly
- Wizard stops with "Could not continue" errors
- Unable to run a trace

1.15.1. Button options aren't displayed properly

Button options aren't displayed properly.

Cause

The options on the **Usage** tab are suppressed by mistake.

Solution

If button options aren't displayed or unexpectedly displayed, check the **Usage** tab to ensure that the correct checkboxes are enabled.

Also check the following:

- The first displayed panel should not have a **Previous** button.
- The last displayed panel should have a **Finish** button.
- If the Previous button is not displaying as expected, check to see if you selected the **Start Node?** option on the **Wizard Info** tab.

  > **Tip**
  >
  > Only the first wizard panel should have the **Start Node?** option selected on the **Wizard Info** tab.

- Do not choose the **Next** button as a **Finish** button. If you do, the wizard will stop executing. As a best practice, disable the **Next** button and enable the **Finish** button in the last panel of the wizard work flow.
- Check to see if any of the options on the **Usage** tab were suppressed by mistake.

Contents

- 1 Cause
- 2 Solution

1.15.2. File variables aren't passed to wizard flow

The wizard stops executing, and the user is returned to the calling application with the following error message at the top of the screen:

`Wizard could not continue. No file variable was passed in.`

Cause

In the **Select $L.file by** subtab on the **File Selection** tab, the option "$L.file passed in" was selected, but no file variable was passed in.

Solution

Either call `wizard.run` with the file parameter passed in correctly, or choose another option in the **Select $L.file by** tab on the **File Selection** tab. For example, select **Create a record of type**.

Contents

- 1 Cause
- 2 Solution

1.15.3. The "Format not found" error occurs

The wizard stops executing, and the user is returned to the calling application with the following error message at the top of the screen:

`Format "test" not found (display,show.rio)`

Cause

An invalid format name was entered in the **Sub Format to Display** field on the **Usage** tab while using the "Request user input" option in the **Wizard Usage** section on the **Usage** tab.

Solution

Enter a valid format name in the **Sub Format to Display** field on the **Usage** tab, so a valid subformat will be displayed. Or, select the **Skip Display** option to bypass attempting to display the form.

Contents

- 1 Cause
- 2 Solution

1.15.4. JavaScript syntax errors occur

The wizard stops executing, and the user is returned to the calling application with the following error message(s) at the top of the screen:

`Process panel <panel name> in RAD wizard.run encountered error in line 5 (wizard.run,<panel name>)`

`Process panel init.expr in RAD wizard.run encountered error in line 5 (wizard.run,init.expr)`

`Cannot evaluate expression (wizard.run,init.expr)`

`Script <UNKNOWN> line 1: ERROR ReferenceError: test is undefined at char 1`

`Unrecoverable error in application: se.call.process on panel call.rad.1`

`Unrecoverable error in application: wizard.run on panel init.expr`

Cause

Invalid syntax was entered in a **JavaScript** tab. The RAD application and panel will give you a hint as to which JavaScript tab the syntax error was entered. Refer to the table below for help in determining where to make the fix.

RAD application/panel	Where to fix
`wizard.run,init.expr`	**JavaScript** subtab of the **File Selection** tab.
`wizard.run,select.file.setup`	Query statement in the **Select $L.file by** subtab of the **File Selection** tab.
`wizard.run,run.expressions`	**Expressions** subtab of the **Actions** tab.
`wizard.run,exit.cancel`	**Javascript Executed on Cancel** subtab of the **Cancel Expressions** tab.

Solution

Correct syntax for JavaScript expressions is discussed in the *JavaScript Programmer's Guide*. When using JavaScript, you must define variables before using them. For example:

`var <name>=new String()`

The correct syntax for assigning a value to a field is as follows:

`Table.field1=value`

To compare the values of two fields to each other, use the following syntax:

`Table1.field1==table2.field2`

Contents

- 1 Cause
- 2 Solution

1.15.5. A query fails to provide correct results

A query fails to provide correct results.

Cause

When entering a query in the **File Selection** tab, all strings such as field names must be enclosed in quotes. If the strings aren't enclosed in quotes, the query will fail to provide the correct results.

For example:

`"agreement.id="+G.new.sla`

Or

`"logical.name=\""+str($group.name.new)+"\""`

Solution

Enclose all strings (such as field names) in quotes when entering a query in the **File Selection** tab.

Contents

- 1 Cause
- 2 Solution

1.15.6. A query returns a list of records rather than a single record

According to the wizard setup, a single record should be returned from the query entered in the **Usage** tab. Instead, a true search is performed.

Cause

Incorrect query syntax is used. For example, `agreement.id in $L.file=168`. Query statements should not include the file variable.

Solution

Use a correct query syntax. For example, `"agreement.id="168`.

Contents

- 1 Cause
- 2 Solution

1.15.7. RAD expression syntax errors occur

The wizard stops executing, and the user is returned to the calling application with the following error message(s) at the top of the screen:

`Process panel <panel name> in RAD wizard.run encountered error in line 2 (wizard.run,<panel name>)`

`Process panel init.expr in RAD wizard.run encountered error in line 2 (wizard.run,init.expr)`

`Cannot evaluate expression (wizard.run,init.expr)`

`Bad arg (2) oper = (wizard.run,init.expr)`

`Bad arg (3) oper index (wizard.run,init.expr)`

`Unrecoverable error in application: se.call.process on panel call.rad.1`

`Unrecoverable error in application: wizard.run on panel init.expr`

Cause

These errors are caused by syntax errors in the wizard panels. The panel name gives you a hint as to where the issue occurred.

Bad arg (x) oper operator y indicates improper syntax of a statement or a missing or invalid parameter that's expected to be supplied to the statement.

Bad arg (1 or 2) oper <operator>: arg 1 indicates that the issue is to the left of the <operator>. If it is arg 2, then the issue is to the right of the <operator>. For example, operator can be defined as follows: = , < , > , ~= , ~< , ~> , etc.

Bad arg (1 or 2) oper <in> arg 1 indicates that the issue is to the left of "in" within the statement. If arg is 2, then the issue is to the right of "in" within the statement.

Bad arg (x) oper <function> indicates that the number of the parameter that was passed to the function is invalid.

The following table will help you find where in the wizard panel the error occurred and where to make the fix.

RAD application/panel	Where to fix
`wizard.run,init.expr`	**Initial Expressions** subtab of the **File Selection** tab.
`wizard.run,select.file.setup`	Query statement in the **Select $L.file by** subtab of the **File Selection** tab.
`wizard.run,get.selection.records`	Query statement in **Selection Criteria** on the **Usage** tab.
`wizard.run,run.expressions`	**Expressions** subtab of the **Actions** tab.
`wizard.run,decide.restart`	**Restart Panel if** field on the **Actions** tab.
`wizard.run,get.message`	**Condition** field of the **Messaging** tab.
`wizard.run,setup.wizard.variables`	Wizard variables on the **Variables** tab.
`wizard.run,find.next.wizard`	**Condition** field on the **Next Wizard** tab.
`wizard.run,exit.cancel`	**Expressions Executed on Cancel** subtab of the **Cancel Expressions** tab.

Solution

Correct syntax for RAD expressions is discussed in the System Language topics in the Service Manager Online Help. You can view this help by clicking **Help** > **Help Contents** if you are using a Windows client, or pressing **F1** if you are using a web client.

The correct syntax for assigning a value to a field is as follows:

`<field> in $L.file=value`

To compare the values of two fields to each other, use the following syntax:

`<field1> in $L.file=<field2> in $L.file`

Any concatenated statement containing a field name and a variable needs to be entered as follows:

`"agreement.id=" +$G.test` where $G.test is numeric or Boolean.

`"agreement.id =\"" + $G.test + "\""` where $G.test is a character.

`"agreement.id='" + $G.test + "'"` where $G.test is a date/time.

Contents

- 1 Cause
- 2 Solution

1.15.8. Records aren't sorted properly

According to the wizard setup, the list of selected records should be sorted by the values in a certain field. When the list of records is displayed, it is sorted by the unique key instead.

Cause

The type of field needed for the query (dbdict name) is invalid.

Solution

Enter a valid sort field in the record type dbdict.

Contents

- 1 Cause
- 2 Solution

1.15.9. Wizard stops with "Could not continue" errors

The wizard stops executing for a variety of reasons, including possible configuration errors. Some of the errors you may encounter are as follows:

- Scenario 1:

  The system displays the `Cannot create record of type "test."` error message.

- Scenario 2:

  The system displays the `Cannot select from record of type "test."` error message.

- Scenario 3:

  The system displays the `Cannot initialize record of type "slas."` error message.

- Scenario 4:

  The system displays the `Invalid selection query: agreement=$G.test.` error message.

- Scenario 5:

  The system displays the `Wizard "test" doesn't exist. Contact your System Administrator.` error message.

- Scenario 6:

  The system displays the following error messages.

  `Display application could not find screen. Query=screen.id="wizard.test.me" and language="ENG"`

  `Unrecoverable error in application: se.call.process on panel call.rad.1`

  `Display application error exit has been taken. $L.ds.ids=L.ds.desc=`

Possible causes

- Cause for scenario 1: The record type entered doesn't exist as a record in the dbdict table.
- Cause for scenario 2: The record type entered doesn't exist as a record in the dbdict table.
- Cause for scenario 3: On the **Usage** tab, **Query for Records of type** has an invalid record type entered.
- Cause for scenario 4: On the **Usage** tab, an invalid field name was used in the **Query for Records** statement.
- Cause for scenario 5: An invalid wizard name was entered on the **Next Wizard** tab. When a nonexistent wizard evaluates to true, the wizard workflow will be caught in an infinite loop and the session thread will be killed.
- Cause for scenario 6: An invalid display screen name was selected on the **Usage** tab.

Solutions

Solution for scenario 1:

Enter a valid record type or dbdict name.

Solution for scenario 2:

Enter a valid record type or dbdict name.

Solution for scenario 3:

Enter a valid record type or dbdict name.

Solution for scenario 4:

Determine and use the correct field names in the statement.

Solution for scenario 5:

Enter a valid wizard name on the **Next Wizard** tab.

Solution for scenario 6:

On the **Usage** tab, enter a valid display screen name or leave the field empty for the default option.

Contents

- 1 Possible causes
- 2 Solutions
  - 2.1 Solution for scenario 1:
  - 2.2 Solution for scenario 2:
  - 2.3 Solution for scenario 3:
  - 2.4 Solution for scenario 4:
  - 2.5 Solution for scenario 5:
  - 2.6 Solution for scenario 6:

1.15.10. Unable to run a trace

Unable to run a trace.

Cause

The cause of this issue is unknown.

Solution

If you receive an error message, use the following to trace the error to get more detail on the message you receive. You can also run a trace when the wizard flow doesn't take the expected path.

To run a trace, follow these steps:

1. Enter the RTM:3 and debugdbquery:999 parameters in the Service Manager sm.ini file.
2. Start a new client connection.

   For complete details, see the instructions in the *Diagnostics and Tuning* white paper.
3. Within the trace, search for the first call to the wizard.run RAD application.
4. Next, follow the flow of the wizard, checking the dbquery statements against the wizard table to see which wizard is called next.
5. Follow the trace to the error message if one was received, or follow the trace to understand which path the application took through the wizard flow.
6. Use print statements in the JavaScript expressions within the wizard panels, to write additional information to the log file. For example, the content of variables that will influence which wizard will be run next or statements as to which wizard is executed at the time.



Note

Remember to remove these debug statements before putting the wizard into production.

Contents

- 1 Cause
- 2 Solution

1.16. Troubleshoot Web Services

The combination of debugging tools and information gathered from faults usually helps you find the root cause of an issue easily. Unfortunately, not all Web Services give sufficient fault messages, which makes debugging the issue more challenging.

This section helps you identify and resolve some of the issues that you might come across while using Service Manager Web Services. Some of these issues include:

- Cannot understand the return codes provided by Web Services
- Cannot manipulate detailed return codes from Document Engine
- The "Doc Engine call failed" error occurs
- The "Invalid or missing file name in XML request " error occurs
- The "can't override getType()" error occurs
- The WSDL2JS utility fails
- Cannot troubleshoot a Web service that's behind a closed firewall
- Max sessions exceeded in Web Services
- Cannot troubleshoot HTTP socket connections
- The "Unable to connect to the remote server" error occurs
- Calling external web services with SSL fails
- Cannot interpret http.log

1.16.1. Cannot create subcategory for a KM record

After you create a KM record by using the RESTful API and set a subcategory for it, the system creates the KM record successfully. However, the system can't locate the subcategory correctly.

Cause

When you are creating knowledge documents by using the RESTful API, the categories field is special and you need to use a slash (/) to distinguish the categories from the subcategories.

Solution

Set The format of the categories field to the "A/B/C" format. For example, `"Service Manager/Support"`.

1.16.2. Cannot understand the return codes provided by Web Services

Cannot understand the return codes provided by Web Services. See the following failure return code and message as an example:

```
message="No (more) records found" returnCode="9" status="FAILURE"
```

Cause

The cause of this issue is unknown.

Solution

Currently the status attribute always contains either "SUCCESS" or "FAILURE." A best practice is to check either the status to see if it has the value “SUCCESS” or to check the return code to see if it is zero. All other values equate to FAILURE. The value of the message attribute is a string which corresponds to the return code value. The Service Manager server global JavaScript method called RCtoString()will convert a particular integer return code value to the corresponding message text. The following are the currently defined values:

Value	Definition
0	Success
1	Bad Length
2	Bad Serial Number
3	Resource Unavailable
4	Unable to Terminate
5	Resource Not Available
6	Resource Expired
7	Specified Name Not Found
9	No (more) records found
10	No messages
11	No query words
12	No stop words
13	No string
14	No such word
15	Not enough memory
16	Already exists
17	Shutdown error
18	Stop words not found
19	Too many documents
20	Unable to open file for output
21	Waiting for resource
22	Word length too long
23	Duplicate file system

Value	Definition
24	Duplicate IPC Key
25	IPC Key Not Found
27	Wrong owner
28	Not authorized
29	Invalid Userid Specified
30	Invalid Password Specified
31	New Password is Invalid
32	Password Expired
33	Authority Revoked
34	Max Attempts to Login Exceeded
35	Max Number of Logins Exceeded
36	Invalid terminal for user
37	Invalid Authorization Code
38	Maximum users exceeded
39	Named user already logged in
40	Not a named user and no floating users available
41	User Already Logged In
42	Forced synchronization
43	IR read count mismatch
44	Seek error
45	24×7: DBLOG error
46	Open error
47	Error closing remote file

Value	Definition
48	Duplicate key
49	Null key
50	All null keys
51	Record modified since last retrieved
52	Record deleted since last retrieved
53	Trigger Error
54	Not supported
55	Record no longer qualifies
56	Query timed out
57	Unable to delete file
58	Partially-keyed or non-keyed query
59	Error occurred in parsing
60	Shared memory version mismatch
61	Distributed Lock Manager can't lock item
62	Refresh not needed
63	Userid expired
64	Userid inactive
65	SQL conversion skipped for this file
66	Query could not be parsed
67	file could not be opened
68	User is not located in LDAP
69	User is not allowed to use ODBC driver
70	Invalid SOAPaction / unrecognized application action

Value	Definition
71	Validation failed
72	User is not allowed to use SOAP API

Contents

- 1 Cause
- 2 Solution

1.16.3. Cannot manipulate detailed return codes from Document Engine

Cannot manipulate detailed return codes from Document Engine.

Cause

No system administrator privileges.

Solution

The System Administrator can manipulate the detailed return code by setting the value of $L.exit in the Document Engine process's final expressions to one of the following:

Action or Error Situation	$L.exit value
record has changed since it was selected	changed
cancel processing the record	cancel
record is locked	locked
Request failed validation	bad.val
record was deleted since it was selected	deleted
exit processing	exit
normal exit	normal
Record should get unlocked	unlock
Sets exit value to menu to return to the menu	menu
Record was added, screen will be refreshed	added
Processing will restart – starting with init of file variable	restart
Processing will proceed with a new state record	newstate
Displayed records will be refreshed	refresh
Displayed joinfile records will be refreshed	refreshjoinfile
Category changes	newcat
Position in record list will be changed	reposition
Record will be reset to original values	resetrec
Mode will be set to close and close processing will start	closestate
Restart processing starting with init of file variable	restart
Mode will be set to add, which goes into the open state	openstate
Initializing values to add record	setupadd

Action or Error Situation	$L.exit value
An undefined action was passed to the document engine	invalid.action
User is not authorized for this action	no auth

Contents

- 1 Cause
- 2 Solution

1.16.4. The "Doc Engine call failed" error occurs

The system displays the following error message:

```
soap_serve - Caught XML API exception scxmlapi(19) - Doc Engine call
failed with cc -1
```

Cause

This error message is issued either when the Document Engine did not attempt to write the record, because the Process called via extaccess doesn't perform a save operation, or if a validation failed and the save could not be performed.

Solution

To fix this issue, ensure that the Process called does perform an action that adds or updates a database record. If it does, add the msglog:1 parameter to the sm.ini and rerun the Web Service. Check the sm.log file for any validation error messages and then either pass the required information or change the extaccess record to add any missing required fields to it. If your are still unsure what's the root cause of the issue after this, add `RTM:3` and `debugdbquery:999` to the `sm.in` i and retest the Web service operation. If your are still unsure what's the root cause of the issue after this, add RTM:3 and debugdbquery:999 to the `sm.ini` file and retest the Web service operation.

1.16.5. The "Invalid or missing file name in XML request " error occurs

The system displays the following error message:

```
<SOAP-ENV:Fault><faultcode>SOAP-ENV:Server</faultcode>
<faultstring>scxmlapi(16) - Invalid or missing file name in XML
request</faultstring><detail><appFaultCode>16</appFaultCode>
<appFaultString>scxmlapi(16) - Invalid or missing file name in XML
request</appFaultString></detail></SOAP-ENV:Fault>
```

Cause

This error message is issued if the binaries can't successfully retrieve the name for the Object to access from the extaccess file. This issue occurs most often when the Object name is in “CamelCase” notation.

Solution

To prevent this issue, don't use “CamelCase” notation (where the name contains compound words or phrases that are joined without spaces, and each word is capitalized within the name.) in the Object Name in the extaccess file. As a best practice, use the name of the dbdicts as the Object Name as well.

If the underlying cause is not the camel case notation, you can modify the SOAP body by adding **filename=”<filename>”** to work around this issue. For example:

```
<soap:Body>
<CreateProblemRequest filename="rootcause"
xmlns="http://<server>:<port>/SM/7">
```

1.16.6. The "can't override getType()" error occurs

The system displays the following error message:

```
getType() in com.peregrine.webservice.ComputerInstanceTypeDevice can't
override getType() in com.peregrine.webservice.common.StructureType;
attempting to use incompatible return type
```

Cause

The ConfigurationManagement WSDL is made up of the device extaccess record in addition to a number of device attribute files (such as computer). The following errors occur when you set the API Caption for the type field in the device extaccess record to “type” or “Type” and then attempt to compile the WSDL using Apache Ant:

```
build_java:
[javac] Compiling 114 source files to C:\Service
Manager\server\webservices\sample\AxisSample\build
[javac] C:\Service
Manager\server\webservices\sample\AxisSample\src\com\
peregrine\webservice\ComputerInstanceTypeDevice.java:225: getType() in
com.peregrine.webservice.ComputerInstanceTypeDevice can't override
getType() in com.peregrine.webservice.common.StructureType; attempting to
use incompatible return type
[javac] found   : com.peregrine.webservice.common.StringType
[javac] required: java.lang.String
[javac]     public com.peregrine.webservice.common.StringType
getType() {
[javac]
[javac] 1 error
BUILD FAILED
C:\Service Manager\server\webservices\sample\AxisSample\
build.xml:184: Compile
failed; see the compiler error output for details.
```

Solution

To avoid this or similar errors, make sure that the name is valid and doesn't conflict with previously defined names when you set up alias names (“API Captions”). All of the common.xsd definitions for data types such as StructureType, ArrayType, have a type attribute, for which Axis manufactures a getType Java method. When it generates a getType method for this new type

property/field, those two methods conflict. It doesn't matter whether you specify “type” or “Type” because Axis uses camel-case naming conventions for its generated method names. Whenever an API caption can cause a conflict with a pre-existing function, change it to be something unique; in this case, for example, make the API caption **CIType**.

1.16.7. The WSDL2JS utility fails

The WSDL2JS utility fails.

Cause

WSDL file has illegal semantics or incompatible schema.

Solution

The WSDL to JS utility executes the SOAP JavaScript record. It reads the provided WSDL with all incorporated schema definitions and creates or updates a JavaScript record in the Service Manager ScriptLibrary table with the objects and methods that can be used for this web service. The generated code can then be called from a custom written JavaScript to consume the external Web Service. If the code generated by WSDL2JS is incorrect or incomplete, contact Customer Support for a new unload of the utility. If the issue is still not solved with the latest version of the WSDL2JS utility, send the WSDL and all imported / invoked xsd schemas to Customer Support together with an unload of the generated JavaScript record. It is very important that the location of the xsd files that are imported or invoked from the WSDL is set correctly, otherwise the WSDL2JS utility will generate incomplete code.



Caution

Every time the SOAP JavaScript record is changed, all existing Web Services generated JavaScripts have to be re-generated by re-running the WSDL to JS utility and all invoking JavaScripts have to be re-compiled.

Contents

- 1 Cause
- 2 Solution

1.16.8. Cannot troubleshoot a Web service that's behind a closed firewall

Cannot troubleshoot a Web service that's behind a closed firewall.

Cause

All SM operation are implemented based on network communications. When the Web service is behind a closed firewall, the system can't locate whether the error is a network failure or a SM system failure.

Solution

Method 1: Use internalURI parameter to load Web Service (WSDL)

The internalURI parameter enables you to load Web Service (WSDL) when your server can't access Internet. You can enter this parameter in the sm.ini file.

Method 2: Check whether a WSDL file that's stored on the local machine works using test data

You can check whether a WSDL file that's stored on the local machine works using test data.

Step 1: Test the WSDL2JS

1. Store the WSDL file locally on the server.

2. Start the WSDL to JS utility and enter, file://<fully qualified path to the file>.wsdl

3. Click **Proceed**.

If the JavaScript file for the Web service is generated without error messages and ends with:

```
lib.SOAP.init();
/// End ----------------
```

... then the WSDL to JS program was able to interpret the WSDL file.

To correctly write the JavaScript functions to call this Web service and generated JavaScript, check the generated JavaScript for the function you want to use, in this case:

```
this.SOAPOperations[ "RetrieveIncident" ]
= new soap_Operation( "RetrieveIncident", "Retrieve","document",
```

```
  "RetrieveIncidentRequest",
"RetrieveIncidentResponse" );
```

The request can be found within that line and refers to the request function further down:

```
function RetrieveIncidentRequest(  )
{
this.$$nsPrefix = "ns";
this.$$attributes = new Array();
this.$$xmlNames = new Array();
this.$$objNames = new Array();
this.$$minOccurs = new Array();
this.getName = getName;
this.getXmlName = getXmlName;
this.setContent = setContent;
this.addContent = addContent;
this.getContent = getContent;
this.isFault = isFault;
this.$$elementChildren = new Array();
this.$$name = "RetrieveIncidentRequest";
this.$$xmlNames[ "RetrieveIncidentRequest" ] =
"ns:RetrieveIncidentRequest";
this.attachmentInfo = new Boolean();
this.$$attributes.push( "attachmentInfo" );
this.attachmentData = new Boolean();
this.$$attributes.push( "attachmentData" );
this.ignoreEmptyElements = new Boolean("true");
this.$$attributes.push( "ignoreEmptyElements" );
this.xmlns = new String("http://servicecenter.peregrine.com/PWS");
this.$$attributes.push( "xmlns" );
this.model = new RetrieveIncidentRequest_IncidentModelType();
this.$$elementChildren.push( "model" );
```

Step 2: Test the request

Once the automatically generated JavaScript code has been saved, write a calling JavaScript to execute the Web Service. The following is a simple example code for IncidentManagement record retrieval:

```
function RetrieveIncident(incident_id)
{
        var IncMgmtSvc = new system.library.IncidentManagement.
IncidentManagement();
        IncMgmtSvc.user="falcon"
        var retrieveReq = new system.library.IncidentManagement.
RetrieveIncidentRequest();
        retrieveReq.model.keys.IncidentID.setValue(incident_id);
        try
        {
                 var retrieveResp = IncMgmtSvc.invoke(retrieveReq);
                if ( retrieveResp.isFault() )
                {
                        throw( "SOAP Fault: " + retrieveResp.
faultstring.getValue() );
                }
                return retrieveResp.model.instance;
        }
        catch( err )
        {
          return( "Error! " + err );
        }
}
retVal=RetrieveIncident("IM1001");
print("Testing the result " + retVal.IncidentID.getValue())
```

1. To test the request, enter `debughttp` in the sm.ini file and restart the server and client.
2. If the file `http.log` exists in the server's `RUN` directory, remove it or remove its contents so that there will be a fresh file to read.
3. Go into the calling JavaScript and click **Execute**. You will most likely get an error message because the Web service you are trying to reach is unavailable.
4. After the execution is complete, open the `http.log` file and search for the following:

   POST /sc62server/ws HTTP/1.1
   accept: application/fastinfoset, text/xml, textl,
   image/gif, image/jpeg, *; q=.2, */*; q=.2
   authorization: Basic ZmFsY29uOg==
   soapaction: Retrieve
   connection: Close
   content-type: text/xml; charset=utf-8
   content-length: 841
   cache-control: no-cache
   pragma: no-cache
   user-agent: Java/1.5.0_08

host: <server>:<port>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/";
xmlns:ns0="http://servername.port_number/SM/7/service_name.wsdl"
xmlns:ns1="http://schemas.xmlsoap.org/wsdl/http/";
xmlns:ns2="http://schemas.xmlsoap.org/wsdl/mime/"
xmlns:ns3="http://servicecenter.peregrine.com/PWS"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"><soap:Body>
<ns3:RetrieveIncidentRequest attachmentData=
"false" attachmentInfo="false"
**ignoreEmptyElements="true">
<ns3:model><ns3:keys><ns3:IncidentID**
mandatory="false"
readonly="false" type="String">IM1001</ns3:IncidentID>
</ns3:keys><ns3:instance><ns3:IncidentDescription
type="Array"/><ns3:Resolution type="Array"/>
<ns3:JournalUpdates
type="Array"/>
<ns3:Solution type="Array"/></ns3:instance>
</ns3:model></ns3:RetrieveIncidentRequest></soap:Body></soap:Envelope>
HTTP/1.1 200
Set-Cookie: JSESSIONID=0405ED23EFF6C9A3874F77796FE4210D;
Path=/sc62server
Connection: close
Connection: close
Content-Type: application/fastinfoset;charset=utf-8
Content-Length: 1323
Date: Wed, 04 Jun 2008 22:09:56 GMT
Connection: close
Connection: closeSet-Cookie: SessionId=16.95.106.150:3487;
Version=1

5. You can copy the bold area into an XML editor such as Altova® XMLSpy® and check whether it is correct XML. If it is, then the request is deemed to be successful (which doesn't necessarily mean that it will return data).

Another method to check the request and response is to run the request through a tool such as tcpmon. To do so, start tcpmon, enter the server name and port of the receiving Web service and connect the invoking JavaScript to tcpmon. Both the request and the response are visible in the tcpmon screen and can be analyzed in an XML editor as well.

Step 3: Test the response

After the request has been submitted successfully, test the response to the request, which is written to the http.logfile. Look for the following text. (The section that's bolded indicates that this is the response message):

```
SOAP-ENV (http://schemas.xmlsoap.org/soap/envelope/
<Envelope><Body>http://servicecenter.peregrine.com/PWSÏ cmn,
http://servicecenter.peregrine.com/PWS/CommonÏ xsd¬http:
//www.w3.org/2001/XMLSchemaÏ xsi(http://www.w3.org/2001/
XMLSchema-instanceð=, RetrieveIncidentResponsex message No
more) records foundx    returnCode@9x schemaRevisionDate
2008-05-30x schemaRevisionLevel@1x status
FAILURE{„… schemaLocation dhttp://servicecenter.peregrine.
com/PWS http://<server>:<port>/sc62server/ws/Incident.xsdð=,
model=, keys},  IncidentIDx typeEString&eth;' IM1001ÿ},
instancexrecordid  IM1001 - x
uniquequery  number="IM1001"ðE ,ð ÿÿÿðà
< #document8ÏSOAP-ENV(http://schemas.xmlsoap.org/soap/
envelope/ð?��Envelope?�� BodyxÍ%http://servicecenter.
peregrine.com/PWSÏ cmn,http://servicecenter.peregrine.com/
PWS/CommonÏ xsd¬http://www.w3.org/2001/XMLSchemaÏ xsi(http://
www.w3.org/2001/XMLSchema-instanceð=,
RetrieveIncidentResponsex message No (more) records foundx
returnCode@9x schemaRevisionDate  2008-05-30x
schemaRevisionLevel@1x status FAILURE{„… schemaLocation
dhttp://servicecenter.peregrine.com/PWS http://geist8440.
americas.hpqcorp.net:13701/sc62server/ws/Incident.xsdð=,
model=, keys},  IncidentIDx typeEStringð' IM1001ÿ},
instancexrecordid  IM1001 - x
uniquequery  number="IM1001"ðF ,ð ÿÿÿÿ
HTTP/1.1 200
Keep-Alive: timeout=60, max=100
Connection: Keep-Alive
Pragma: requestnum="185"
Content-Encoding: gzip
Content-Type: application/fastinfoset;charset=utf-8
Transfer-Encoding: chunked
Date: Wed, 04 Jun 2008 22:09:56 GMT
```

1. Copy the section mentioned above from the http.log file into a text file and assign it a name such as responsetest.xml. If you used tcpmon to get the information, you can use the XML response as is. If it came from the http.log, you will need to modify the special characters in the log to correct XML syntax.
2. Change the calling JavaScript to override the invoke function to read and interpret the contents of the responsetest.xml file. The following is the section of the code needed to do that.

```
// Temporarily override the "invoke" function to replace it
with
// a function which reads an XML response from a file
<ServiceObject>.invoke = function( ) {
var resultObj = new Object();
resultObj.responseObj = null;
var resultXML = new XML();
resultXML.setContent( "c:\\<path>\\<responsetest.xml>",
true );
try
{
lib.SOAP.deserialize( "<name of the generated JavaScript>",
resultXML.getDocumentElement(), resultObj );
}
catch( e )
{
print( "Error deserializing response: " + e.toString() );
return null;
}
try
{
this.soapEnvelope = resultObj.soap_Envelope;
this.soapBody = resultObj.soap_Envelope.soap_Body;
if ( this.soapEnvelope.soap_Header != undefined )
{
this.soapHeader = this.soapEnvelope.soap_Header;
}
else
this.soapHeader = null;
return resultObj.soap_Envelope.soap_Body.getContent();
}
catch( e )
{
print( "Error extracting Response Object: " + e.toString() );
return null;
}
}
```

3. Change the line of the calling JavaScript that invokes the Web Service from <ServiceObject>.invoke to simply invoke to call the invoke function defined within that calling JavaScript.
4. Click **Execute** to run this modified JavaScript. If it finishes without errors, the response is deemed successful.

If any of the above tests fail to complete, contact Service Manager Customer Support and provide the WSDL file, the request and response xml text files with any error messages, and the `sm.log` and `http.log` files with debughttp turned on.

1.16.9. Max sessions exceeded in Web Services

The Service Manager server runs out of available sessions.

Cause

If a Web Services request contains "connection: keep-alive" or it uses HTTP/1.1 without a connection header, the Service Manager server will keep the session alive for a predefined interval that's defined by setting the "webservices_sessiontimeout" parameter in the `sm.ini` file. If a Web Services client doesn't reuse the session for subsequent requests by providing valid headers, the Service Manager server creates a new session for each subsequent request and quickly run out of available sessions.

Solution

To avoid running out of available sessions, try the following solutions one by one.

Solution 1

Set the HTTP header "connection:close" so that the Service Manager server will not keep a Web Services session open after the current request is finished.

Solution 2

Utilize the Web Services session persistence by doing one of the following to reuse the existing Web Services session on the Service Manager server.

1. Use connection: keep-alive. If the connection header is missing, it defaults to "keep-alive" for HTTP/1.1.
2. The Web Services client needs to supply a session cookie with the same user log-in information that created the session.

> **Note**
>
> Even with Web Services session persistence, each SOAP API request is stateless, so that requests are handled independently between one another.

1.16.10. Cannot troubleshoot HTTP socket connections

Cannot troubleshoot HTTP socket connections.

Cause

The Service Manager server attempts to keep an HTTP socket connection open as long as possible, but the protocol requires that it must close if the server returns a SOAP fault. If there is no successful authentication, it must return a SOAP fault.

Solution

Redirected ports

To ensure the client has the correct hostname and port number, a SOAP client application can direct requests to the TCP port number used by the sm -httpPort instance, but must be able to recognize SOAP header values in the initial response:

- redirectServerHost
- redirectServerPort

The server returns these SOAP header values identifying the dynamically allocated TCP host and port number for the spawned process. During the client/server session, subsequent SOAP requests must be directed to the same hostname and port identified in the initial response.

TCP ECONNRESET messages

If a client/server connection using a spawned child thread terminates, the sm -httpPort child thread receives a TCP ECONNRESET message. The child thread responds to this by self-terminating to ensure that orphaned child thread doesn't collect on the server. However, poorly-designed client applications, or other third-party SOAP tools, that don't gracefully close a connection could cause the server process to see a TCP ECONNRESET message, and that also terminates the server thread.

Contents

- 1 Cause
- 2 Solution
  - 2.1 Redirected ports
  - 2.2 TCP ECONNRESET messages

1.16.11. The "Unable to connect to the remote server" error occurs

The most common error occurs when your Web Services client application fails to obtain a response, or you receive this error message:

Server Error in <name> Application

The underlying connection was closed: Unable to connect to the remote server

Cause

The Web Services client may be directing SOAP requests to the wrong host or to the wrong TCP port number. Service Manager generates WSDL files that contain the hostname and TCP port number for the Service Manager server instance receiving the request. The Web Services client application may use the hostname and port number used during application development, but the production hostname and port might be different each time the application runs if they are dynamically allocated.

If the server instance generating the WSDL is different from the hostname or port number receiving the client application requests, the client/server connection will fail.

Solution

Follow these rules to ensure successful client/server SOAP communication.

- Ensure that your Web Services request is not running against a common port with heavy server traffic. Otherwise, tracking request and response messages will be difficult.
- Type the following at the command line to generate debug logs:

  sm -httpPort:unique portnumber -sslConnector:0 -debughttp:1 -log:../logs/debug.log

  where

  `-httpPort` identifies a dedicated port for Web Service client connections

  `-log` defines a path to store the logs for this connection.

  > **Note**
  >
  > It is easier to troubleshoot errors if each SOAP client application connects to its own TCP port number.

- Examine the HTTP.LOG file for response messages. You can use this information to determine where a Web Service client connection failure occurs.

Contents

- 1 Cause

- 2 Solution

1.16.12. Calling external web services with SSL fails

Calling external web services with SSL encryption from Service Manager fails. The sm.log file contains an exception like this:

```
JRTE E java.security.PrivilegedActionException:
com.sun.xml.messaging.saaj.SOAPExceptionImpl: Message send failed
RTE E Error calling method: doSoapRequest  in class:
com/hp/ov/sm/server/utility/SoapClient Exception
(com.sun.xml.messaging.saaj.SOAPExceptionImpl:
java.security.PrivilegedActionException:
com.sun.xml.messaging.saaj.SOAPExceptionImpl: Message send failed)
RTE E SCException caught in soapRequest(): Error calling method:
doSoapRequest  in class: com/hp/ov/sm/server/utility/SoapClient Exception
(com.sun.xml.messaging.saaj.SOAPExceptionImpl:
java.security.PrivilegedActionException:
com.sun.xml.messaging.saaj.SOAPExceptionImpl: Message send failed)
RAD E Error calling method: doSoapRequest  in class:
com/hp/ov/sm/server/utility/SoapClient Exception
(com.sun.xml.messaging.saaj.SOAPExceptionImpl:
java.security.PrivilegedActionException:
com.sun.xml.messaging.saaj.SOAPExceptionImpl: Message send failed)
```

Additionally, a sm_<process identifier>_stdouterr.log was created. The relevant stack trace is:

```
Caused by: javax.net.ssl.SSLException: Received fatal alert:
unexpected_message
at com.ibm.jsse2.j.a(j.java:21)
at com.ibm.jsse2.j.a(j.java:32)
at com.ibm.jsse2.qc.b(qc.java:5)
at com.ibm.jsse2.qc.a(qc.java:645)
at com.ibm.jsse2.qc.h(qc.java:809)
at com.ibm.jsse2.qc.a(qc.java:106)
at com.ibm.jsse2.qc.startHandshake(qc.java:586)
at com.ibm.net.ssl.www2.protocol.https.c.afterConnect(c.java:15)
at com.ibm.net.ssl.www2.protocol.https.d.connect(d.java:39)
at
sun.net.www.protocol.http.HttpURLConnection.getOutputStream(HttpURLConnect
ion.java:1090
<http://www.protocol.http.httpurlconnection.getoutputstream(httpurlconnect
ion.java:1090/> )
at com.ibm.net.ssl.www2.protocol.https.b.getOutputStream(b.java:33)
at
com.sun.xml.messaging.saaj.client.p2p.HttpSOAPConnection.post(HttpSOAPConn
ection.java:323)
```

Cause

The remote endpoint is not supporting Server Name Indication (SNI), which is activated by default as of JRE 1.7. The issue is related to the remote endpoint (in this case, the web services server), and whether it supports SNI extensions during the SSL handshake or not. When Service Manager uses JRE 1.8, it starts to use SNI extensions during the SSL handshake. If the remote endpoint doesn't, the web service will fail with the error message provided above.

Solution

Activate SNI at the remote endpoint. This is the recommended solution. If the remote endpoint doesn't support SNI extensions, and SNI can't be activated, add the following JVMOption<n> parameter either to the `sm.ini` file, or to the start command of the servlet(s) in the `sm.cfg` file:

`JVMOption2:-Djsse.enableSNIExtension=false`

This will disable the SNI security feature in Java.

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1.17. Troubleshoot SCAuto SDK

This section helps you identify and resolve some of the issues that you might come across while using ServiceCenter Automatic (SCAuto) Software Development Kit (SDK). Some of these issues include:

- Listener-to-application connection fails
- Internal event: RAD application not invoked
- External event: SCAuto application not invoked
- No internal event: SCAuto application event occurs
- No external event: RAD application invoked

Contact Software Support

If you need assistance in the resolution of your problem, make sure you have the following data available before calling Software Support:

- Core files
- Error logs produced by your program or SCAuto application with the `-debug` option (scsubmit, scfiled).
- Input and output files for scfiled errors
- Standard output and error files for manually executed scsubmit commands
- Service Manager log and Scheduler logs (if specified)
- Service Manager msglog for affected applications or services
- Service Manager print of agent status (Event Services menu)
- Unloaded Event records relevant to the errors
- Unloaded filter specifications (if relevant)

1.17.1. Listener-to-application connection fails

If the connection between SCAuto Listener and the SCAuto application fails, make sure that Listener is running, that the TCP/IP specifications are correct, and that there is connectivity between the server and client.

Cause

The connection between Listener and the SCAuto application (for example, sdktest) fails.

Solution

Do the following:

1. Make sure Listener is running on the Service Manager server platform.
   To check, run the Service Manager status command:
   - If Listener is not running, start it from the status option `start schedulers`, and then select `scauto.startup`.
     If the start fails, the Listener daemon writes log messages to the **sm.log** file. Check the log for any error messages. Also, check the TCP/IP specifications for your platform. A service name other than that specified for Service Manager server is required. The Listener keyword specified in the file must match this name.
   - If Listener is running, check the service name specified in the file on the Service Manager platform for the Listener keyword.
     If the keyword is not specified, Listener defaults to the service name scauto. This name is useful when checking TCP/IP specifications.
2. Make sure the TCP/IP specifications are correct on the SCAuto platform.
   Make sure the host and service are specified, and that they match the host and service name of Listener.
   To verify the specifications, you can use either of the following:
   - *Local files*
     On Linux, these files are located in the following directories:
     `/etc/hosts`

     `/etc/services`
   - *Name server*

1.17.2. Internal event: RAD application not invoked

If an internal event occurs but Rapid Application Development (RAD) application isn't invoked, check Event Scheduler and the review schedule record specifications.

Cause

You have an event in the `event.in` file, but RAD application isn't invoked. For example, you have a **pmo** event, but the problem isn't opened.

Solution

Do the following:

1. Make sure the Event Scheduler is running on the Service Manager server platform.
   To check, run the Service Manager status command:
   - If the Event Scheduler isn't running, start it from the status option start schedulers, and then select event.startup.
     If the start fails, review error messages and retry. If errors persist, call Software Support.
   - If the Event Scheduler is running but not processing, stop the Event Scheduler, build a new schedule record, and restart.
2. Review schedule record specifications.
   For details, see Troubleshoot Event Services.

1.17.3. External event: SCAuto application not invoked

If an external event occurs, but the external SCAuto application is not invoked, check SCAuto Listener and the SCAuto application.

Cause

You have an event in the **event.out** file, but the external SCAuto application is not invoked. For example, you have an scfiled event, but no event in output file.

Solution

Check the following:

1. Make sure there is connectivity to Listener.
   For instructions, see Listener-to-application connection fails.
2. Make sure the SCAuto application (for example, `scfiled`) is active and retrieving the event:
   If the SCAuto application is not running, check the **scfiled.log** for errors.
   Correct errors and restart, using appropriate command with the debug option, if available.
   The debug option `(-debug)` enables you to determine if the event was retrieved, or whether the retrieval failed:
   - If the event was retrieved, it is an SCAuto application failure.
   - If event was not retrieved, check the SCAuto application for connect and sequence number specification:
     Example:
     `scauto_connect (...,events, user), scauto_query (id, startseq,.), scfiled -evlist -user -checkpoint`
     If you do not specify these fields correctly, events are not retrieved.
     The sequence number of the event to be retrieved must be equal to or greater than your specification. Also, you must specify the events you will service on your connect statement or `scfiled` parameter. If you specify anything other than all for your event and user specifications, make sure the event in the event.out file matches your criteria.

1.17.4. No internal event: SCAuto application event occurs

If an internal event does not occur, but an external SCAuto application event occurs, check SCAuto Listener and the SCAuto application.

Cause

You do not have an event in the event.in file, but an external SCAuto application event occurs. For example, a pmo event is sent, but is not in Service Manager.

Solution

Check the following:

1. 1 Make sure there is connectivity to Listener.
   For instructions, see Listener-to-application connection fails.
2. Make sure the SCAuto application (for example, `scfiled`) is active.
   If the SCAuto application is not running, check the **scfiled.log** for errors. Correct the errors and restart, using appropriate command with the debug option, if available.
   The debug option enables you to determine if the failure is an SCAuto application failure or a connection failure. Common application failures are invalid event structures in the scfiled input `<infile>` or invalid specifications on the `scauto_create (id,...)` statement. Your application should check all return codes and log any errors.

1.17.5. No external event: RAD application invoked

If an external event doesn't occur, but the Rapid Application Development (RAD) application invokes, see Event Services.

Cause

You don't have an event in the event.out file, but the RAD application invokes. For example, the system sends an email, but there is no email event for it.

Solution

Review Troubleshoot Event Services.

1.18. Access logs

The log files are available at the following locations:

- Service Manager server log file (sm.log):

  By default, the log file is located in `<Service Manager home>\Server\logs`.

- Service Manager Windows client log file:

  Go to **Windows** > **Preferences** > **Service Manager** > **Logs**. The **Log file** field specifies the log file path.

- Service Manager web client log file:

  By default, the log file is located in the Web application server's home directory: `<Web application server home directory>/sm.log`.

- Smart Analytics log files:

  By default, you can find the Smart Analytics servers log files under corresponding folders in which different Smart Analytics components are installed, such as `<SmartAnalytics Installation>\IDOL`, `<SmartAnalytics Installation>\ImageServer`, and `<SmartAnalytics Installation>\Content1`.

Related topics

- For information about the Service Manager server log file, see Server log file.
- For information about the Service Manager client log file, see Client log file.
- For information about the Smart Analytics log files, see Check Smart Analytics log files.
- To contact support, see Contact support.

1.19. Contact support

You can contact support at Support and services.

Before you contact support, ensure that you have performed the following:

- You have performed the general checks. See General check.
- You have collected the relevant data about the problem you will send to the Customer Support Service:
    1. A description of the problem, including timestamps if applicable.
    2. A description of your environment.
        1. Is this a production or test environment?
        2. Is it a new implementation - install or upgrade?
        3. Any environment changes before encountering the issue?
        4. Has the observed behavior ever functioned correctly before the issue?
    3. Exact and unambiguous steps to reproduce the problem.
    4. Expected behavior.
    5. A screenshot of your system version information:
        1. Sign in to the Windows Client.
        2. From the System Navigator, click **System Administration** > **Base System Configuration** > **Monitoring** > **Version Information**.
    6. The `sm.log` file.
    7. All `sm.ini` and `sm.cfg` files from all servers.
    8. Any screenshots that help to describe the problem.

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